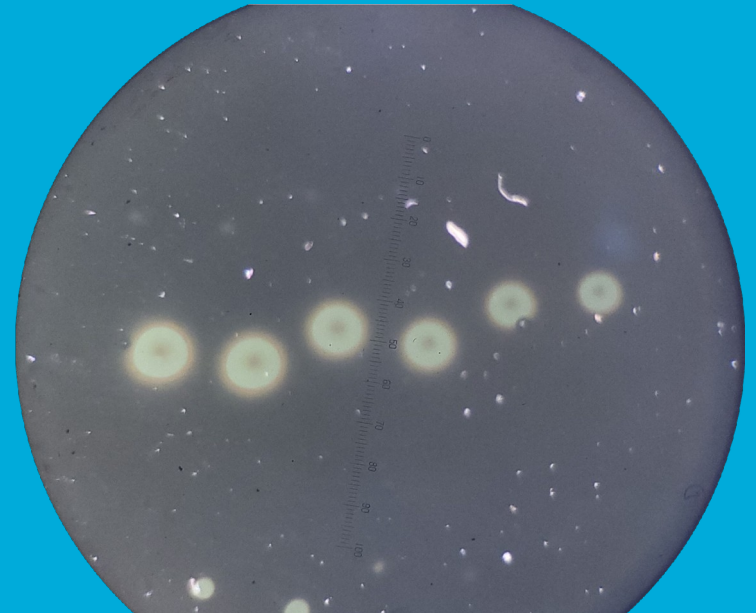


Optical Analysis of GRIN Lenses using TMM Calculation

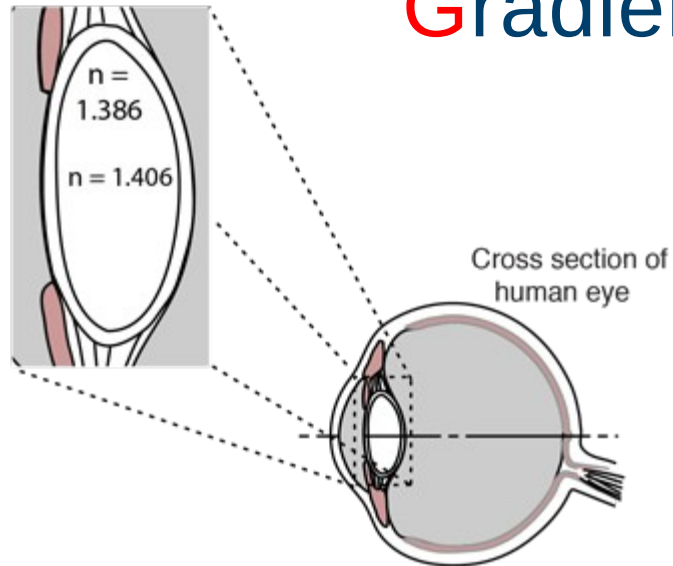
Soeke Sabelus



Introduction

$$n = \frac{c_0}{c_m}$$

Gradient Refractive Index Lens



Source: [1]

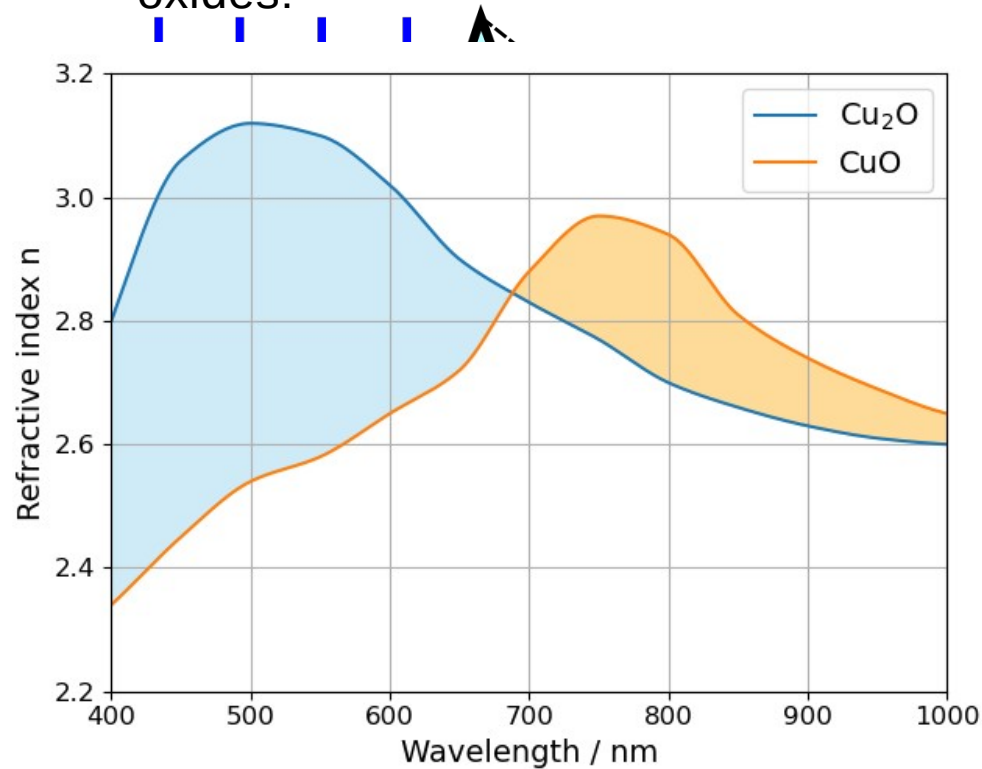
GRIN rod-lenses for fibre couplers



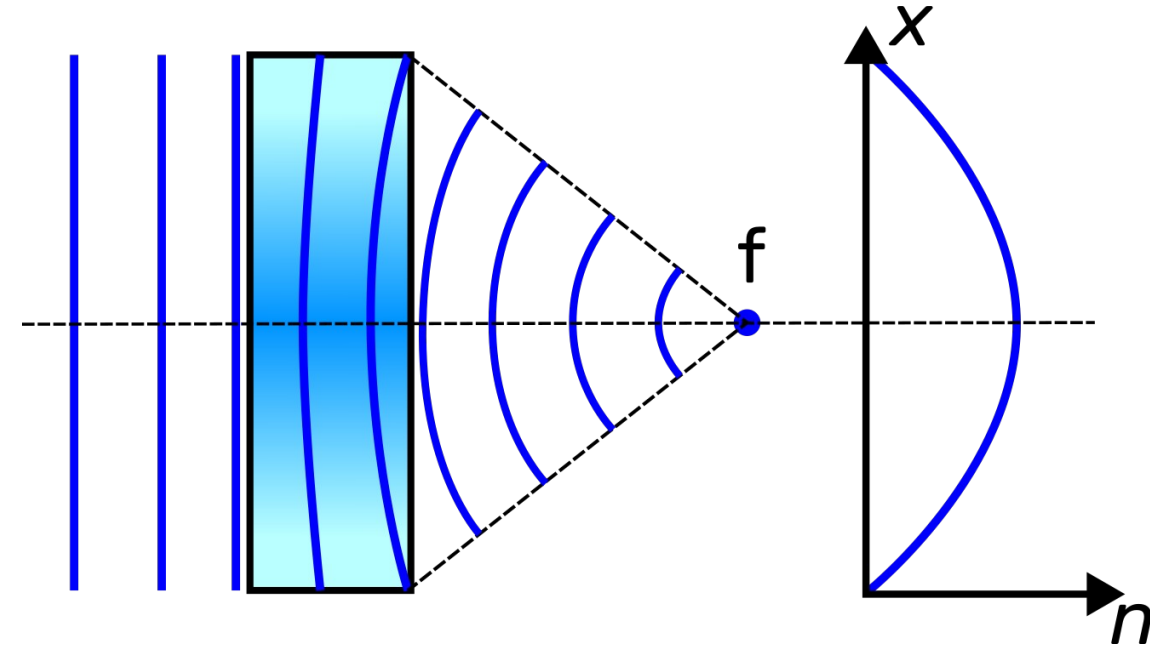
Source: [2]

Theory: Optical Lenses (Wave Optics)

Refractive index of 100nm thick
oxides:



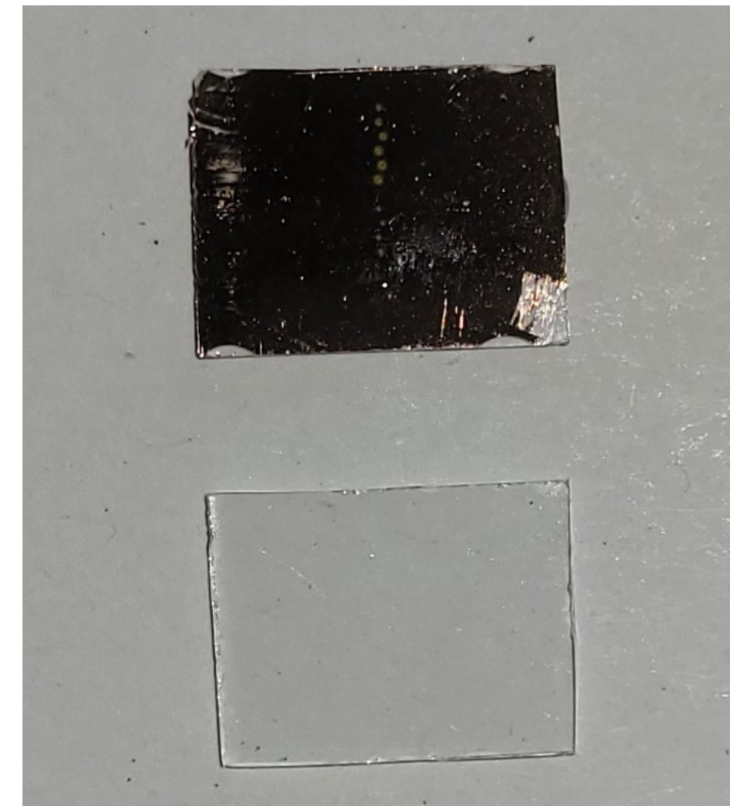
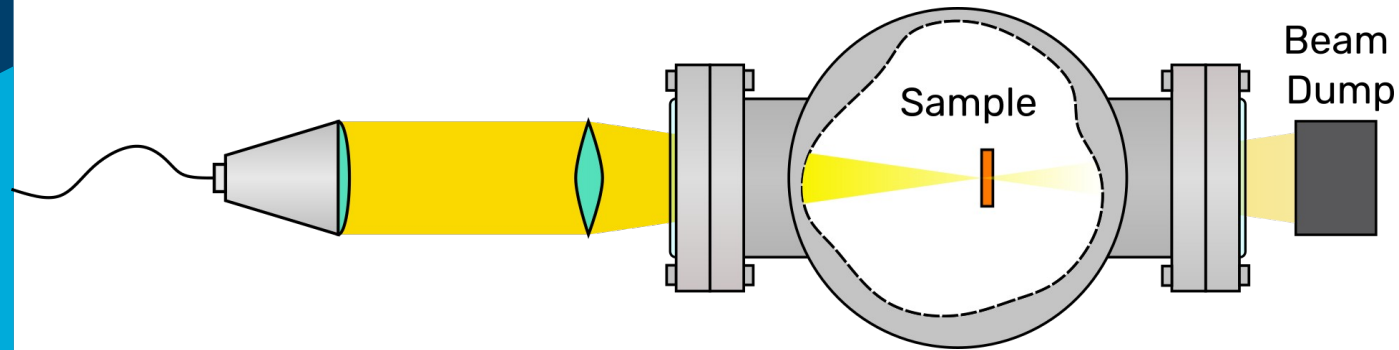
Gradient refractive index (GRIN) lens:



→ Phase difference due to refractive index
gradient Δn

Sample Preparation

- 1) Evaporate Cu on glass substrate (~50 nm)
- 2) Laser annealing in 50 mbar O₂

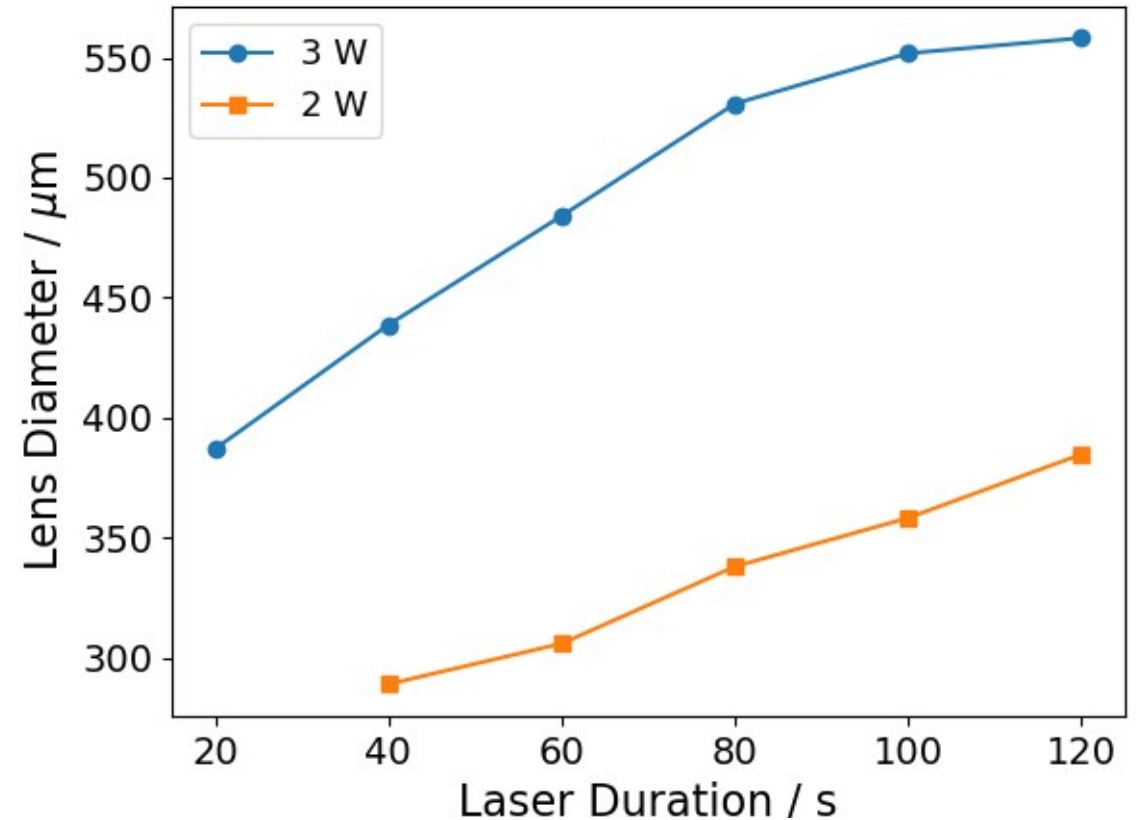
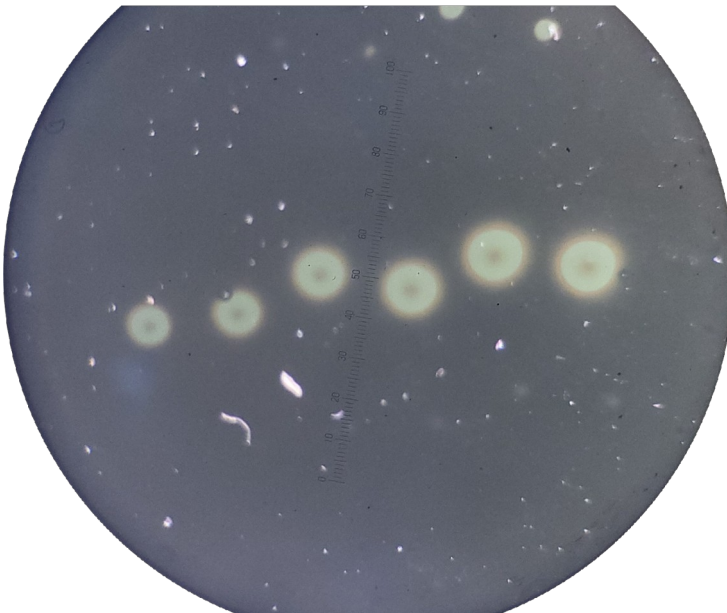


Sample (top) and
substrate glass (bottom)

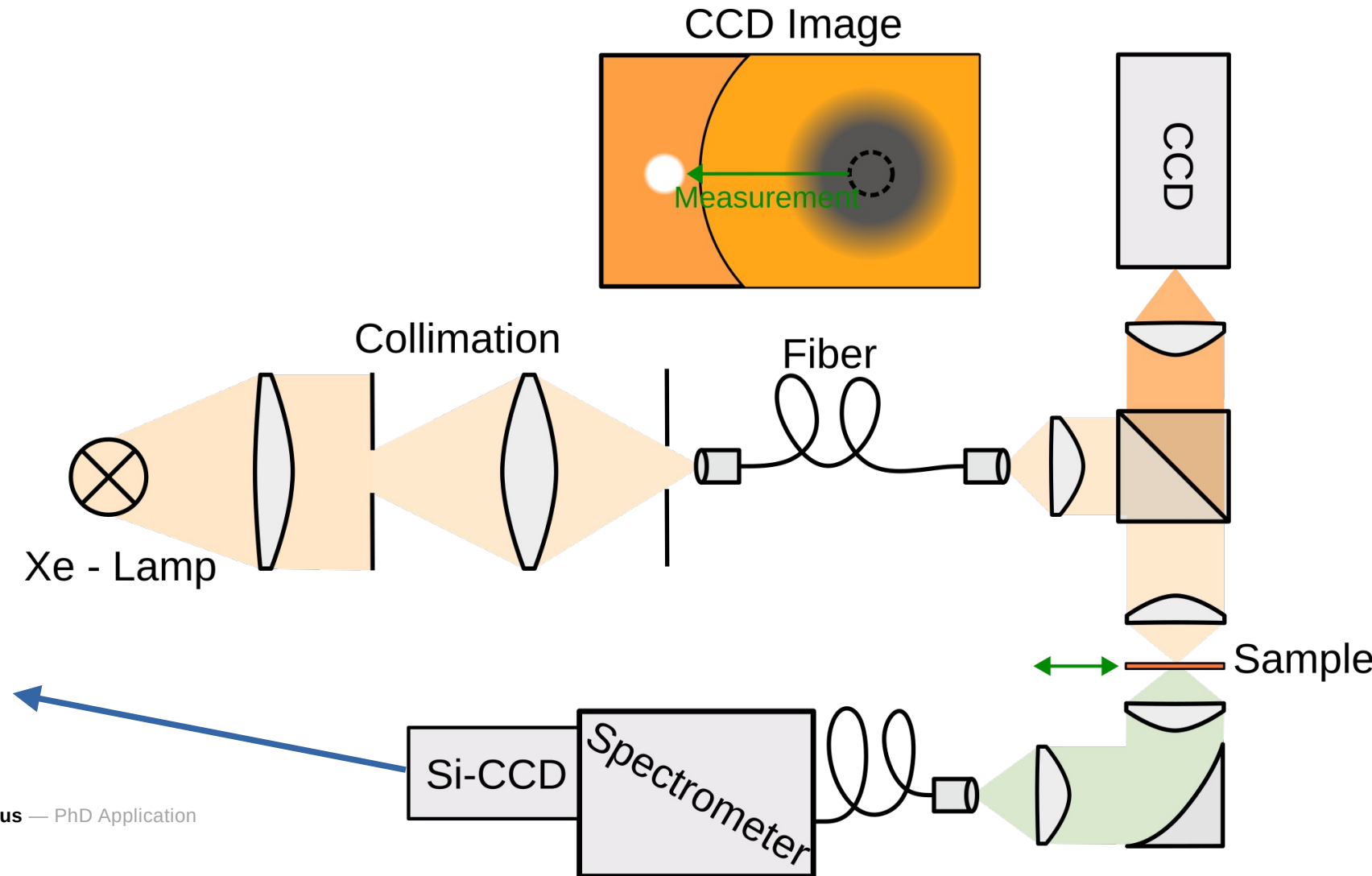
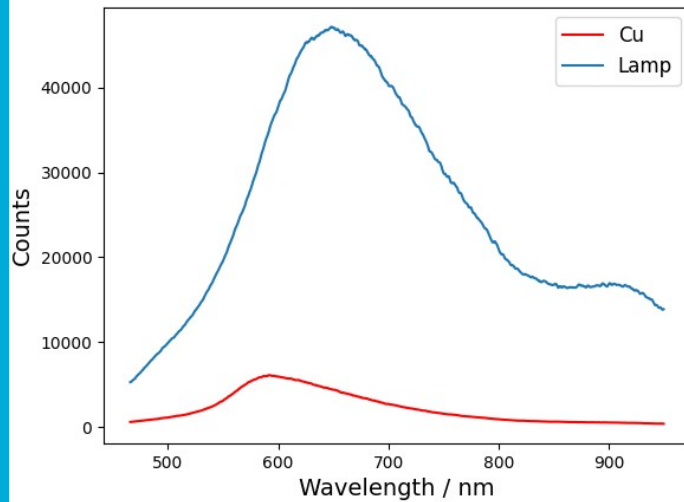
- 3) Analysis of optical properties through transmission spectra

Effect of Pulse Duration on Lenses

- Size increases with pulse length
- Color difference at longer illuminations in the center of lens
- 3W series: (20s → 120s)

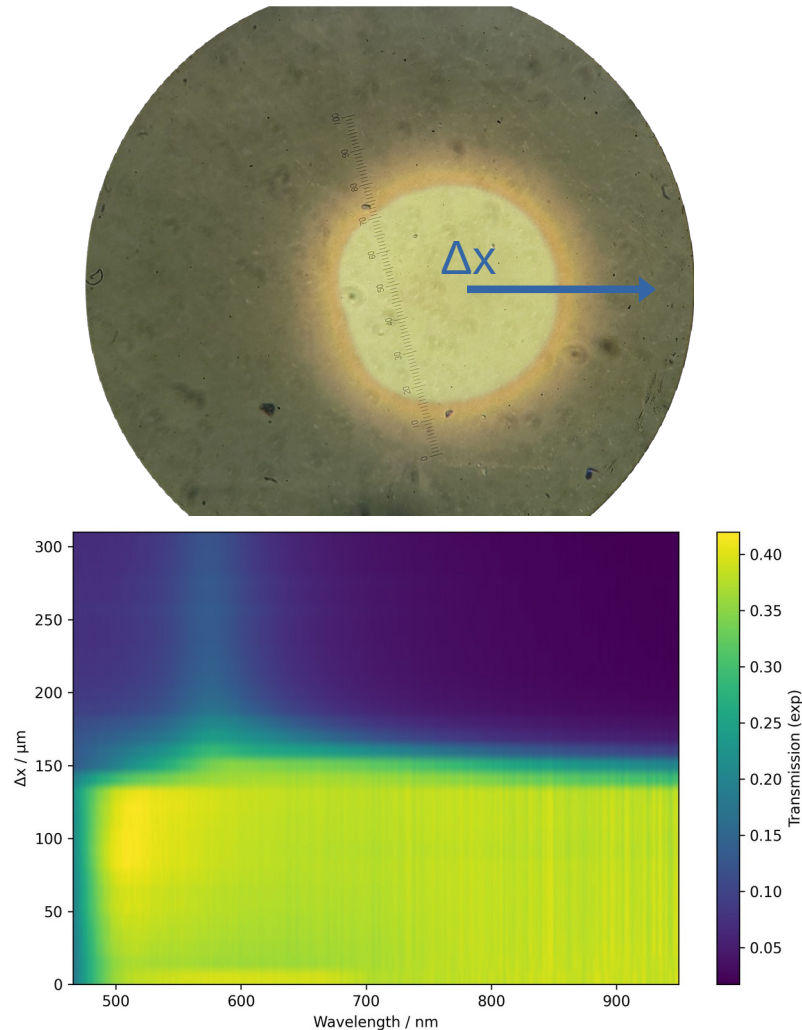


Transmission Spectrum Measurement Setup

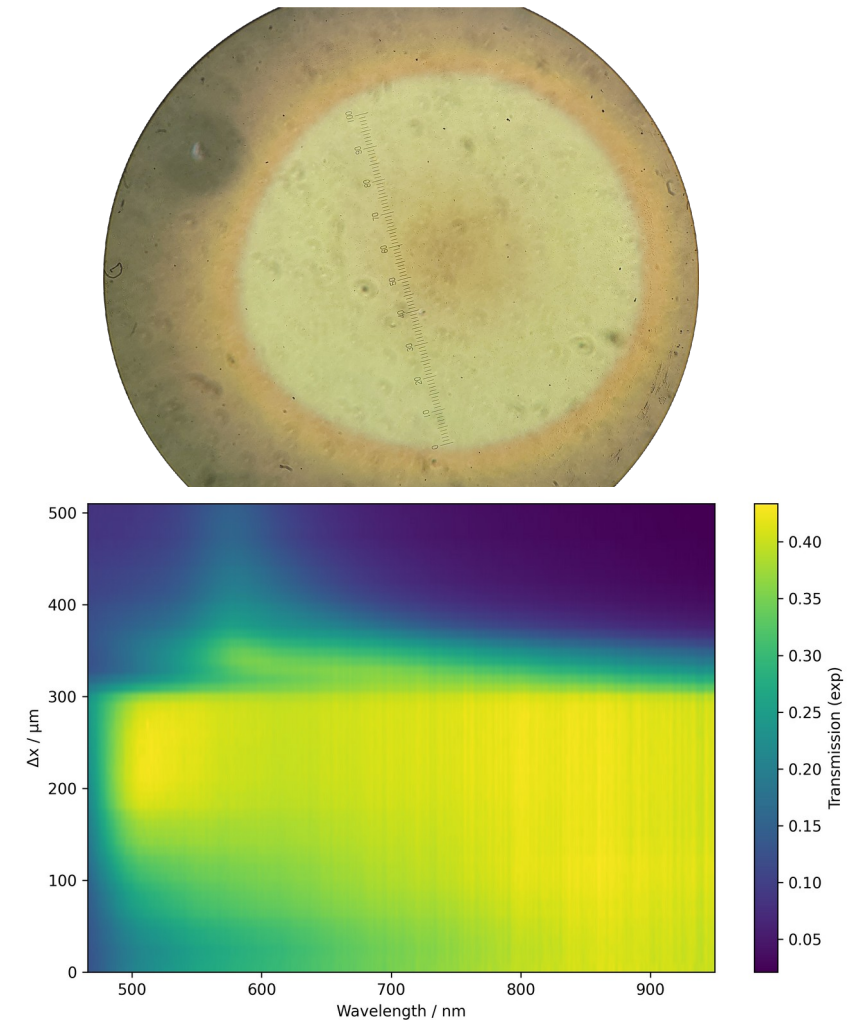


Example Lenses at Different Power and Duration

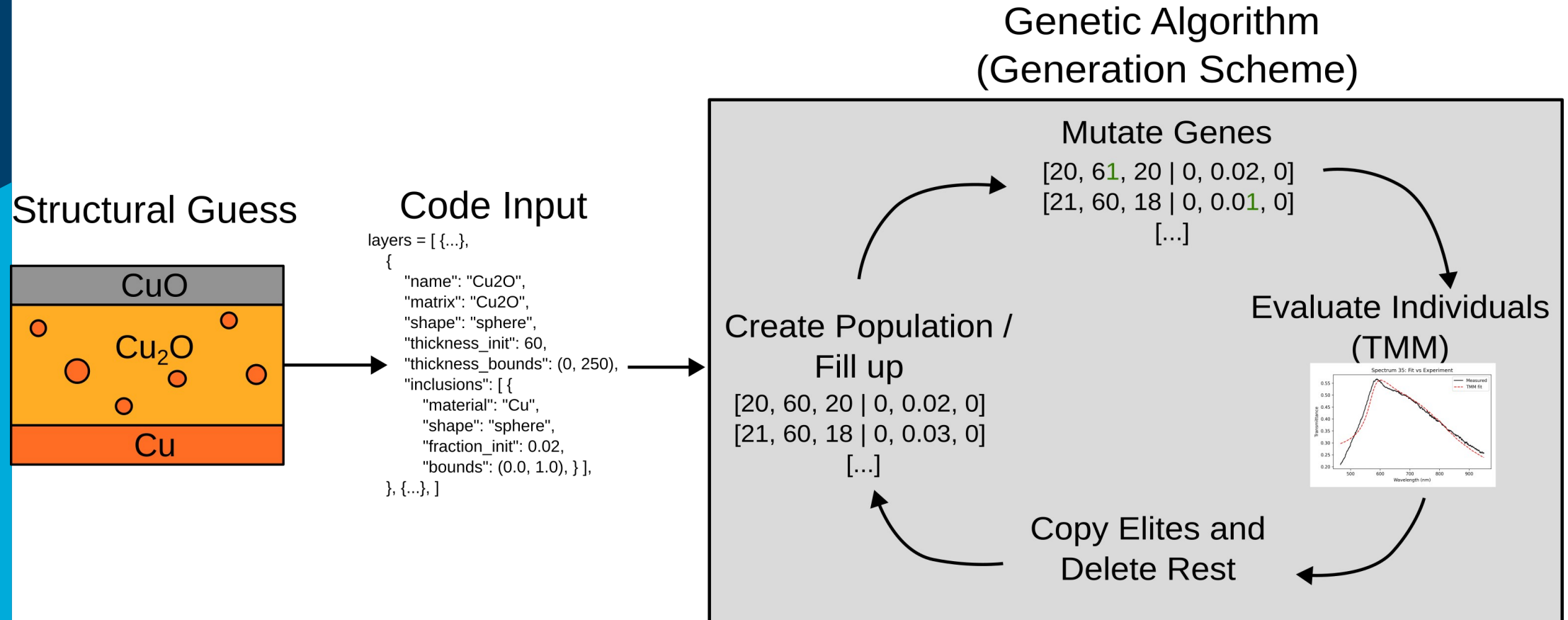
Lens 1: 2W laser power, 60s illumination



Lens 2: 3W laser power, 120s illumination

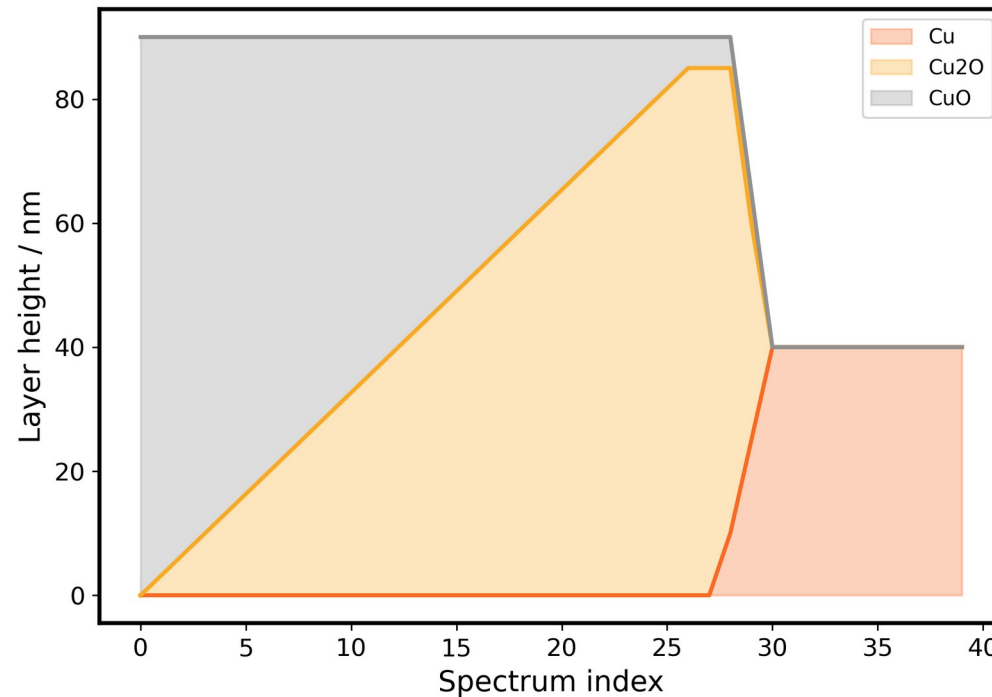


Fitting with Transfer Matarix Method (TMM)

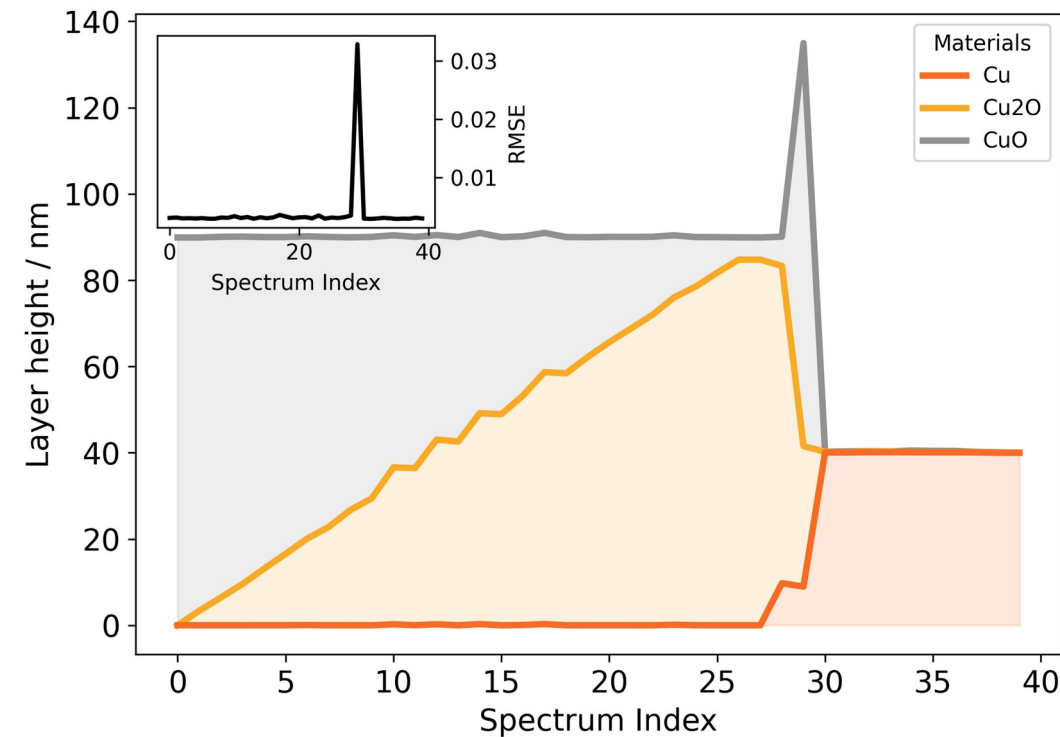


Benchmarking of the Genetic Algorithm (GA)

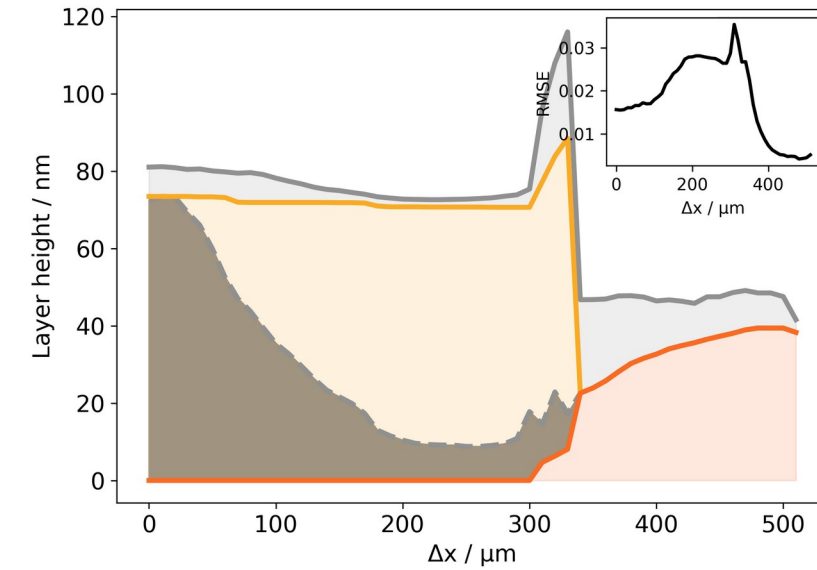
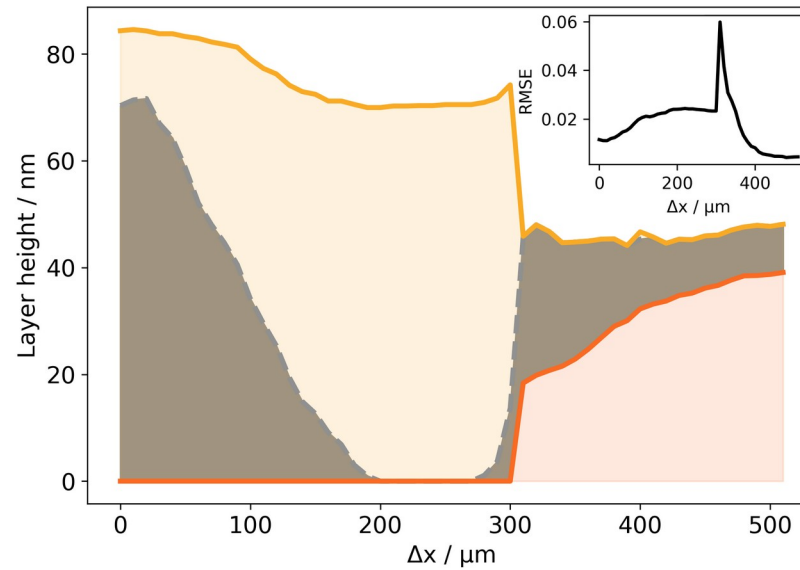
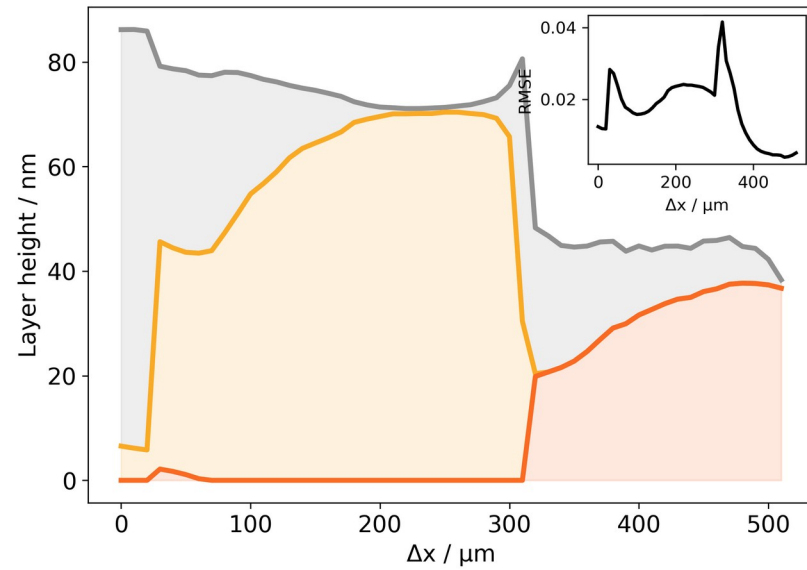
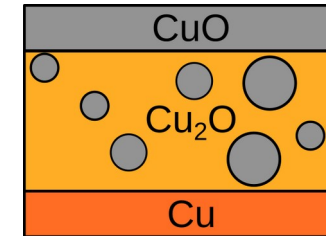
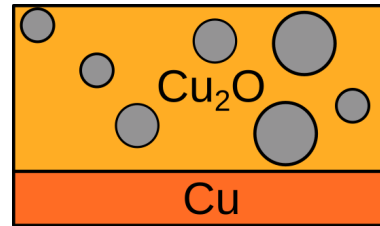
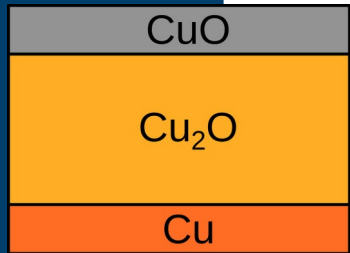
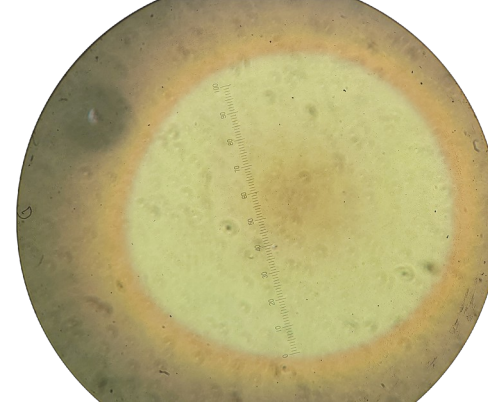
– Generate theoretical transmission:



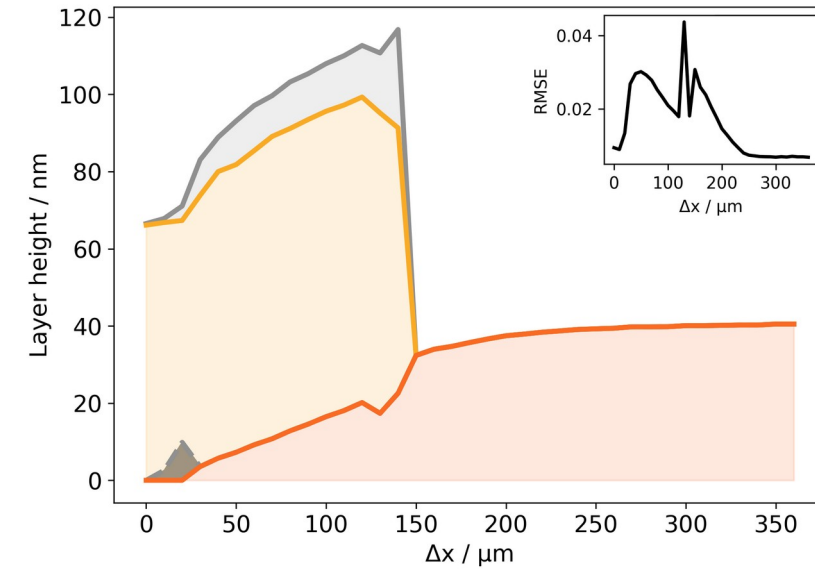
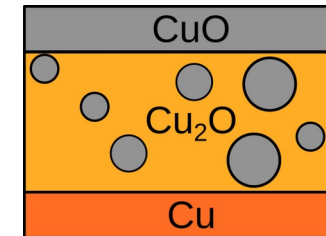
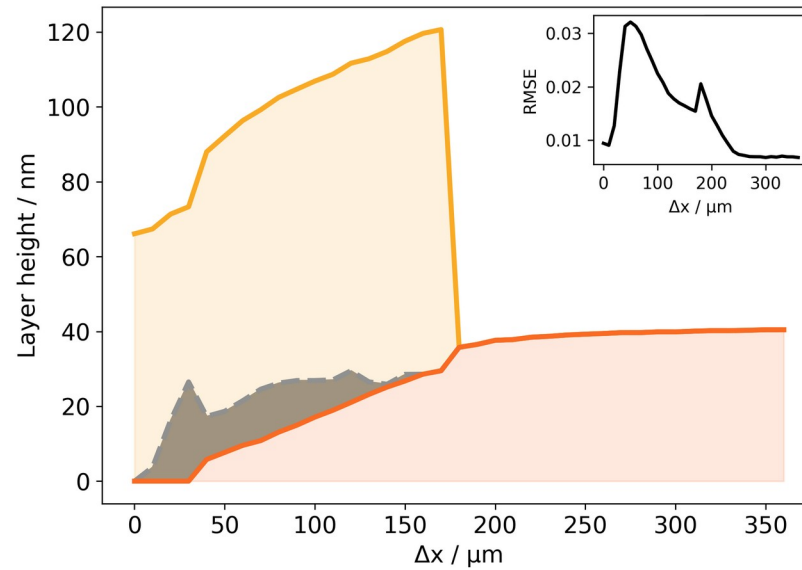
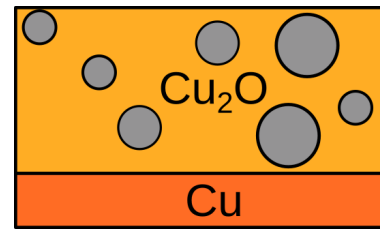
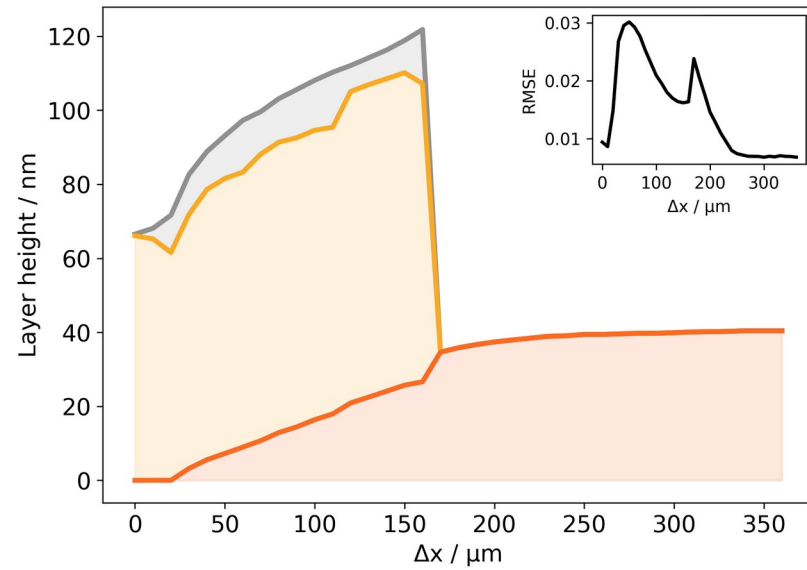
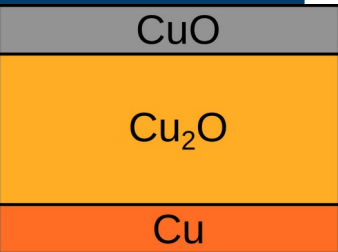
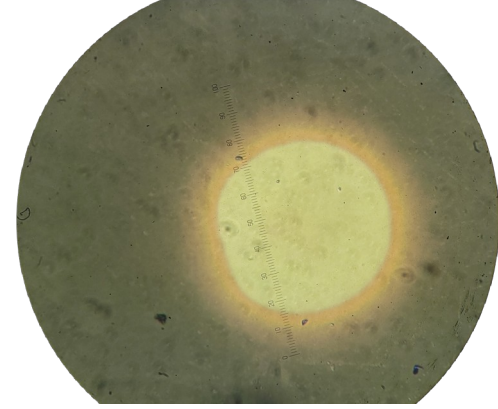
– GA result to simulated data:



Lens 3W, 120s Fitting Results

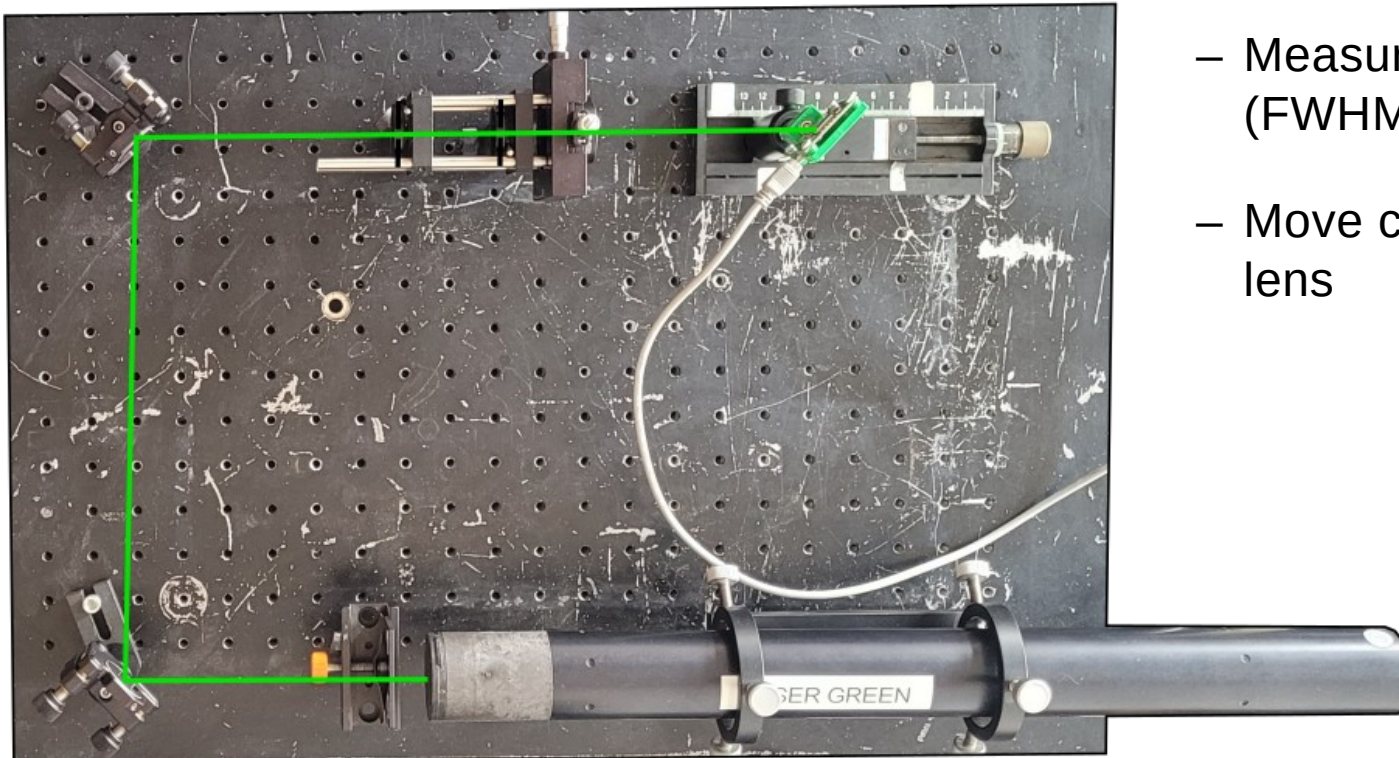


Lens 2W, 60s Fitting Results



Optical Properties of the GRIN Lenses

– Setup:

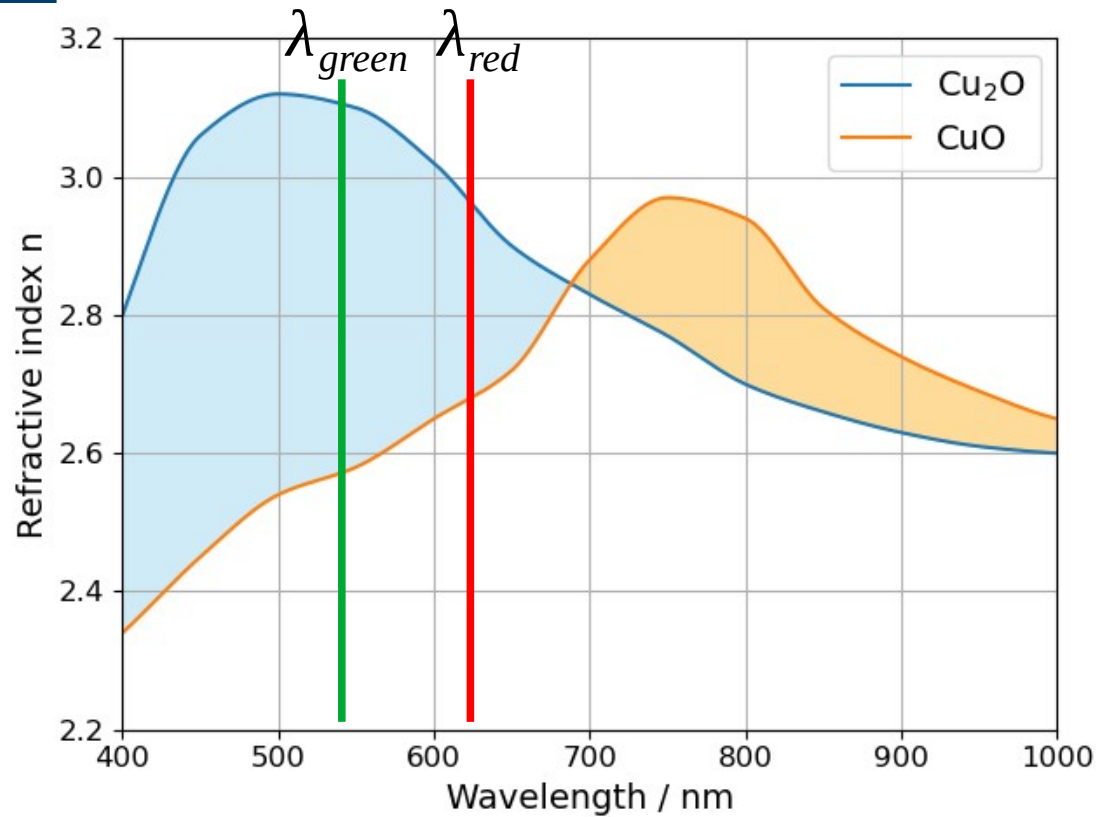


Goal:

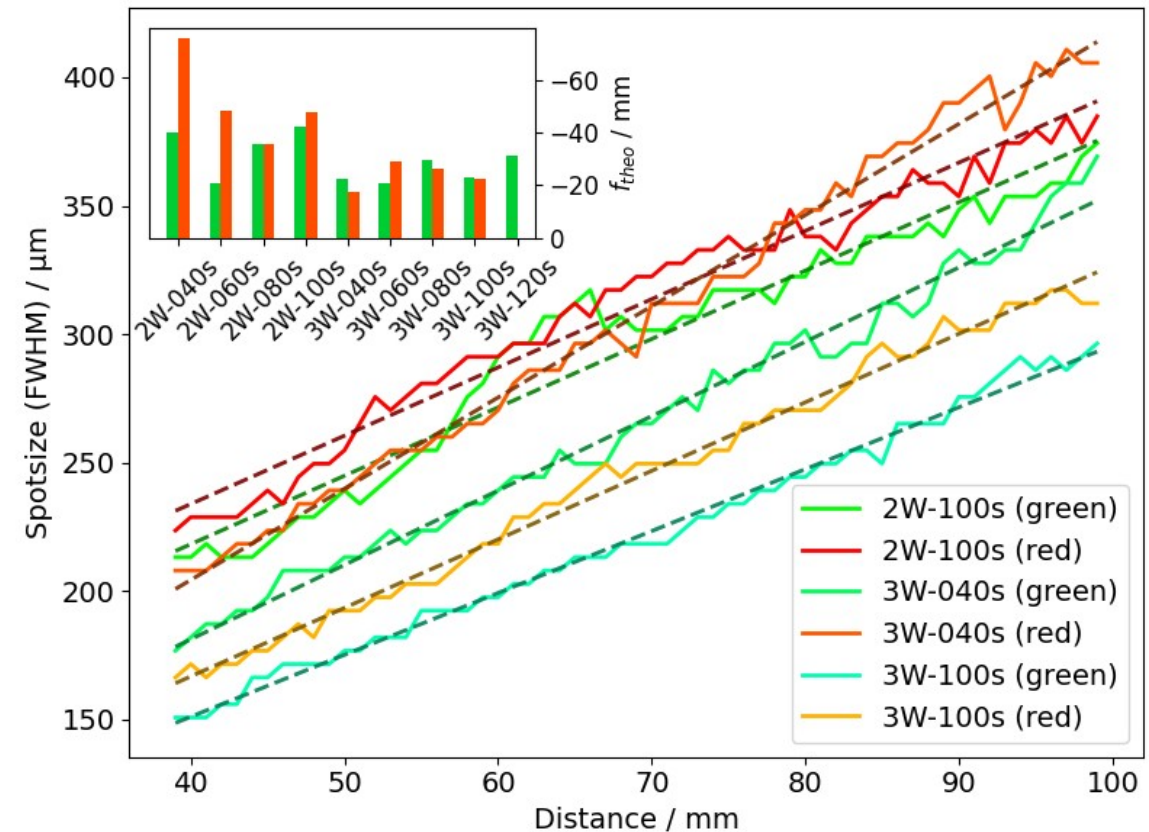
- Measure full width at half maximum (FWHM) using camera
- Move camera to map out focus of lens

Optical Properties of the GRIN Lenses

– He-Ne laser divergence



– GRIN lens divergence



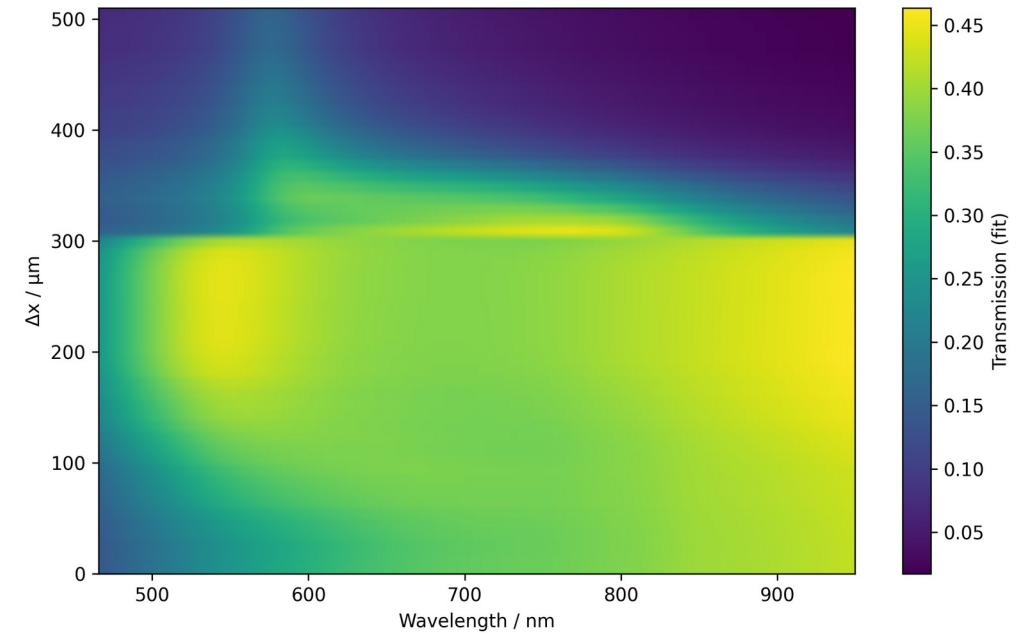
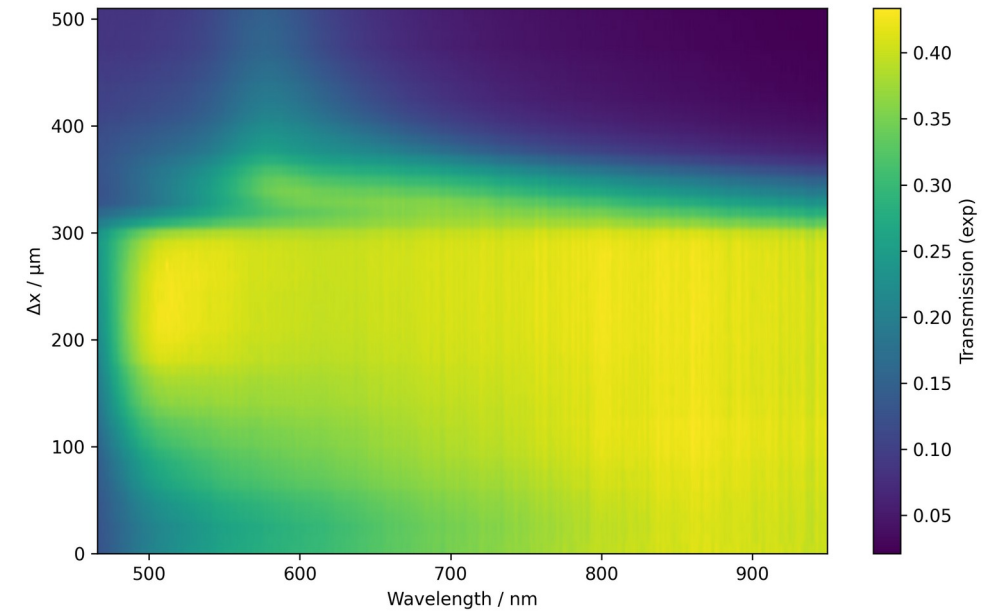
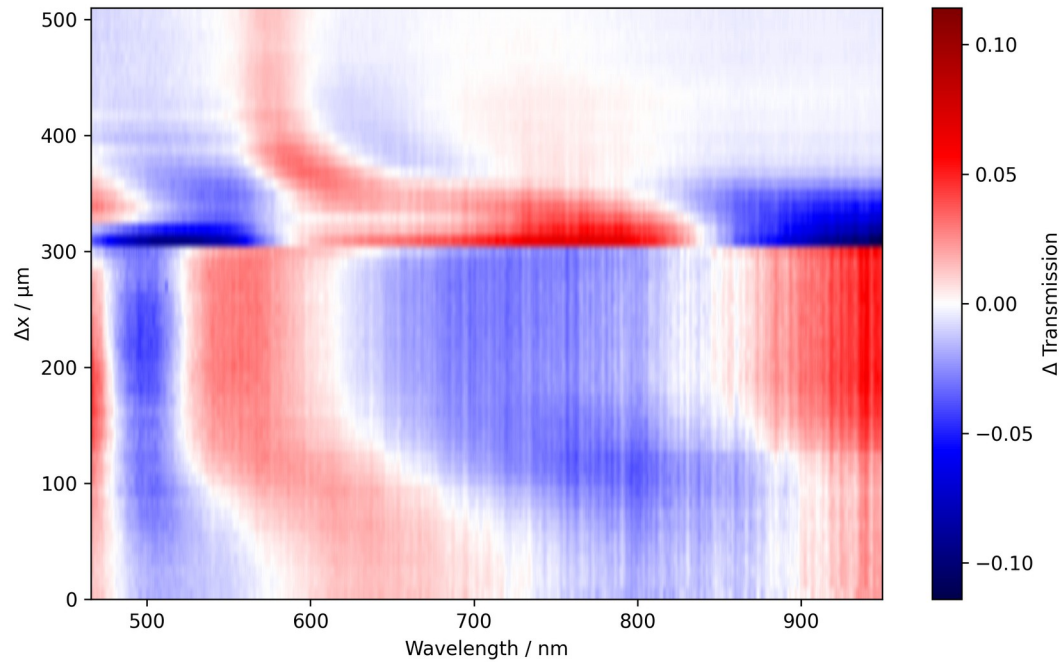
Sources

- [1] R. Nave, HyperPhysics – Light and Vision, <http://hyperphysics.phy-astr.gsu.edu/hbase/vision/eyescal.html>
- [2] Grintech GmbH, Jena, <https://www.grintech.de/produkte/grin-stablinsen/>

Thank you for listening! Any questions?

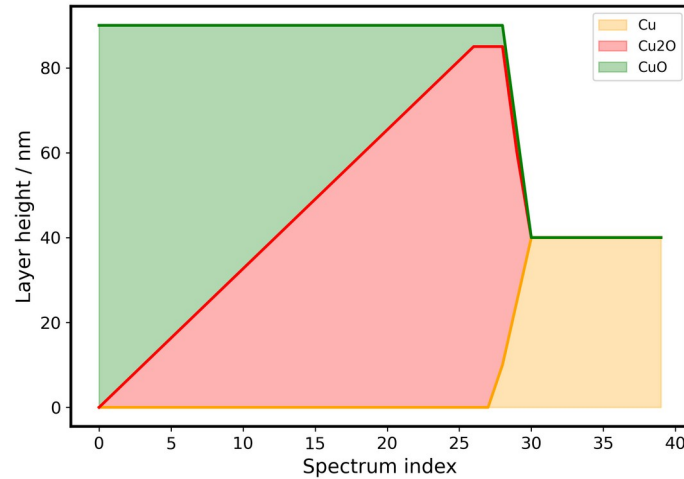
Lens 120s 3W comparisson

– Top experiment, bottom simulation

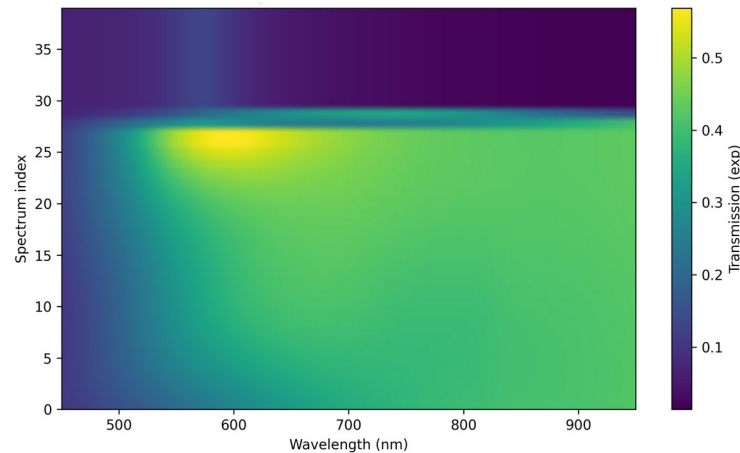


Benchmarking of the Genertic Algortihm

- Simulate Spectra (that resemble a GRIN lens):

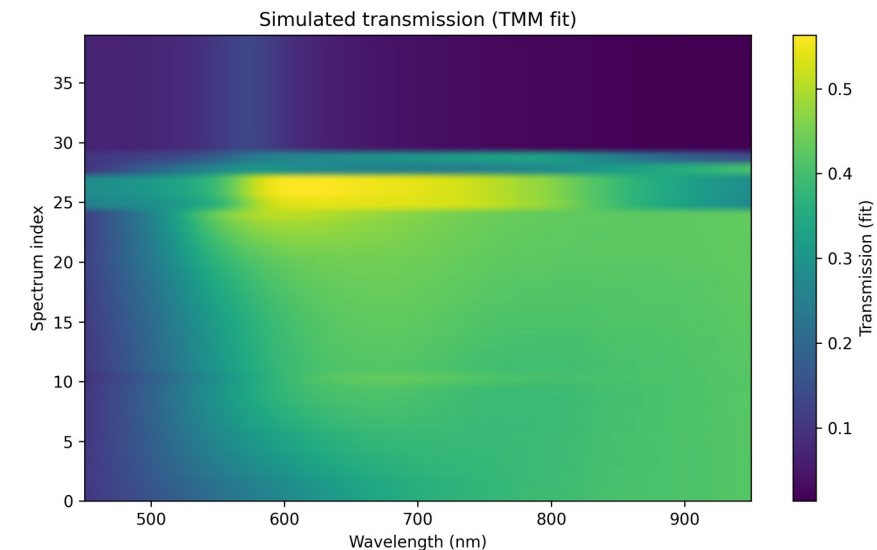
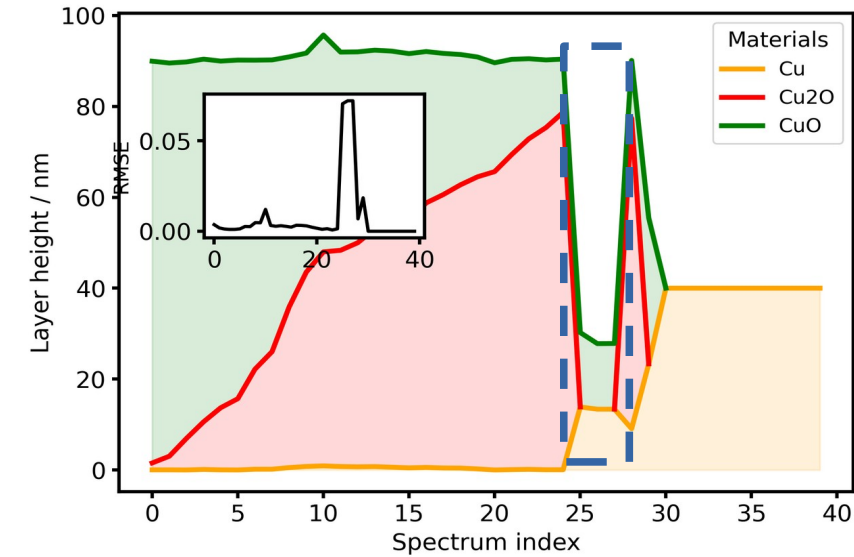


Fit with GA



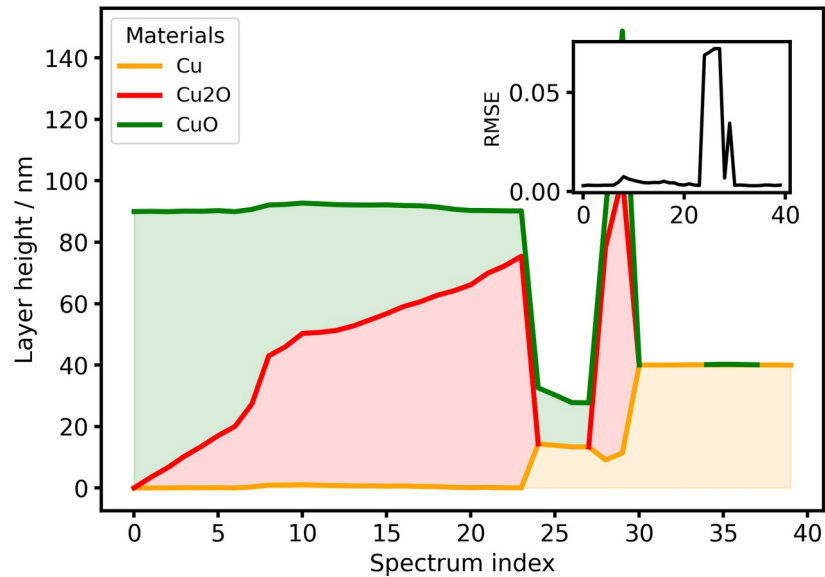
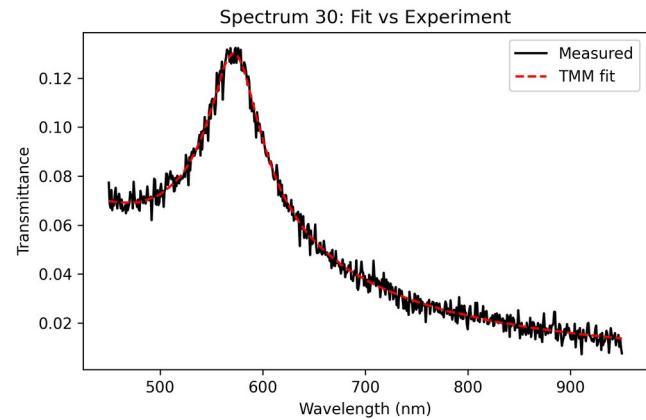
lication

Local minimum

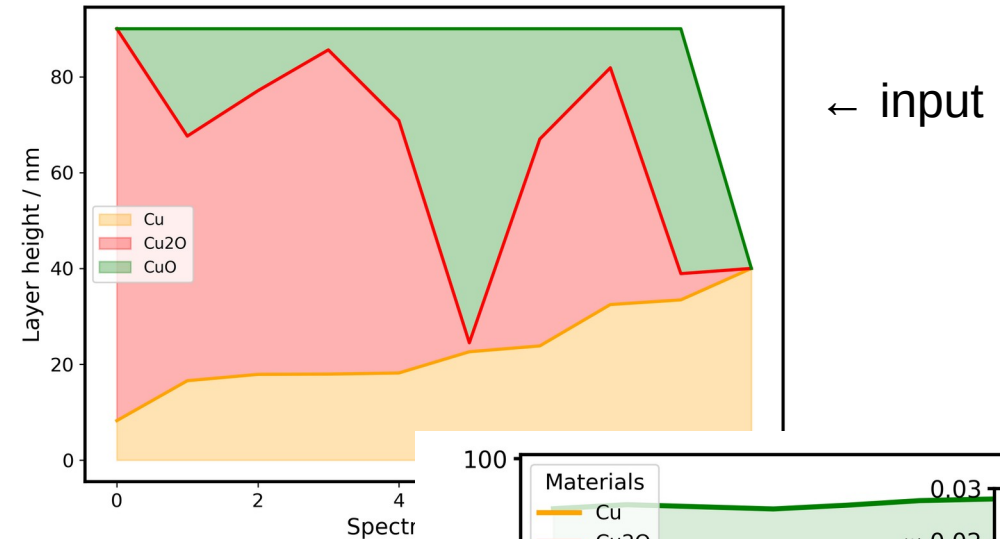


Benchmarking with Noise and Random Layers

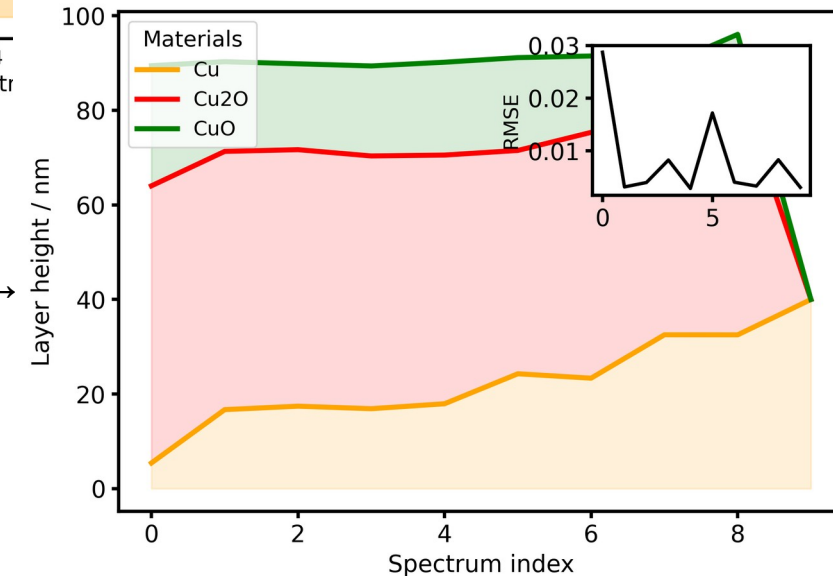
– With 1% noise added:



– Random oxide structure (and noise):

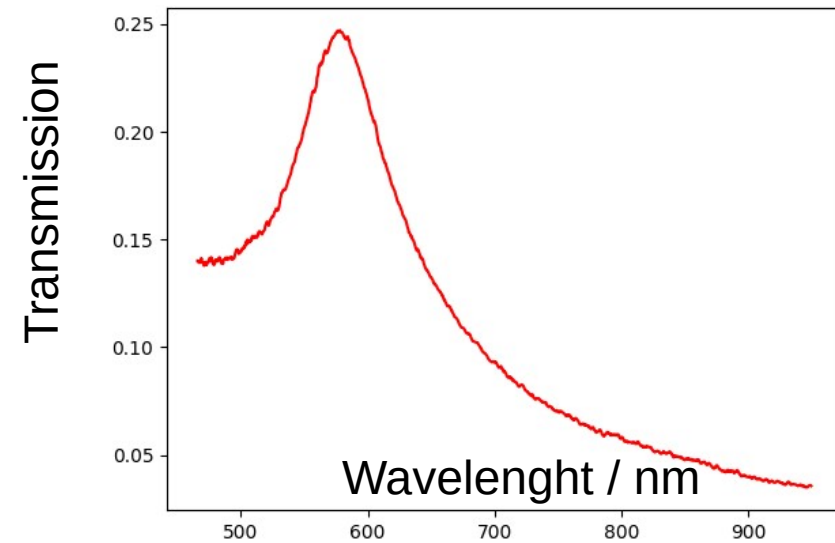
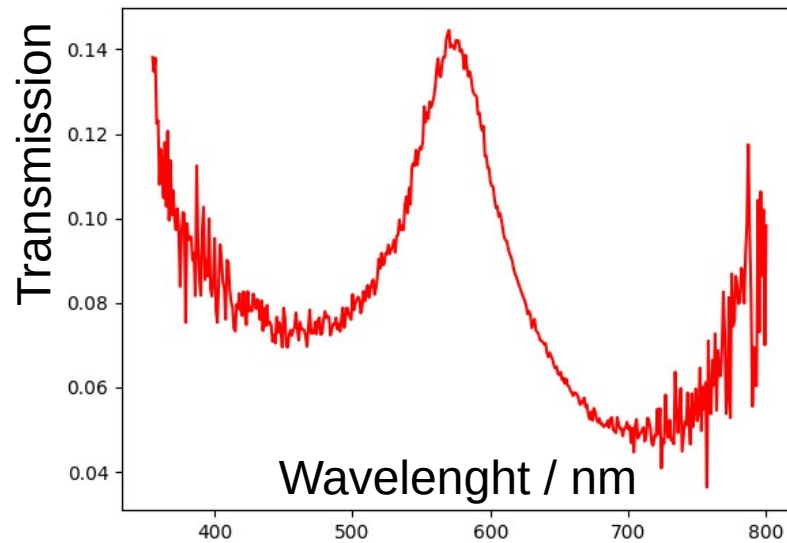


Fitting result →



Influence of Spectrum Scaling

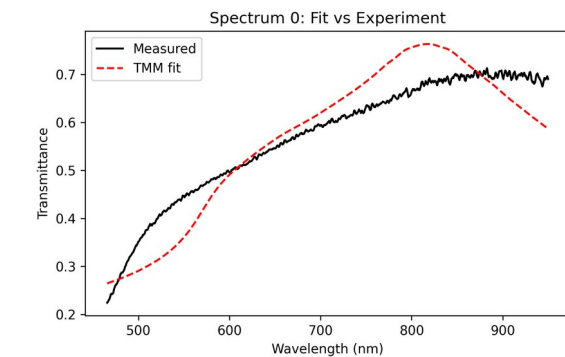
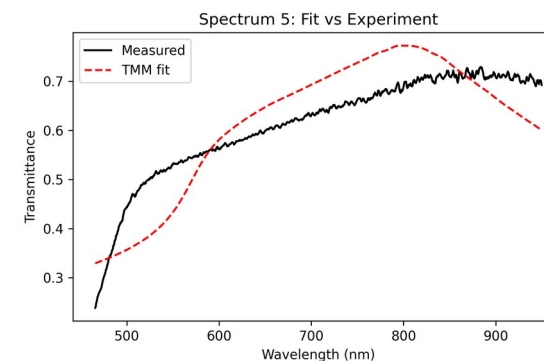
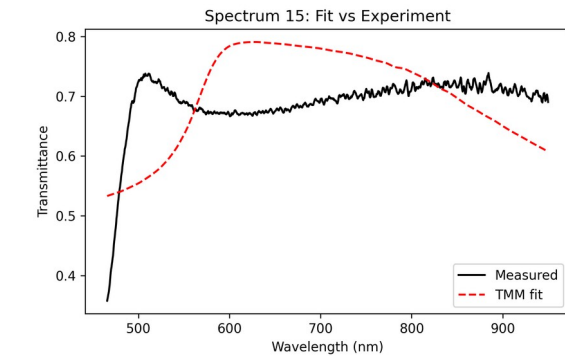
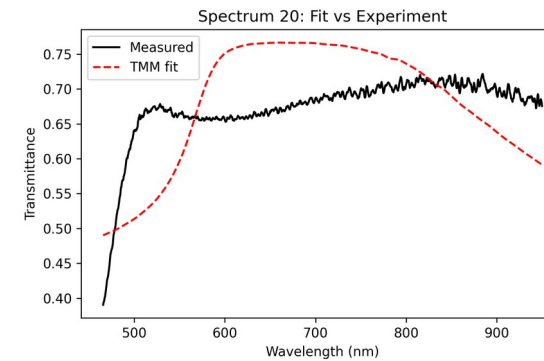
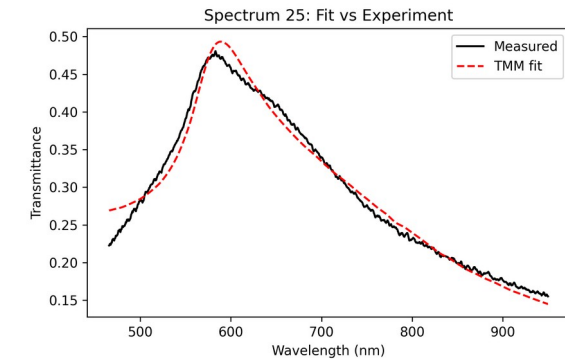
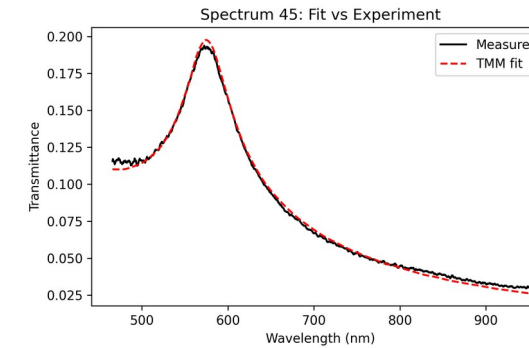
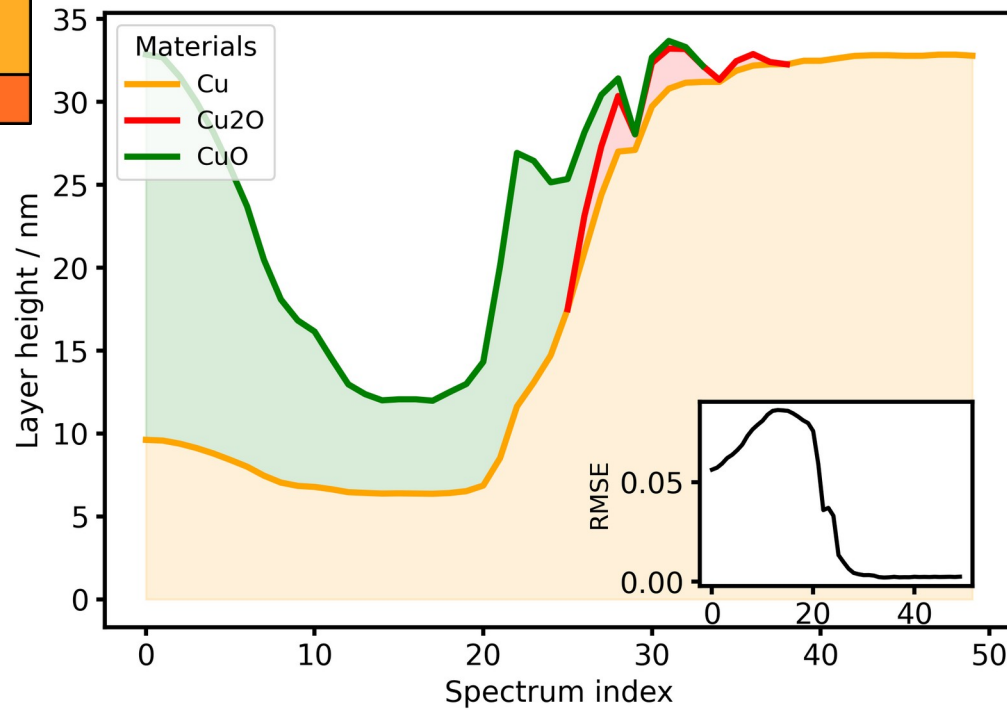
- Measurement of calibrated spectrometer (left) and transmission from WiTec Setup (right)



Problem: Amplitudes do not match!

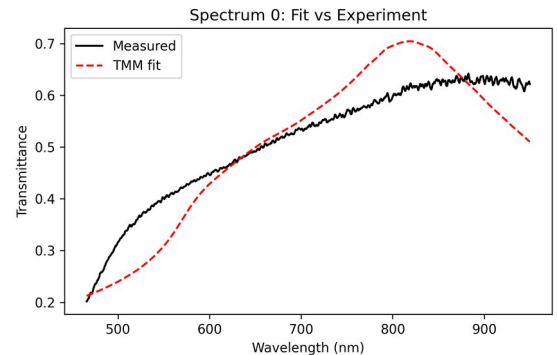
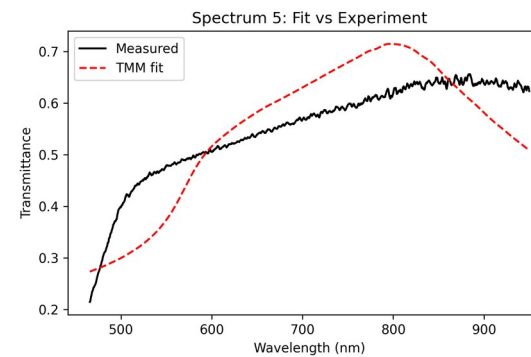
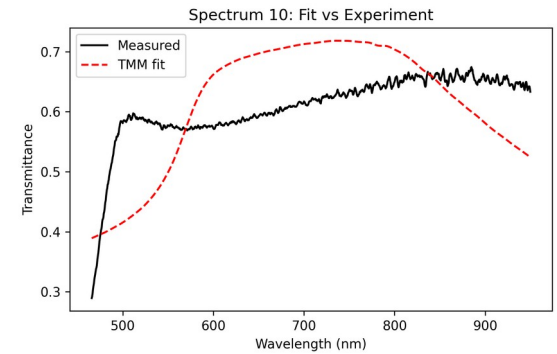
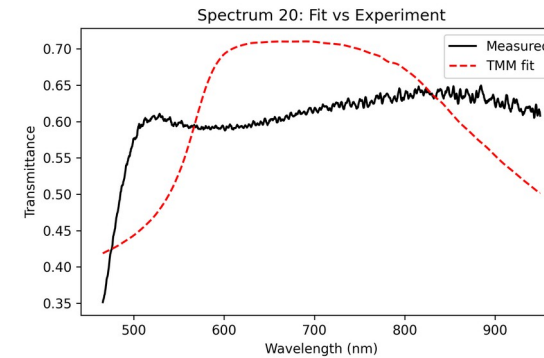
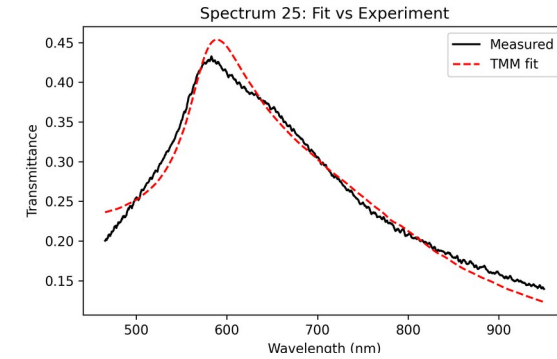
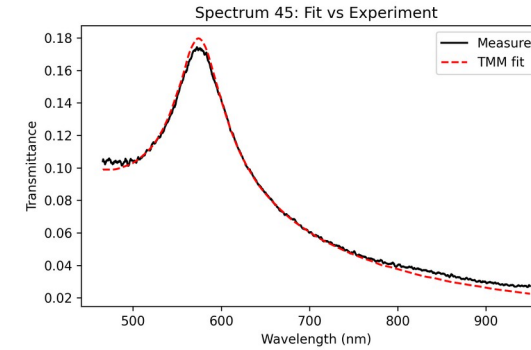
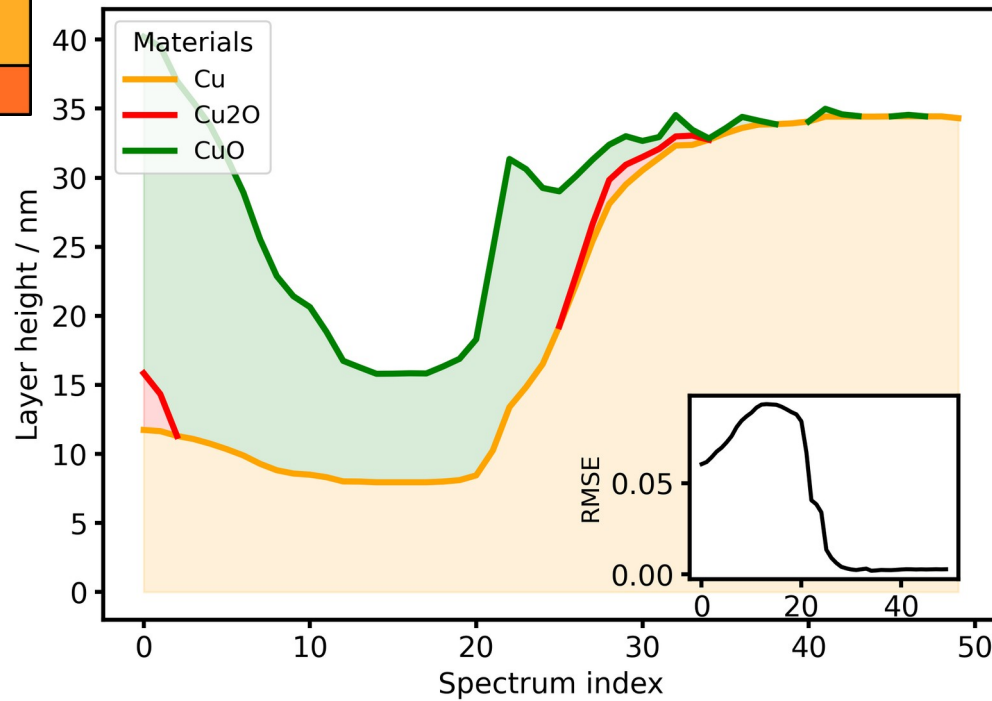
Influence of scaling Parameter on Fit (Lens 3W, 40s)

– Unchanged scaling:



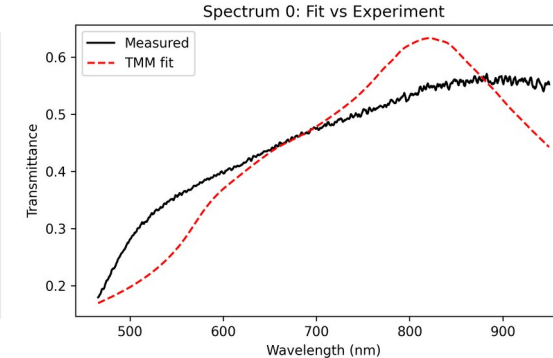
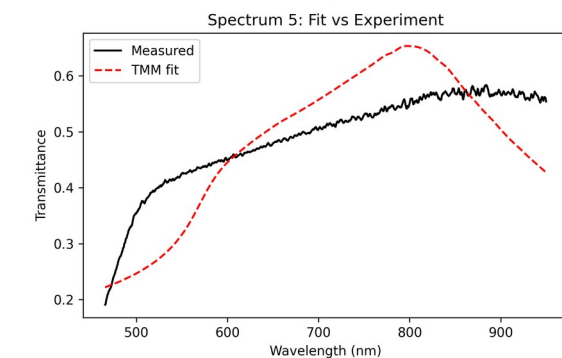
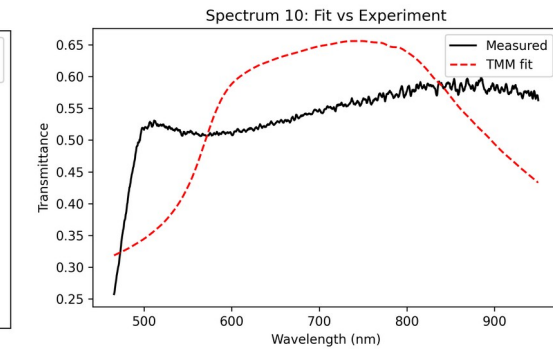
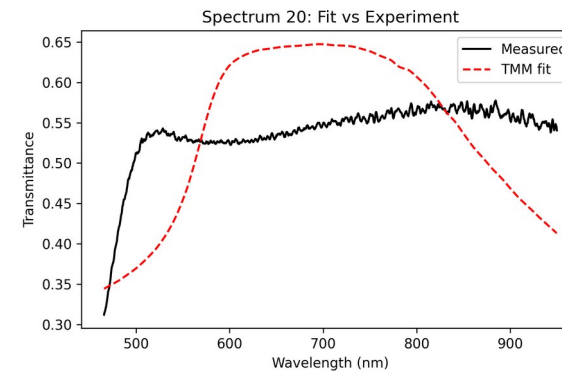
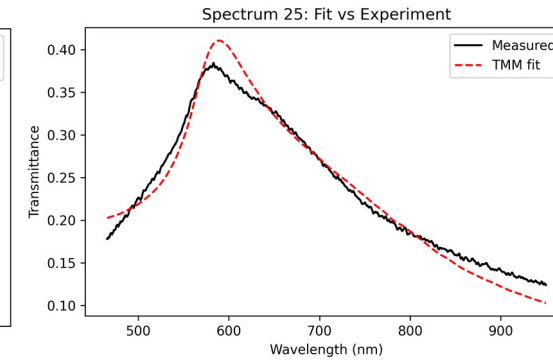
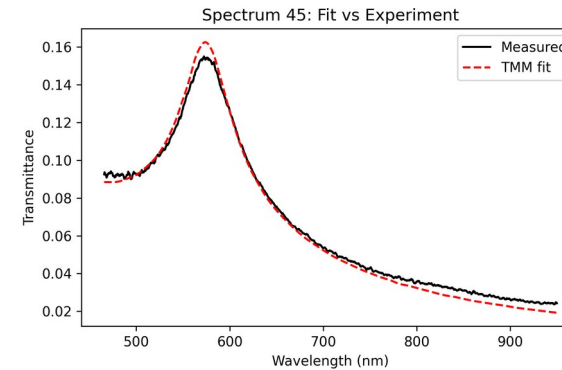
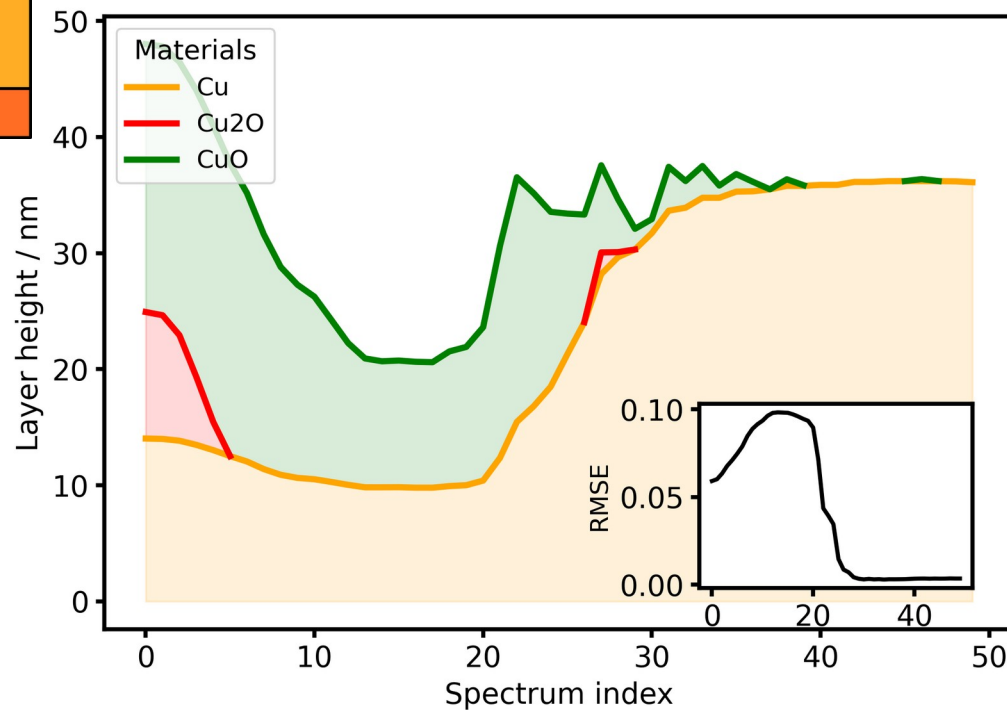
Influence of scaling Parameter on Fit (Lens 3W, 40s)

– 0.9x scaling:



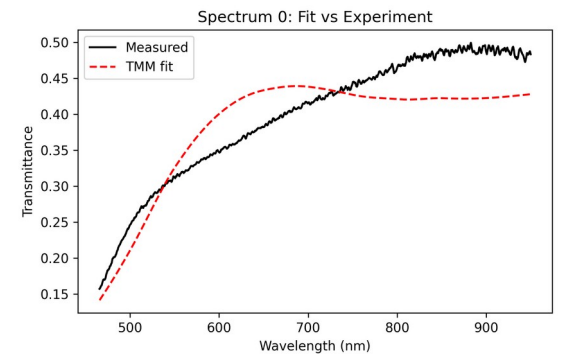
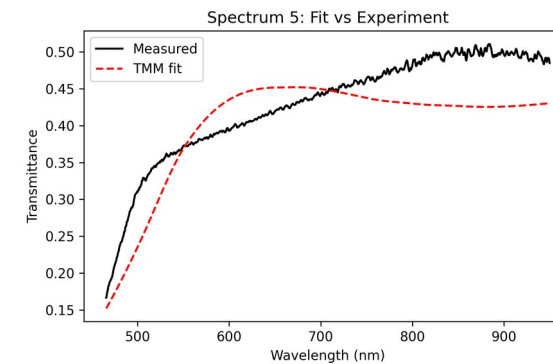
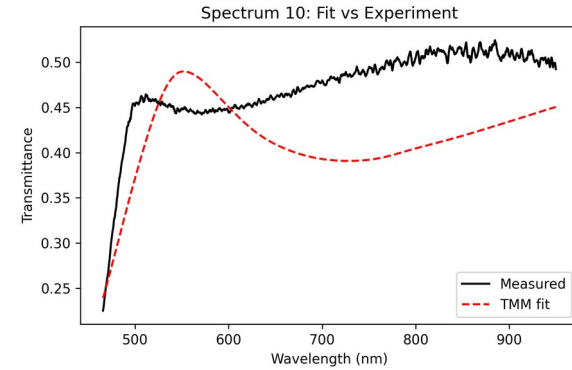
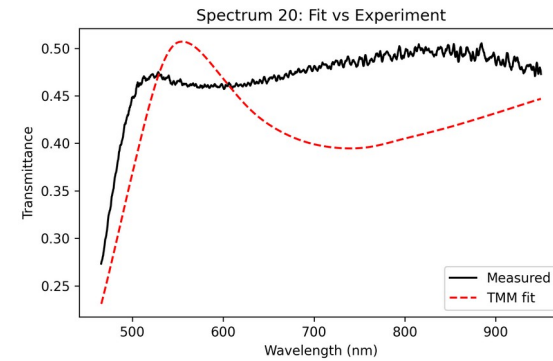
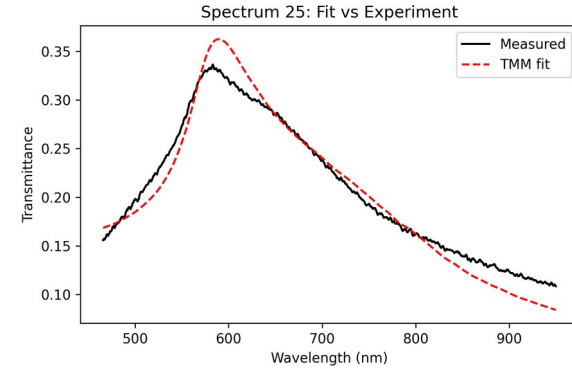
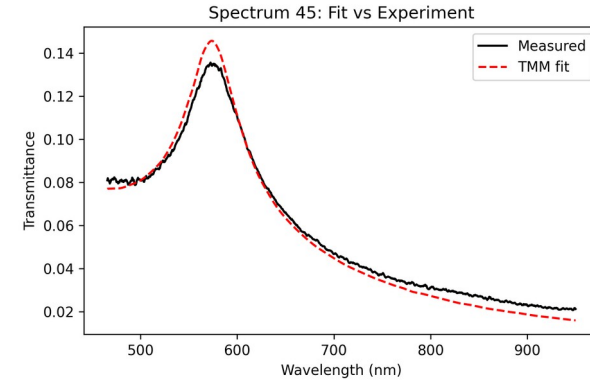
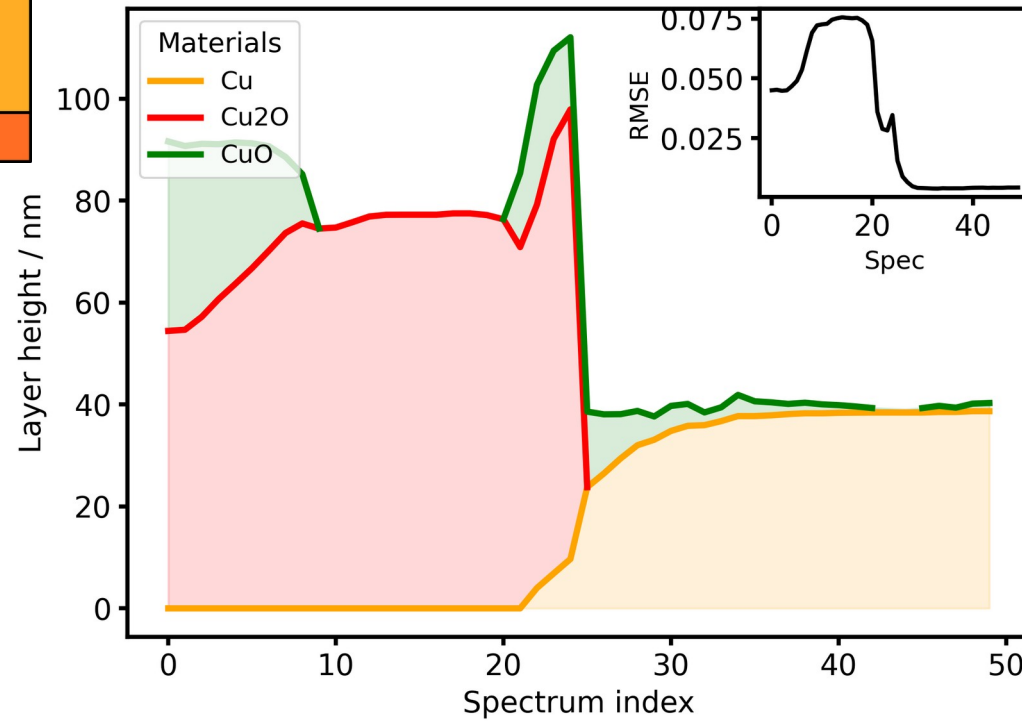
Influence of scaling Parameter on Fit (Lens 3W, 40s)

– 0.8x scaling:



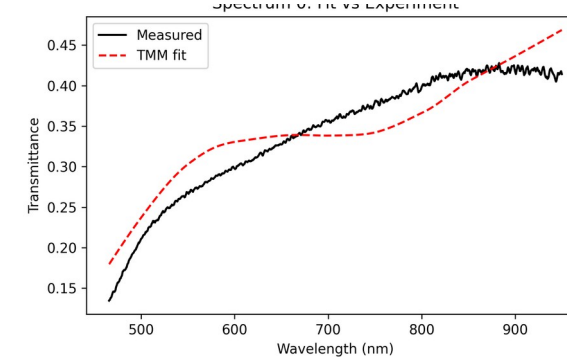
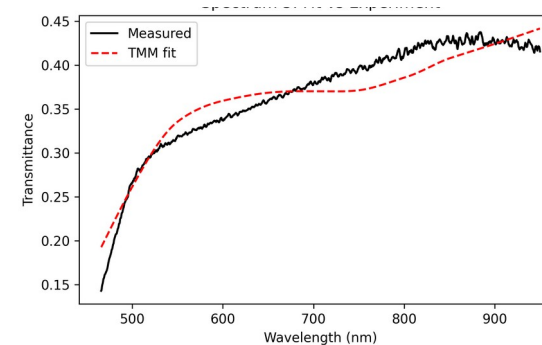
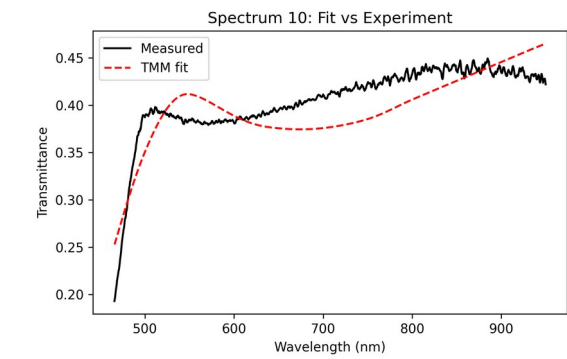
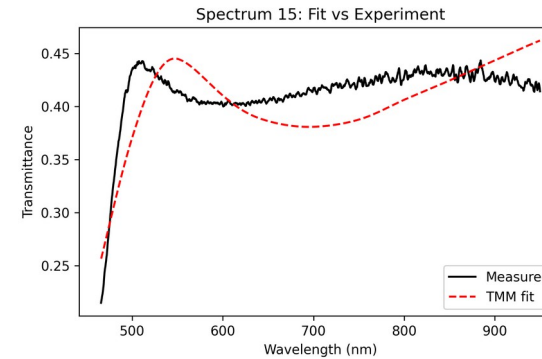
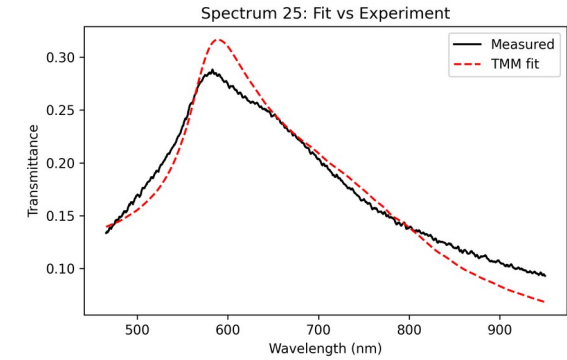
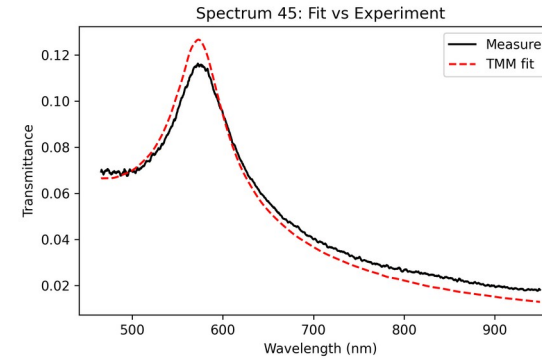
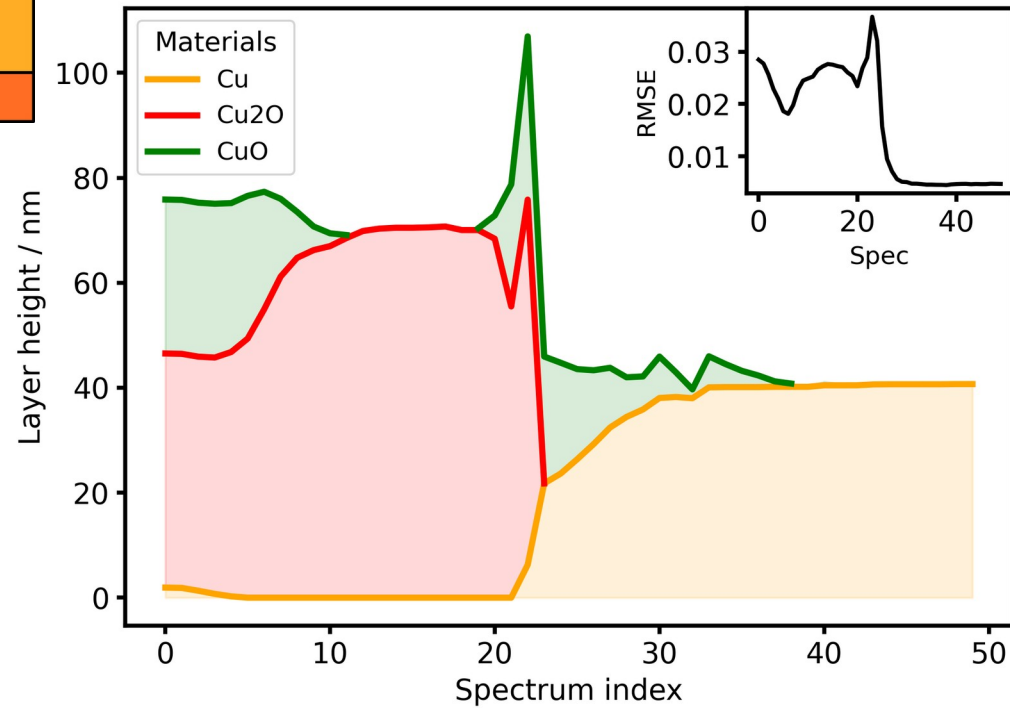
Influence of scaling Parameter on Fit (Lens 3W, 40s)

– 0.7x scaling:



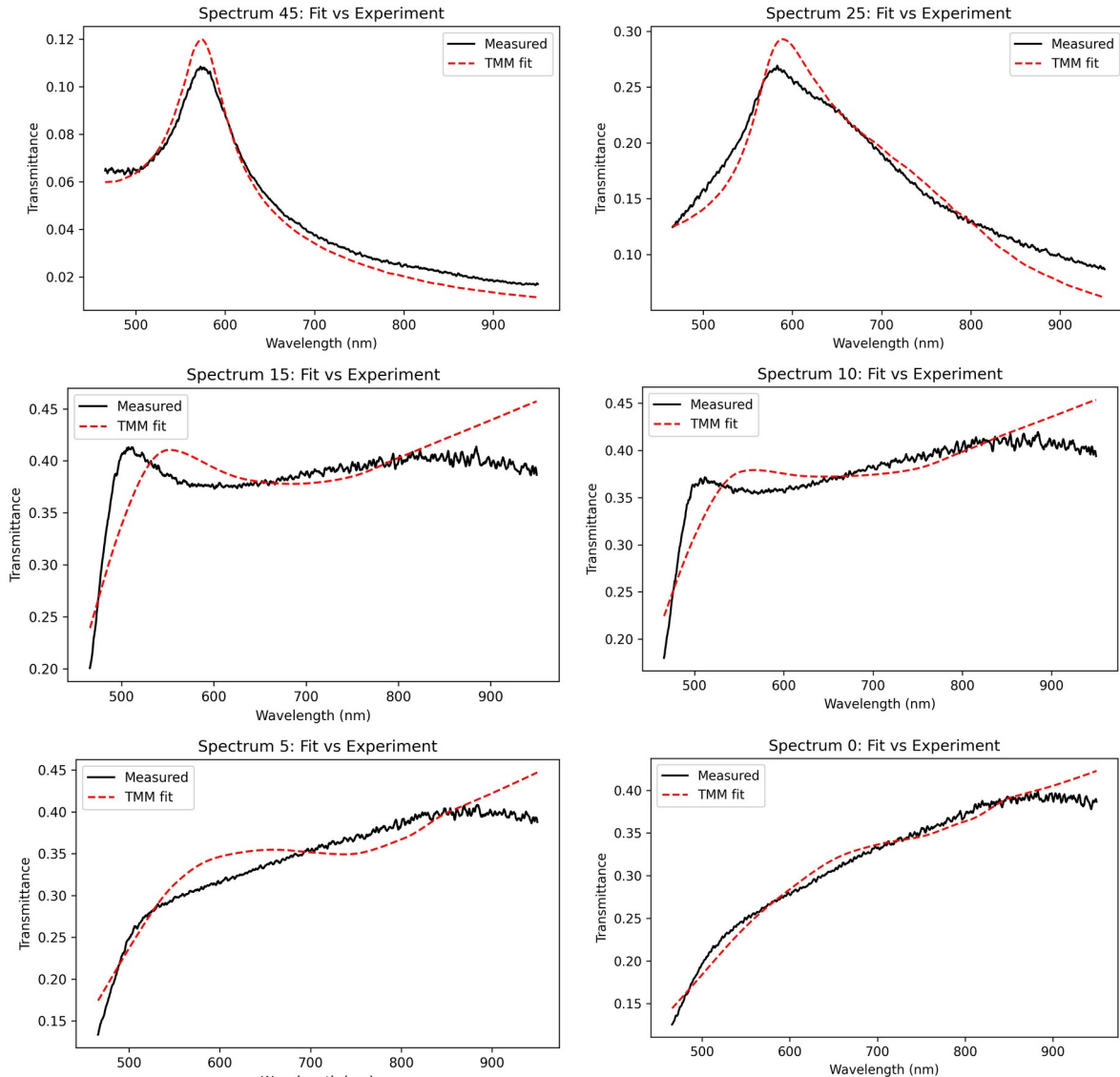
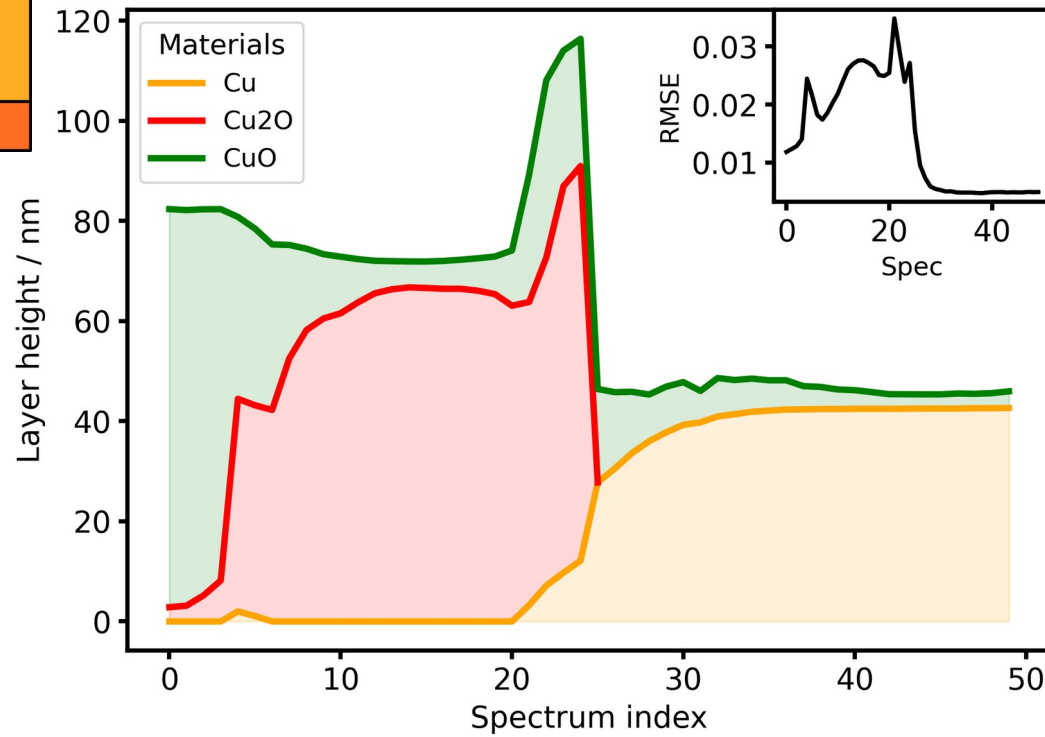
Influence of scaling Parameter on Fit (Lens 2W, 40s)

– 0.6x scaling:



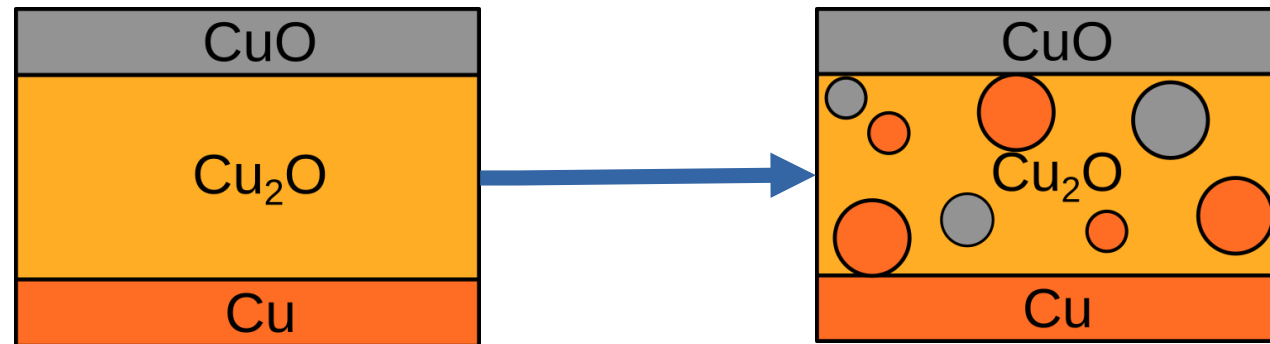
Influence of scaling Parameter on Fit (Lens 3W, 40s)

– 0.56x scaling:

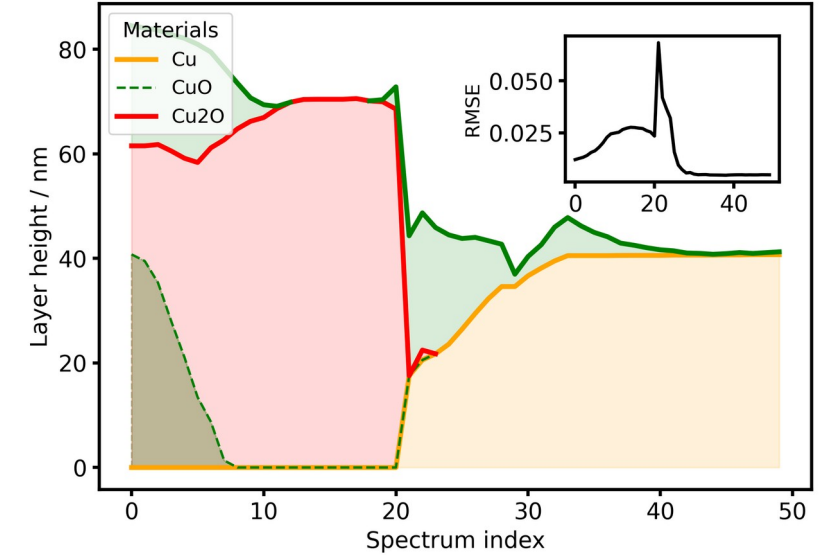
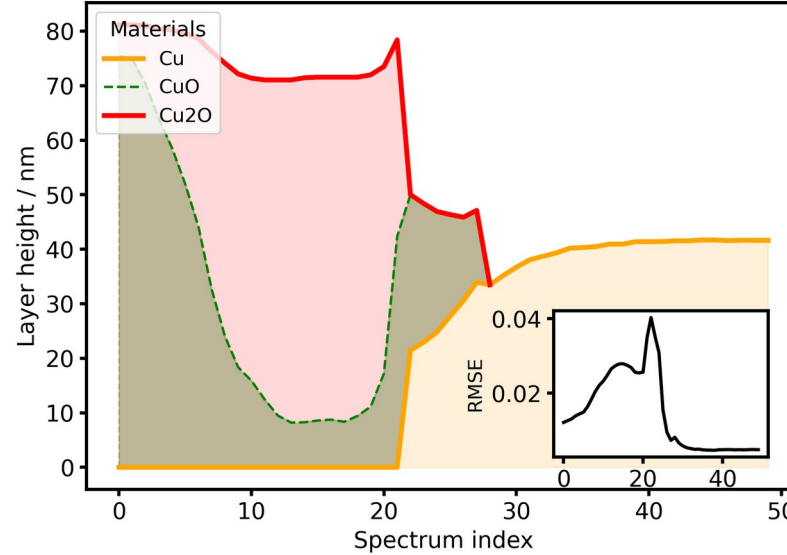
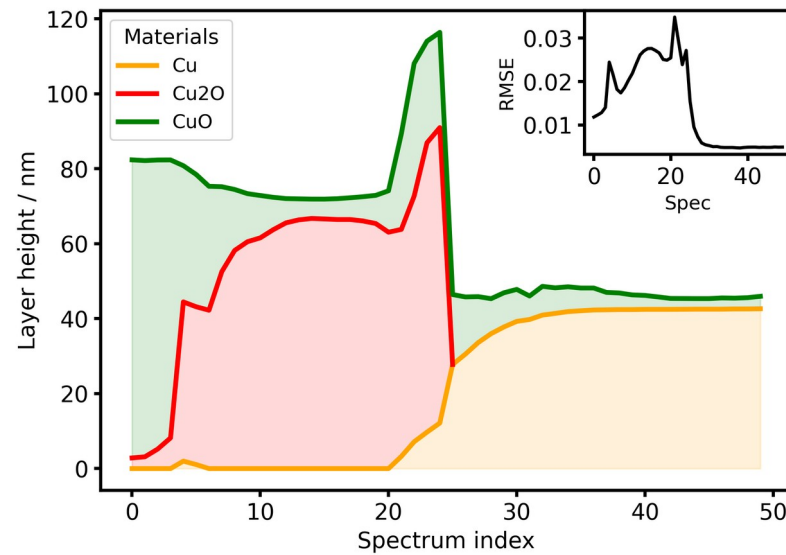
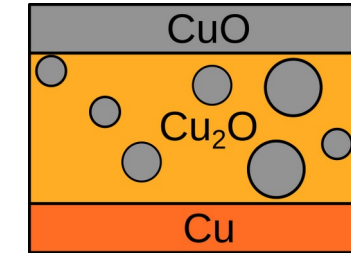
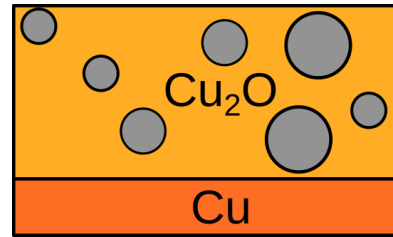
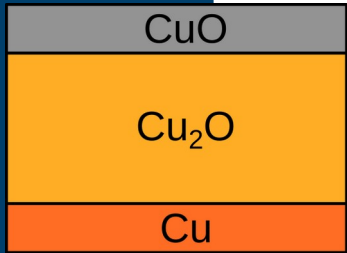
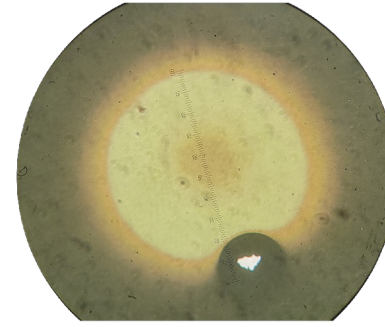


Optimizing the Fitting

Brüggemann Mixing, Smart Mutation Scaling, Clamping Layer Height Bounds

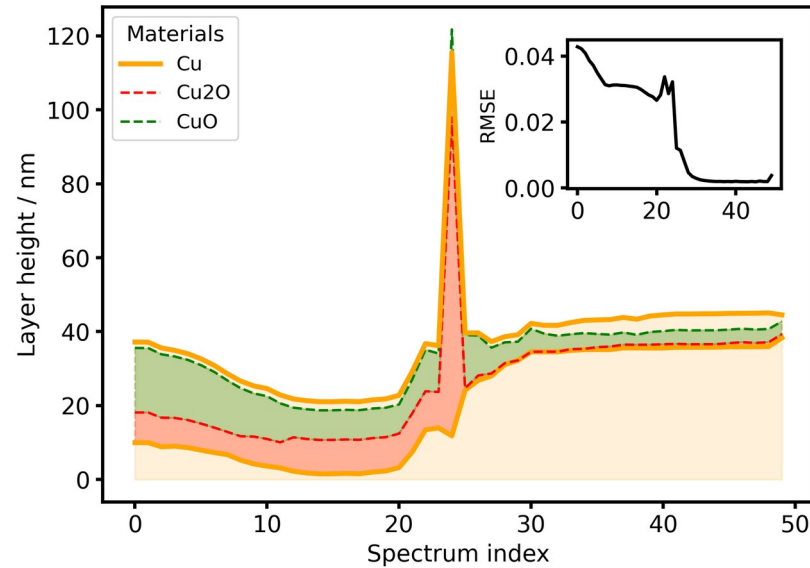
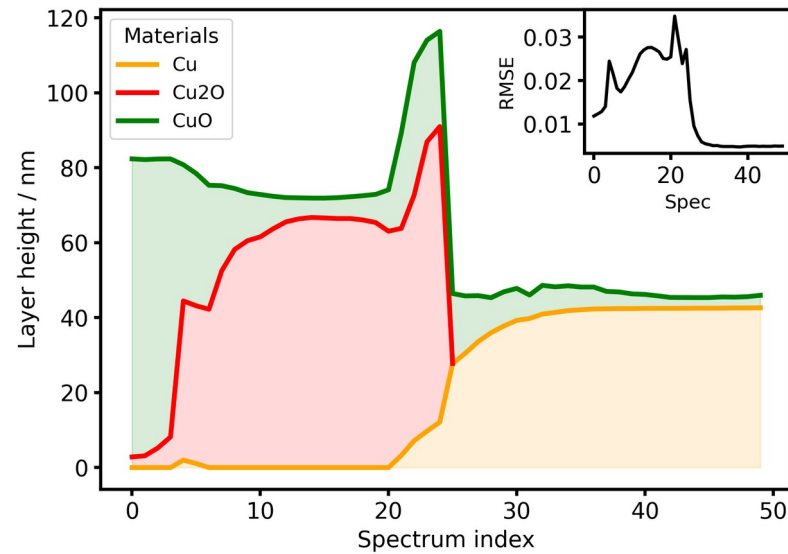
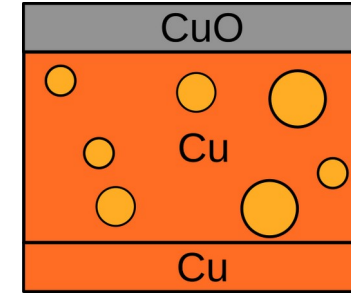
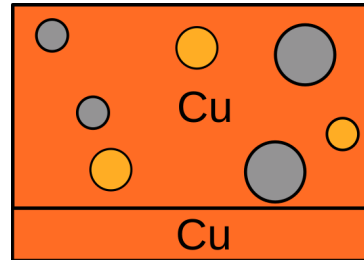
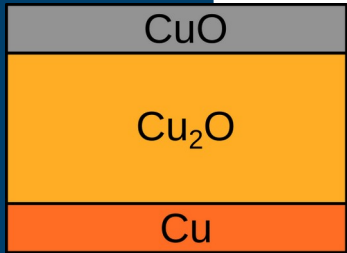
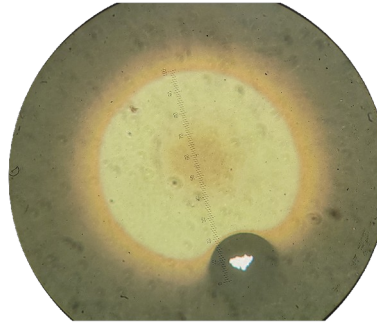


Results of different Fitting Schemes: (Lens 3W, 40s)

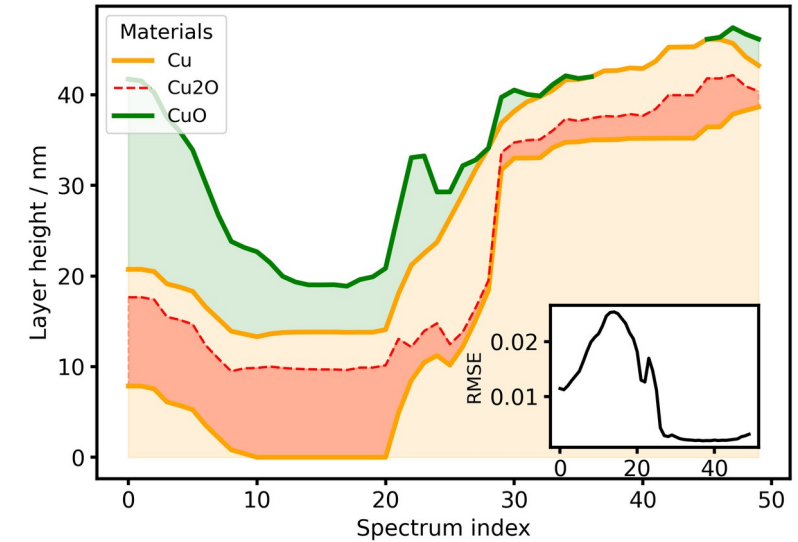


No improvements over no-mixing scheme!

Results of different Fitting Schemes: (Lens 3W, 40s)



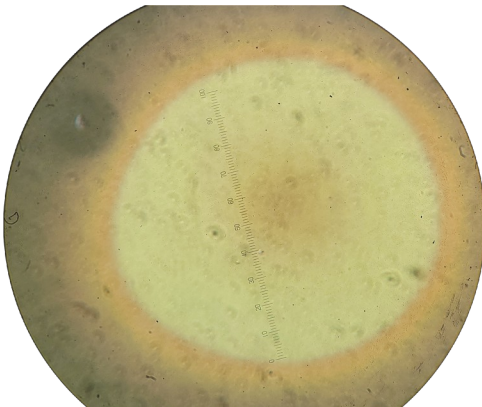
Spike from local minimum



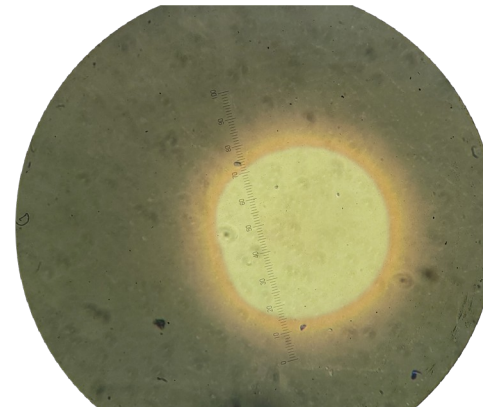
Better RMSE, but different
height profile!

Best Results for other lenses

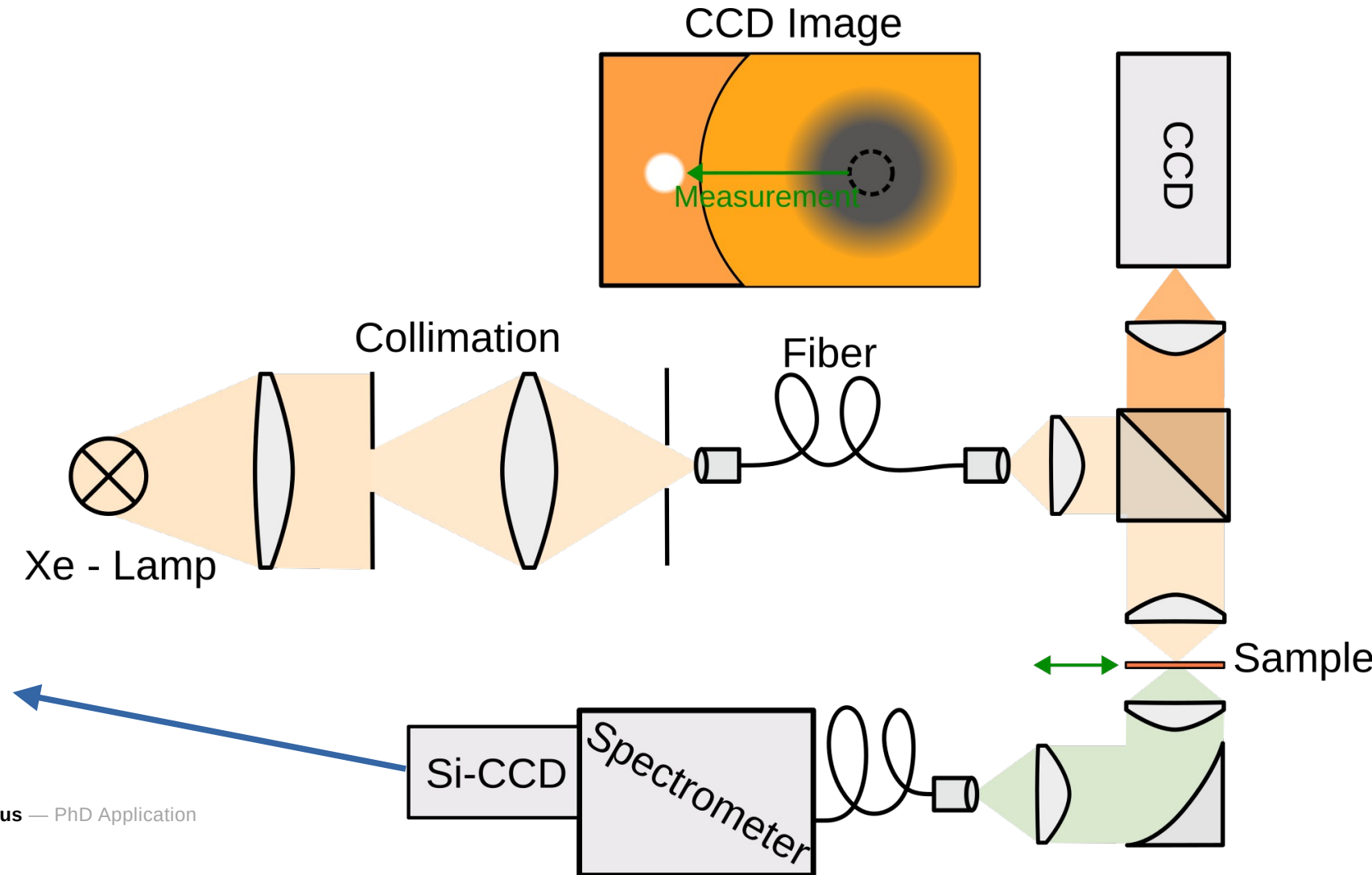
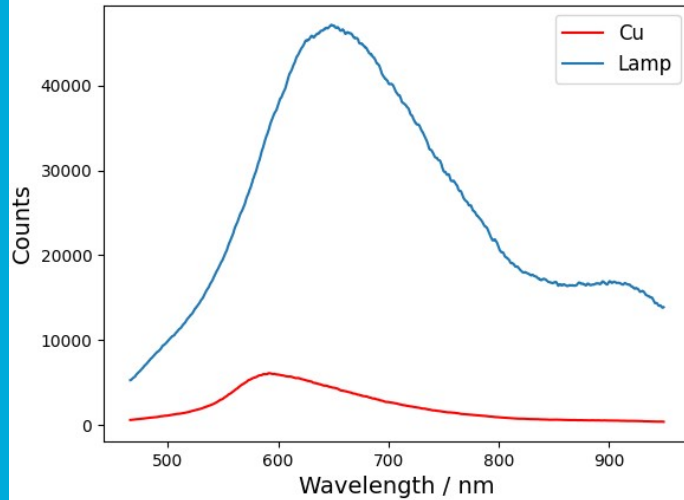
3W 120s



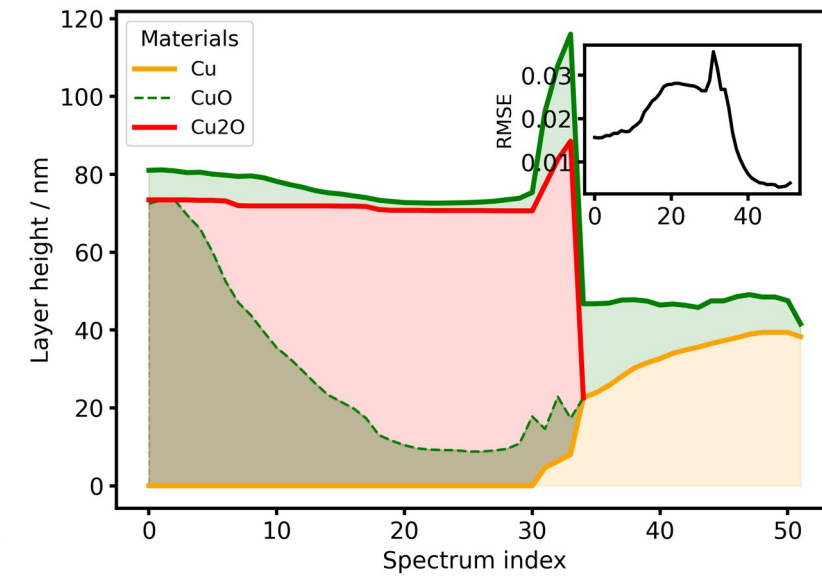
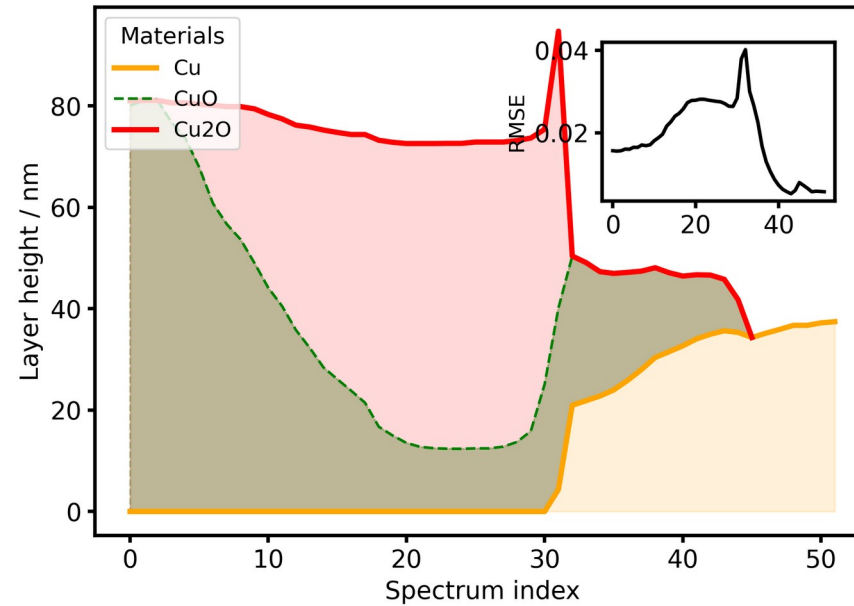
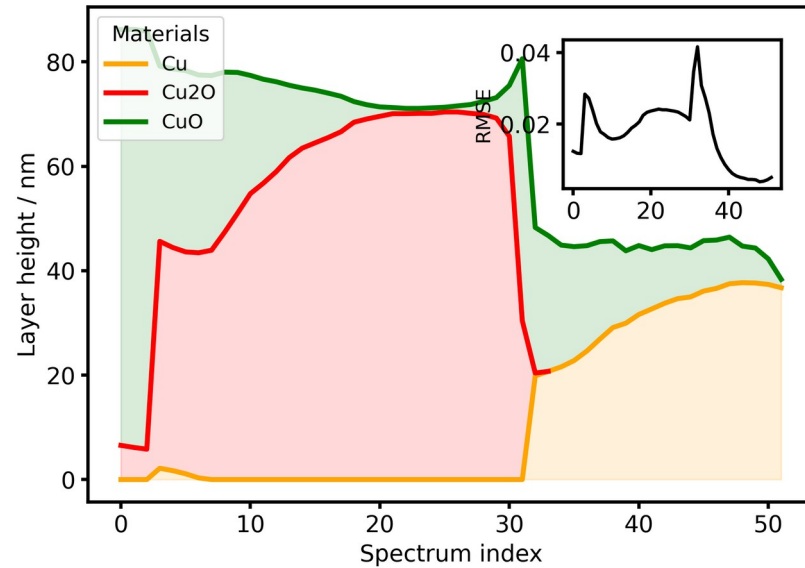
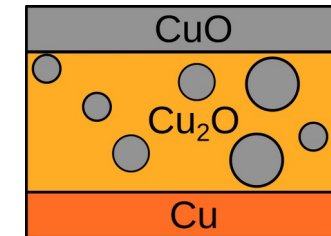
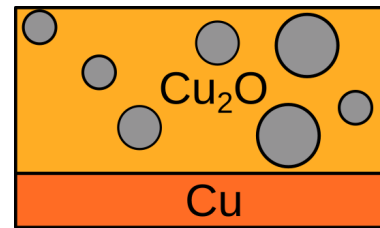
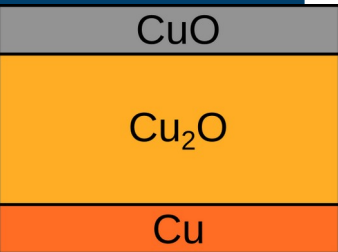
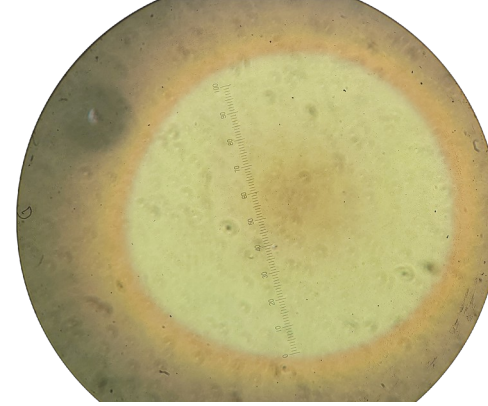
2W 60s



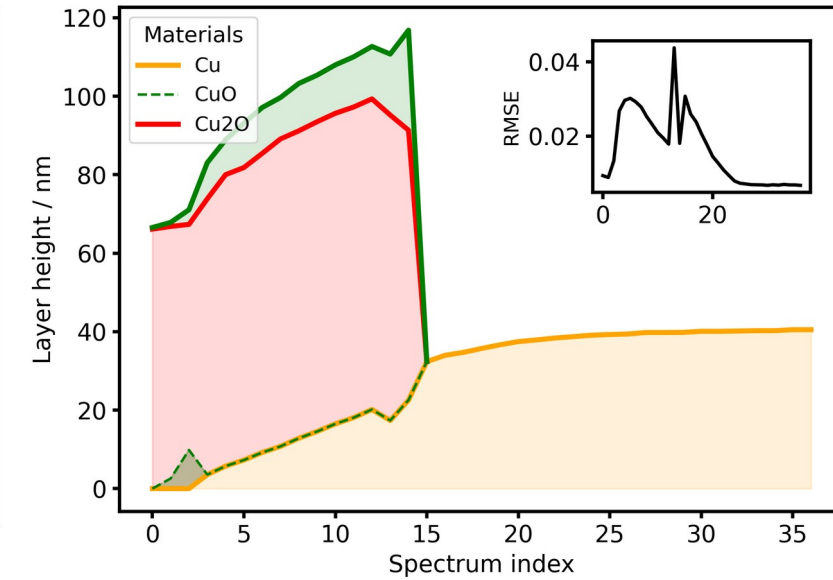
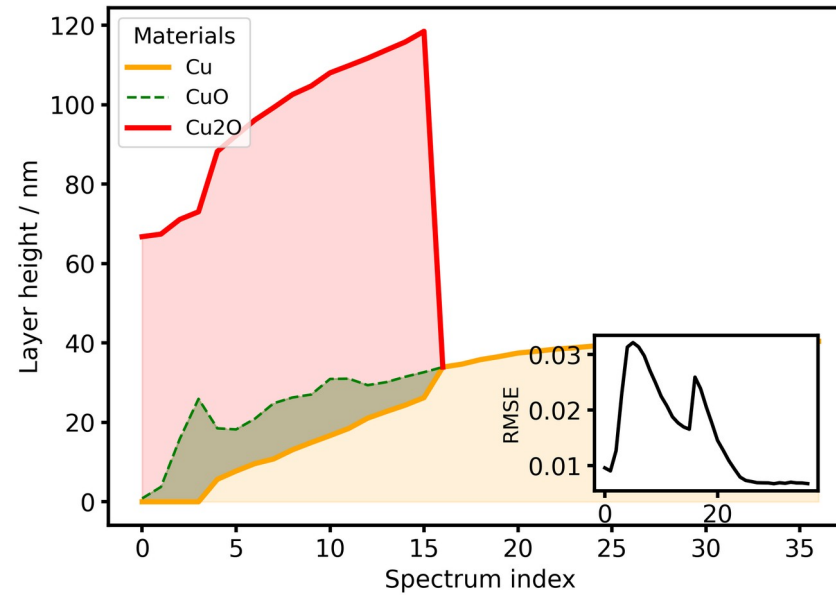
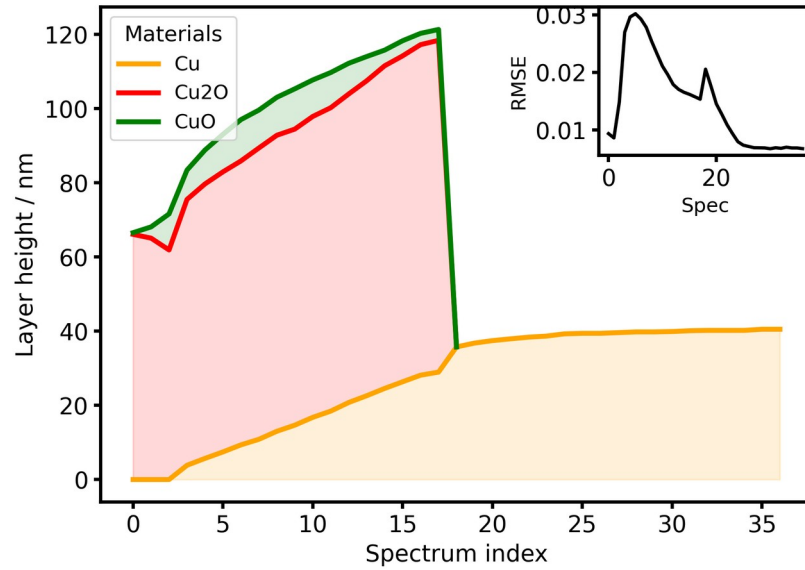
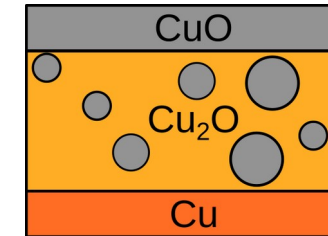
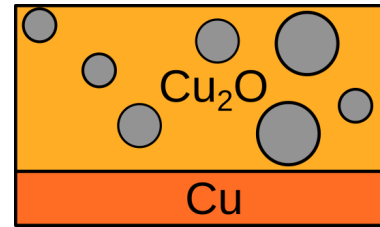
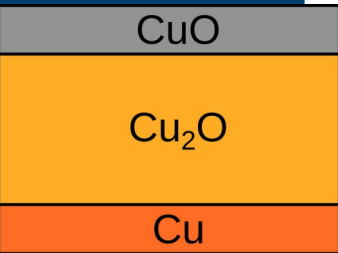
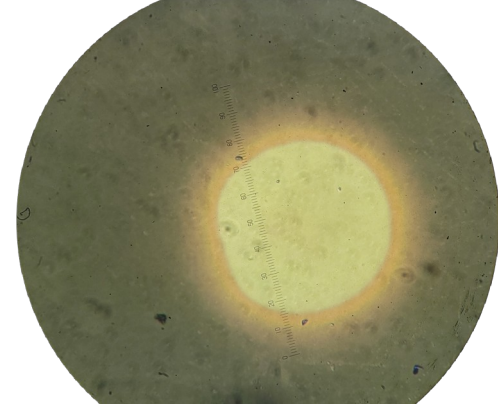
Transmission Spectrum Measurement Setup



Lens 3W, 120s Fitting Results



Lens 2W, 60s Fitting Results



What's next?

- Do a power variation series (constant illumination time)
- Optimize goodness of GA by re-fitting simulated data
- Try out larger magnification objective, to reduce spatial averaging

Thanks for listening!

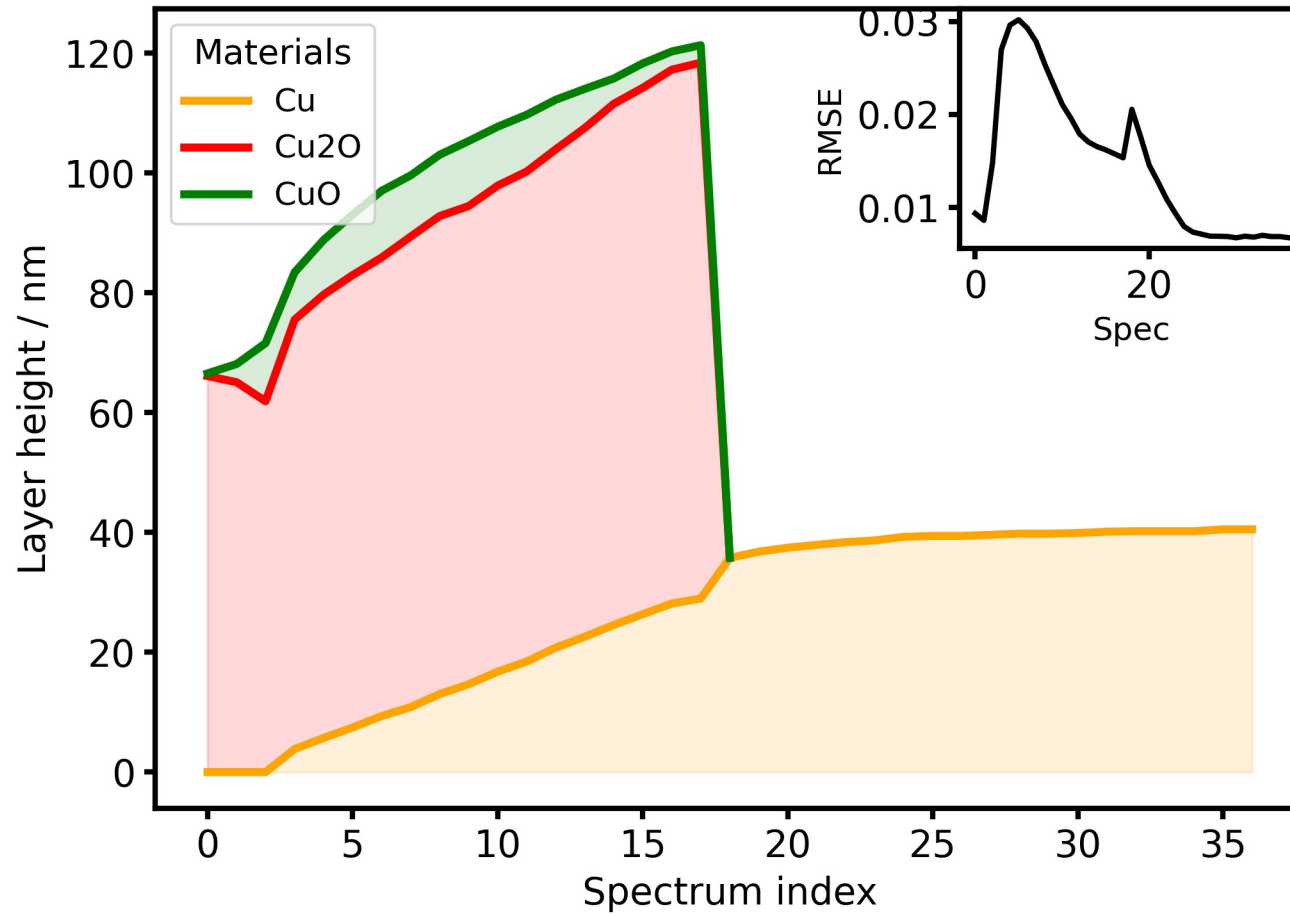
Lens: 50 mBar O₂; 2W; 60s

Hier Bild von Linse einfügen

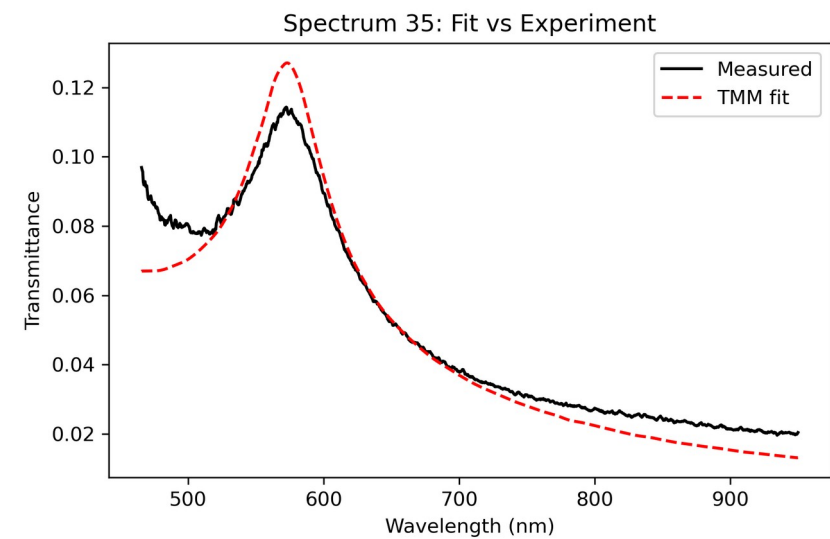
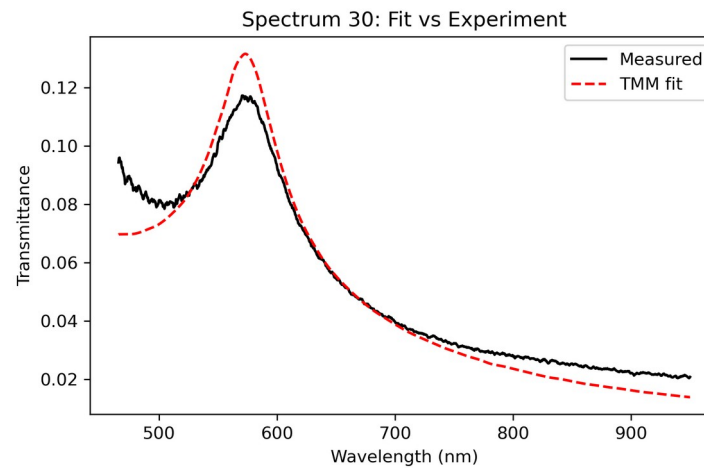
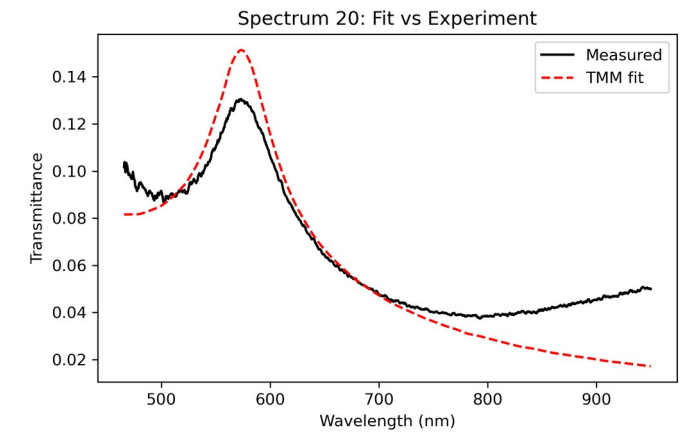
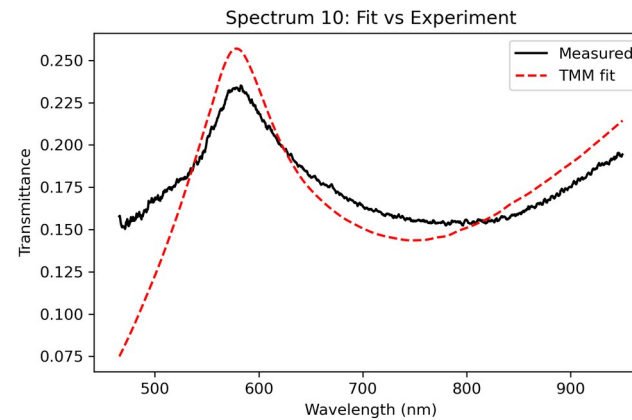
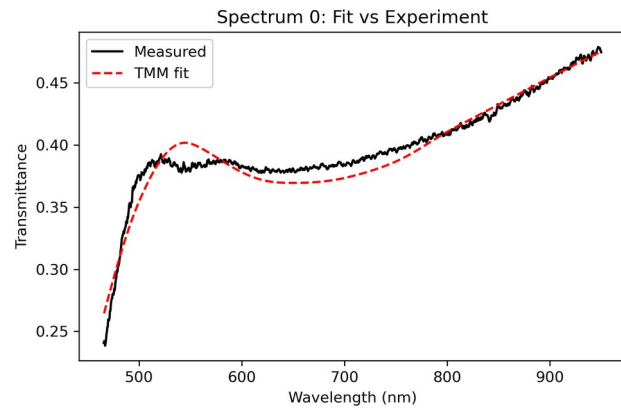
Platzhalter-
Logo

Für mehr als drei Logos (oder wenn der Platz unten rechts nicht ausreicht)

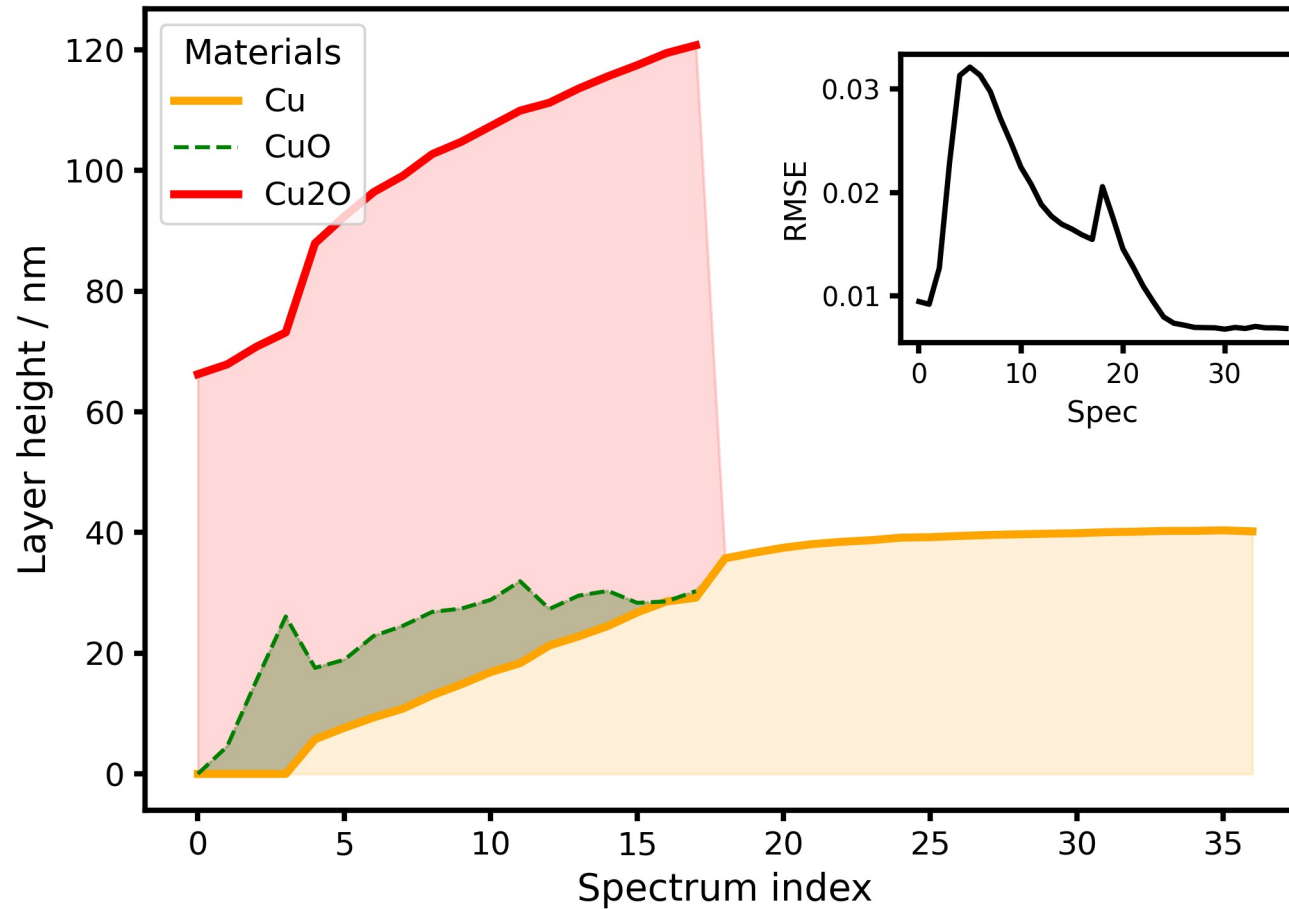
Fitting: Cu – Cu₂O – CuO



Fitting: Cu – Cu₂O – CuO

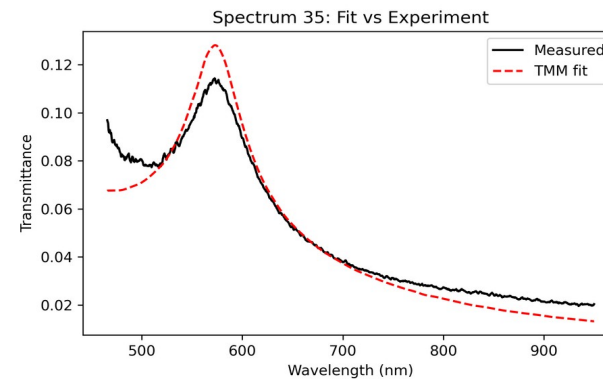
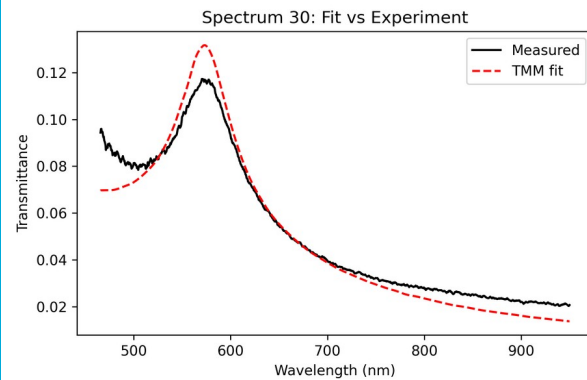
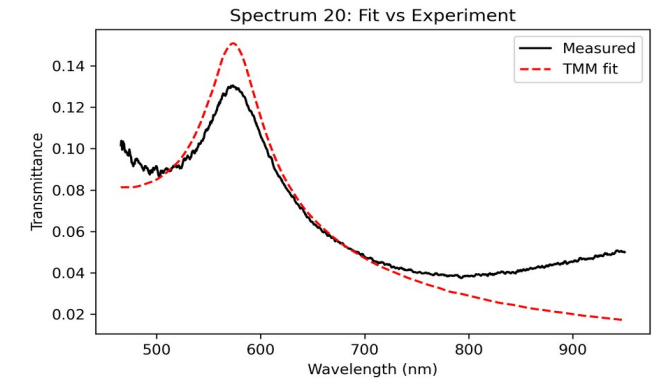
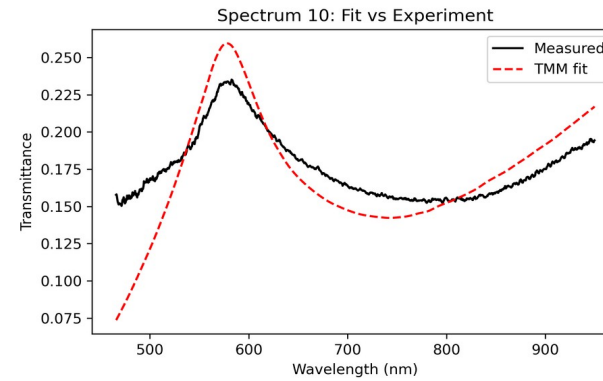
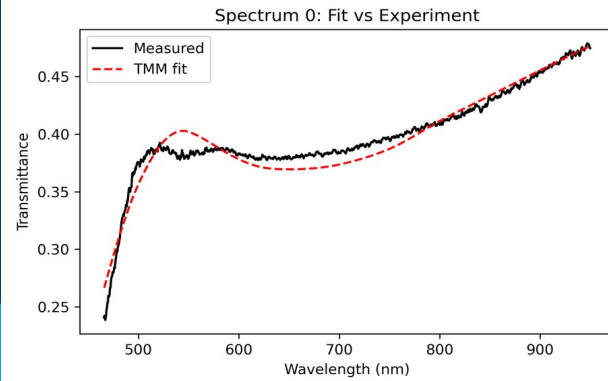


Fitting: Cu – Cu₂O + CuO (sphere)



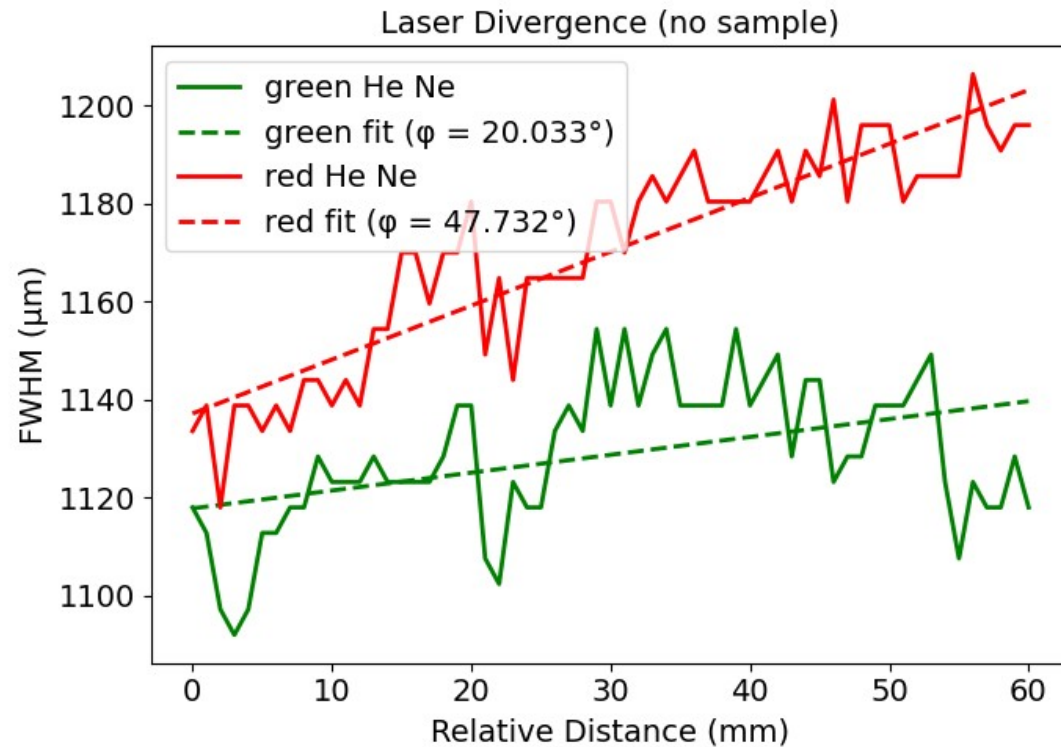
- No difference to individual layer model!

Cu – Cu₂O + CuO (sphere)

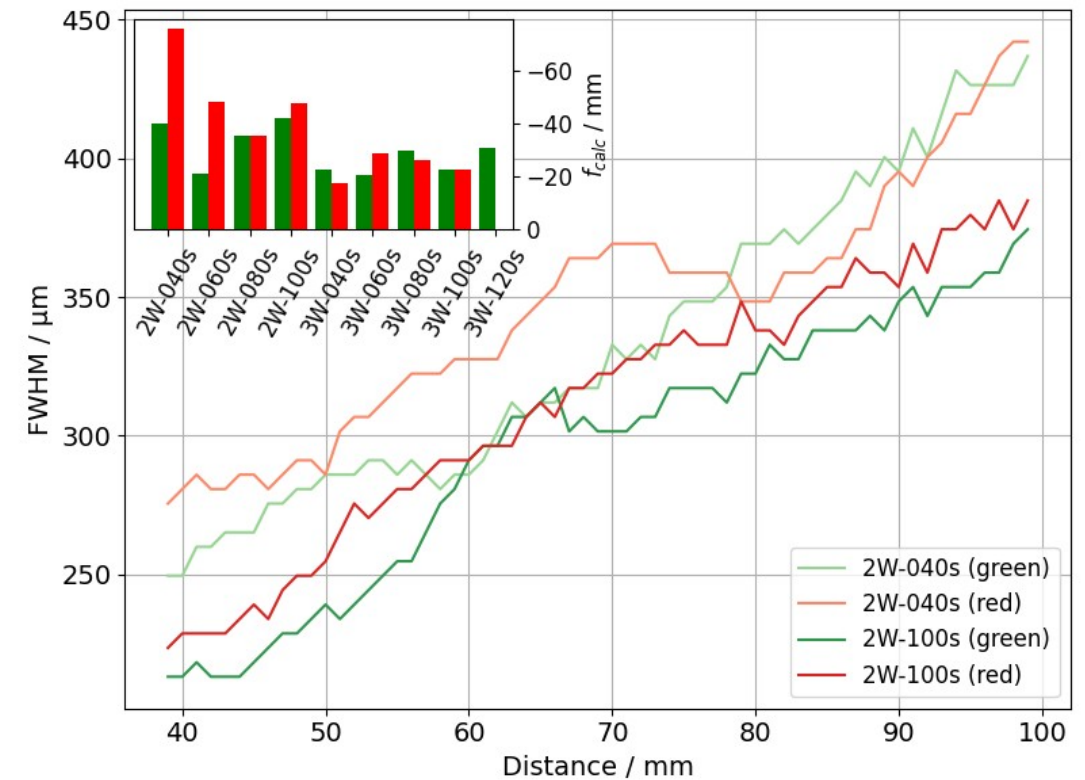


Optical Properties of the GRIN Lenses

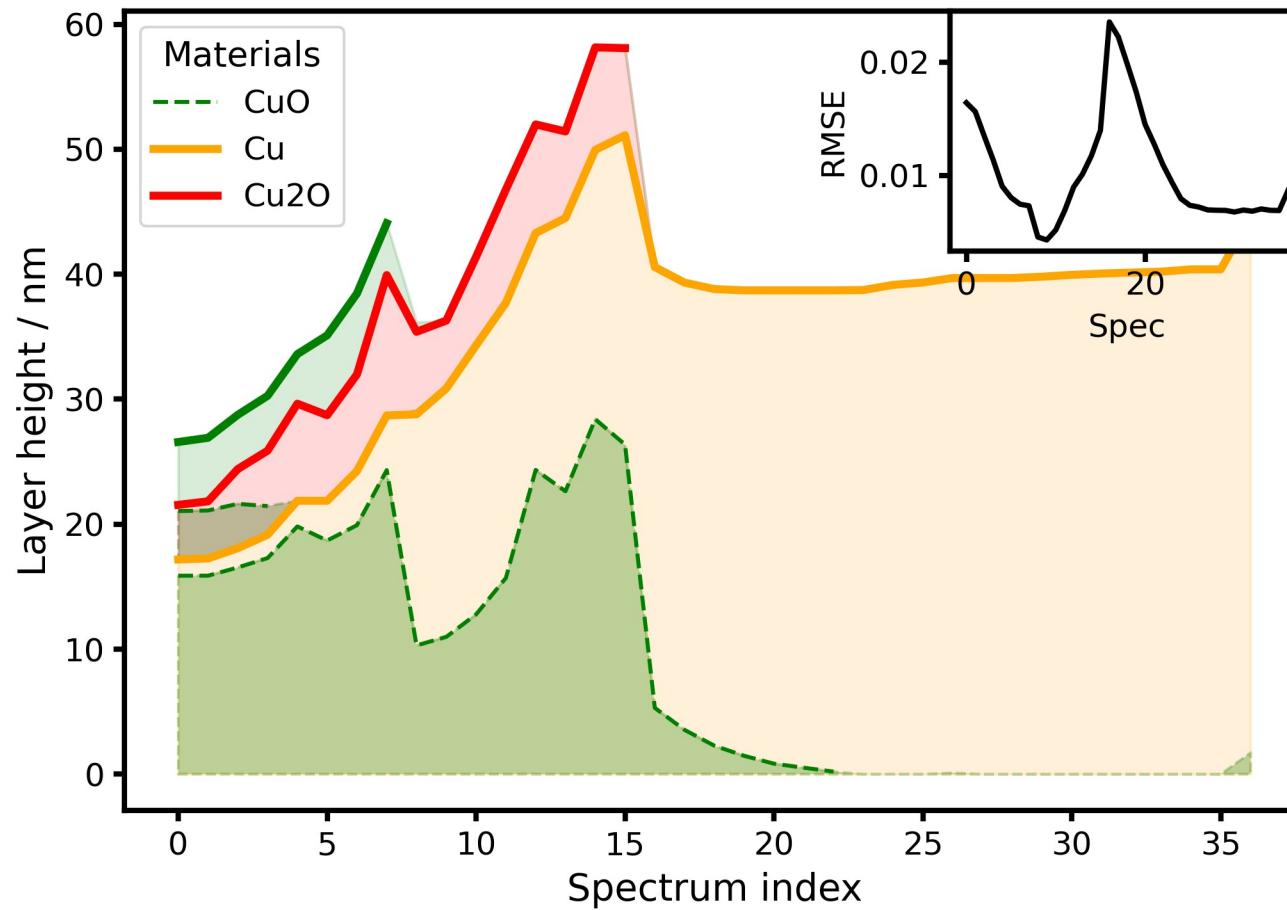
– He-Ne laser divergence



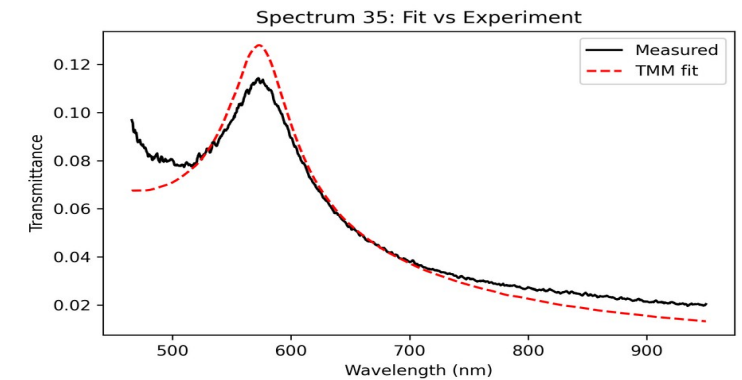
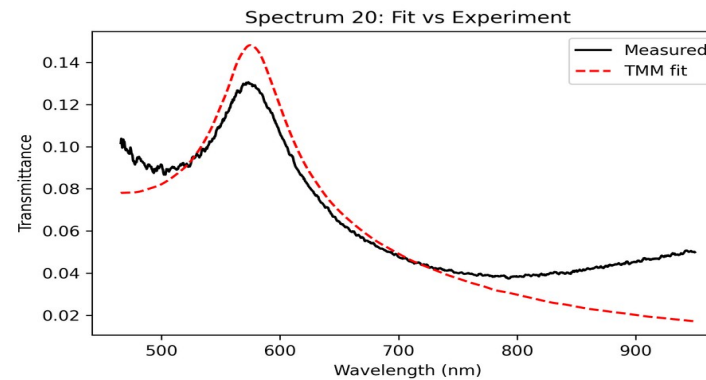
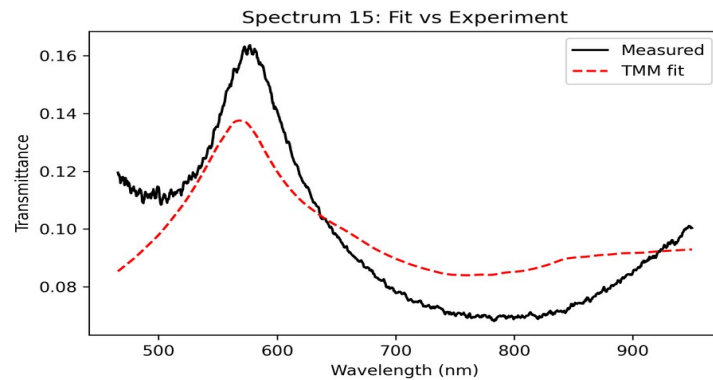
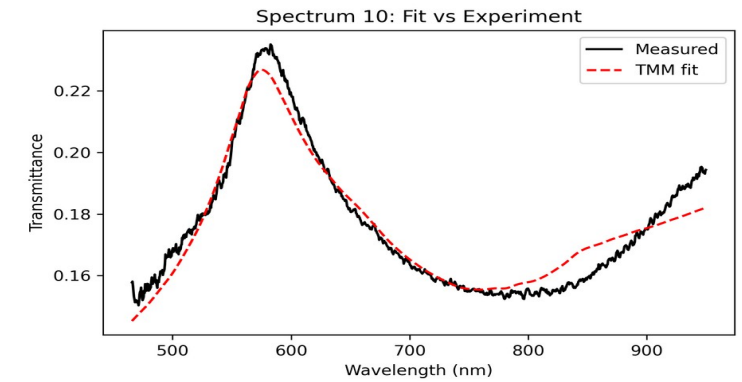
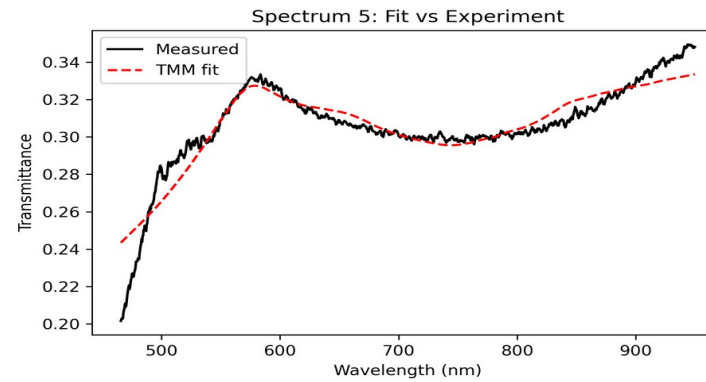
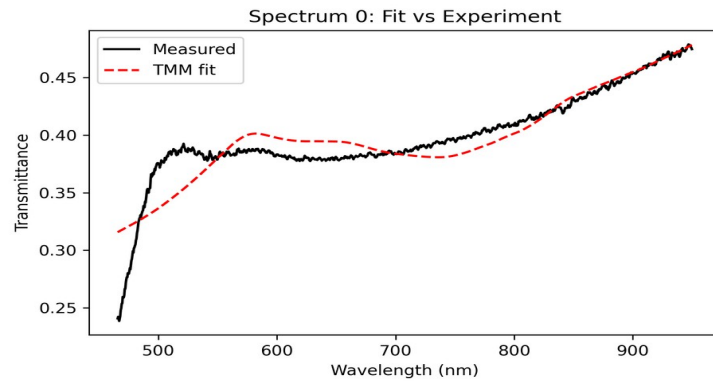
– GRIN lens divergence



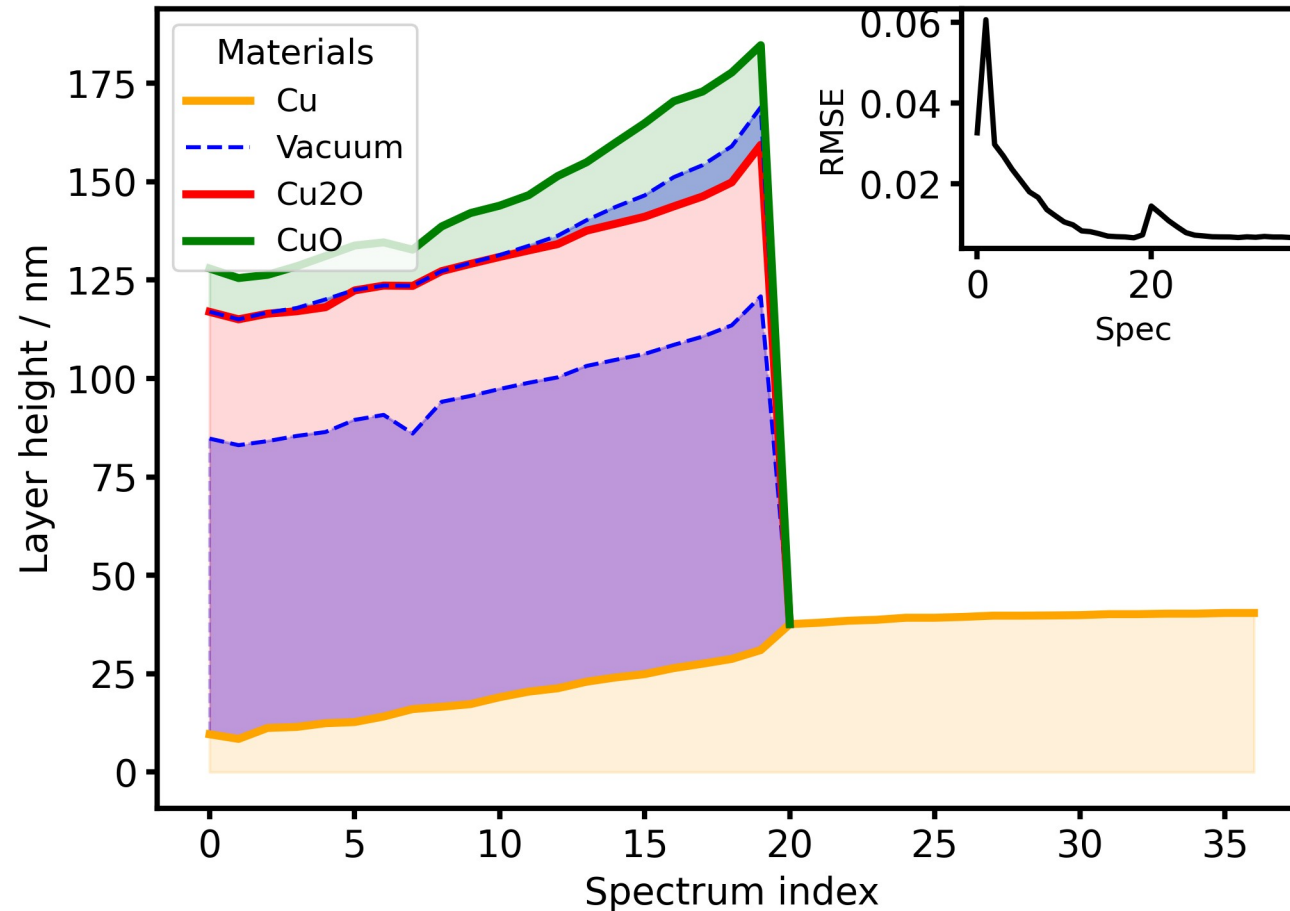
Cu – Cu₂O + CuO (sphere) – CuO



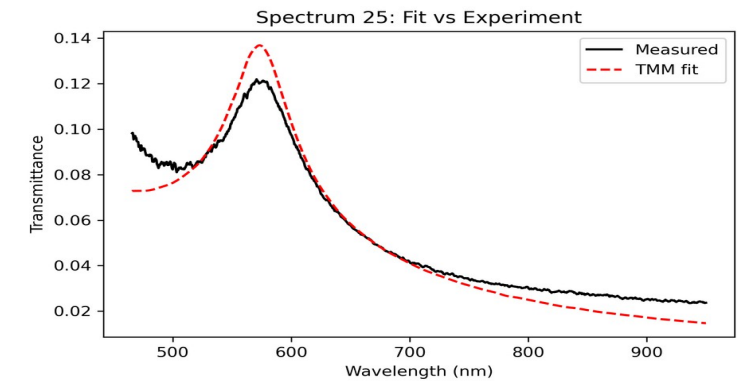
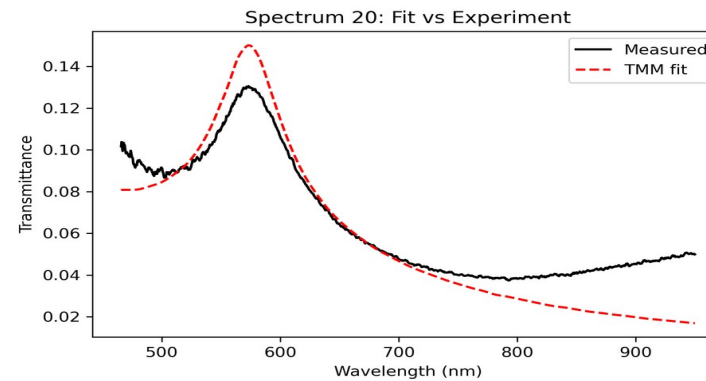
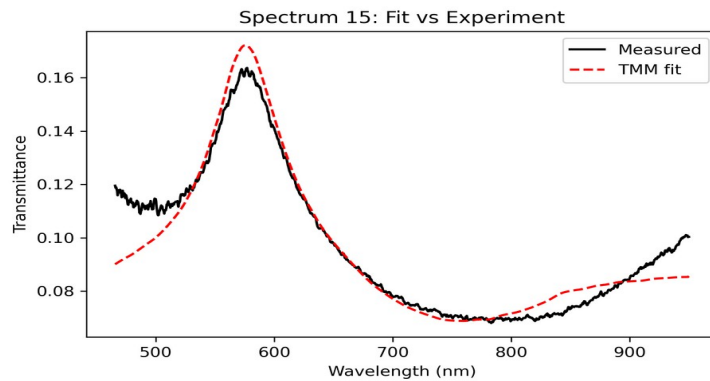
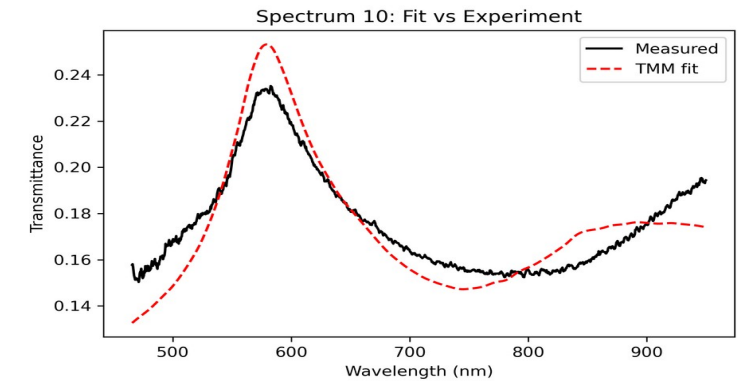
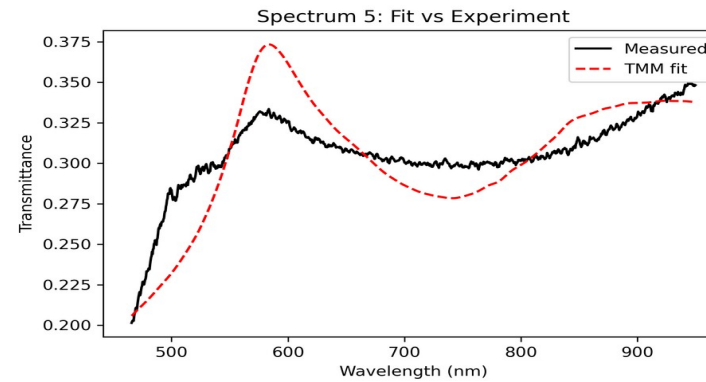
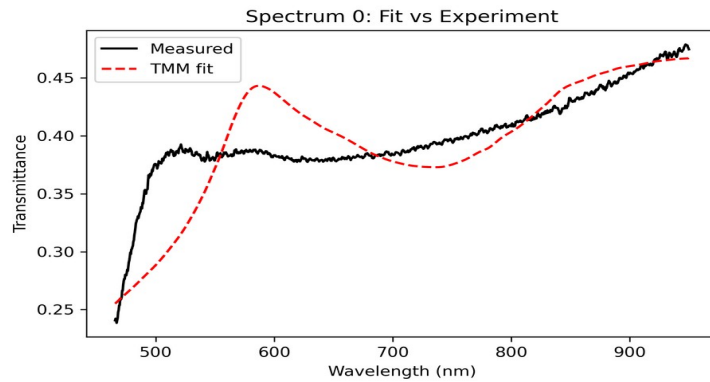
Cu – Cu₂O + CuO (sphere) – CuO



Cu – Cu₂O + Vacuum (sphere) – CuO + Vacuum (sphere)

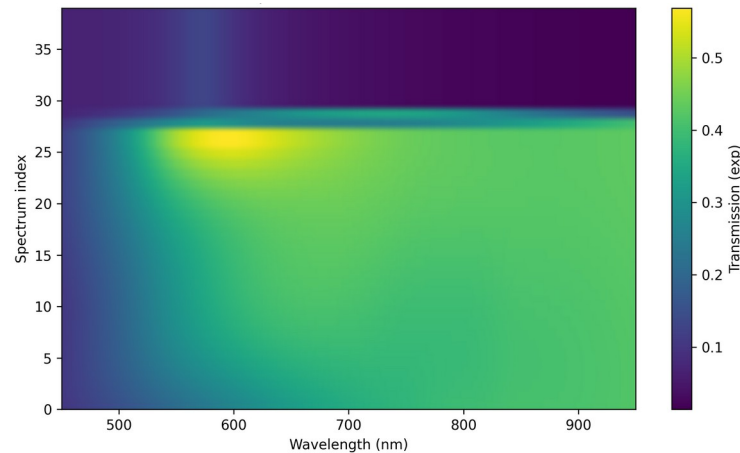
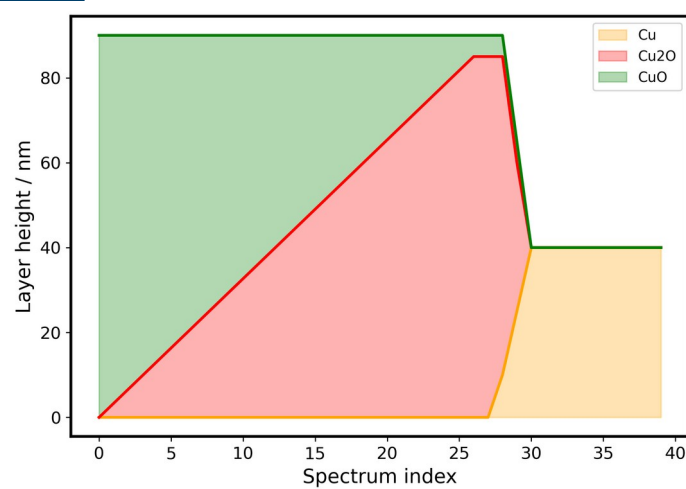


Cu – Cu₂O + Vacuum (sphere) – CuO + Vacuum (sphere)



Benchmarking of the Genertic Algortihm

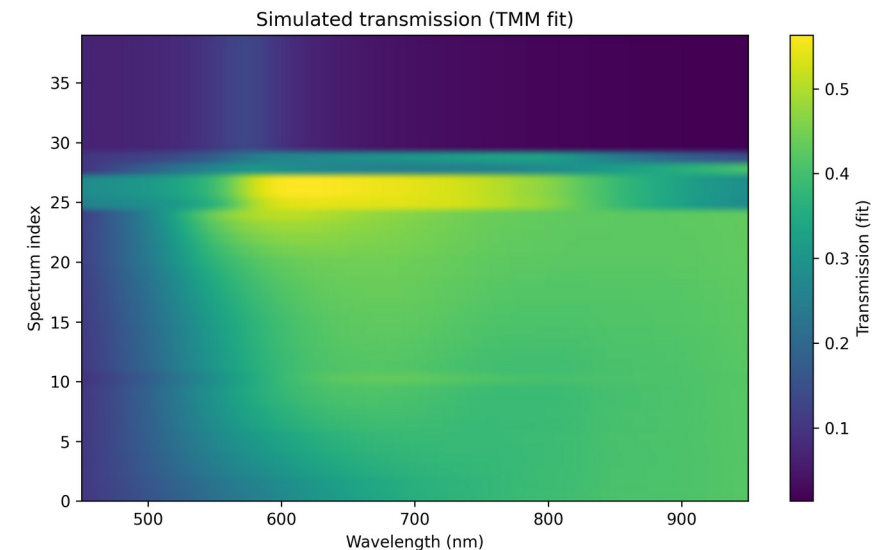
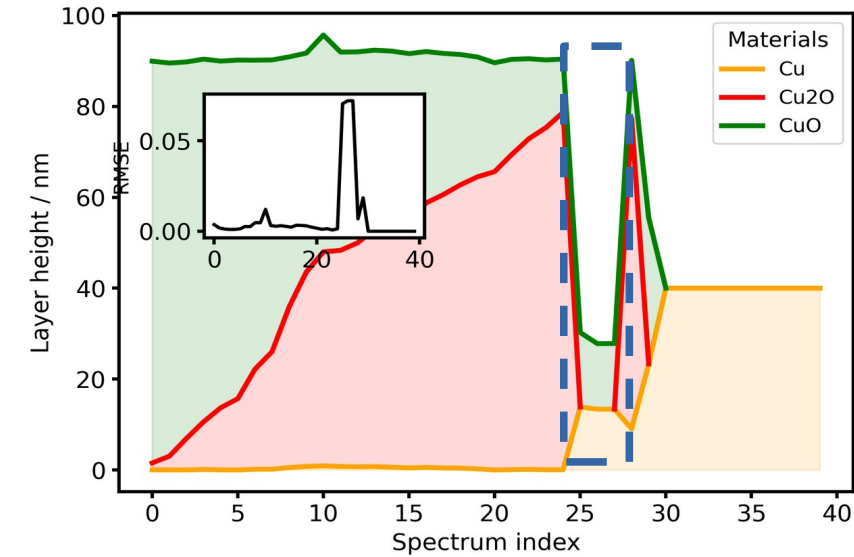
- Simulate Spectra (that resemble a GRIN lens):



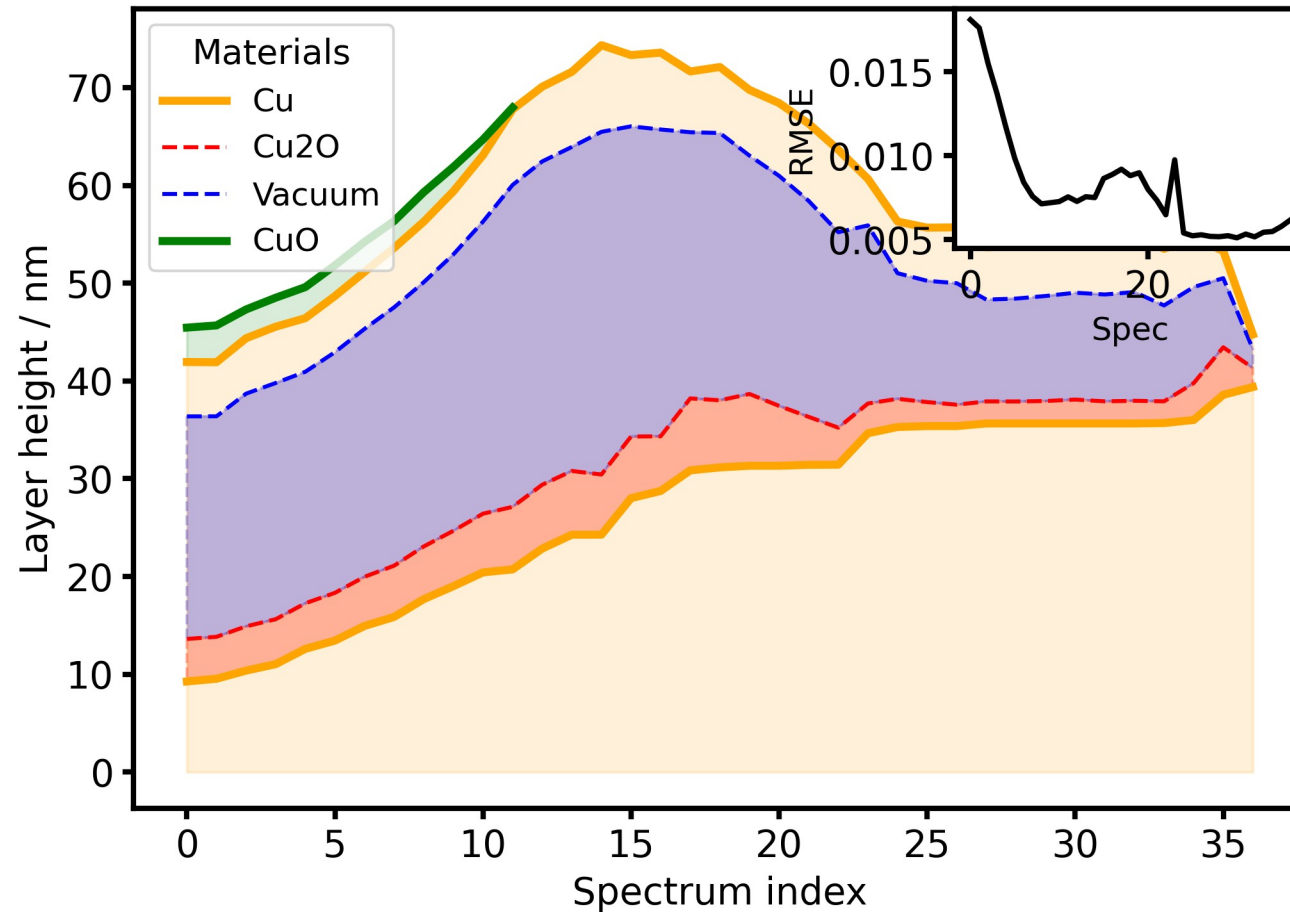
Fit with GA



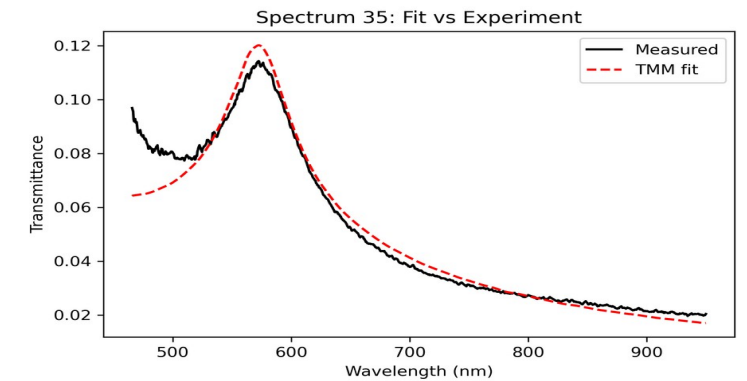
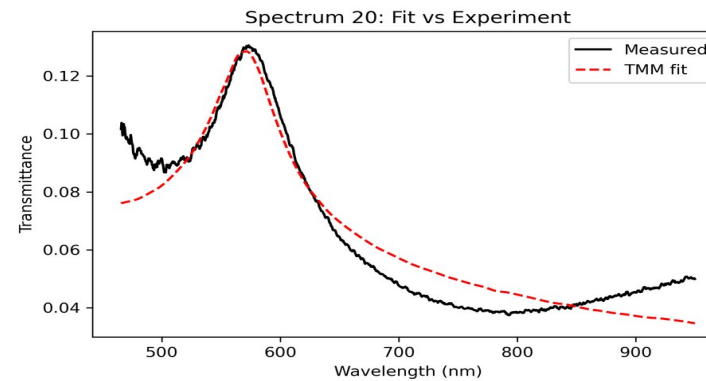
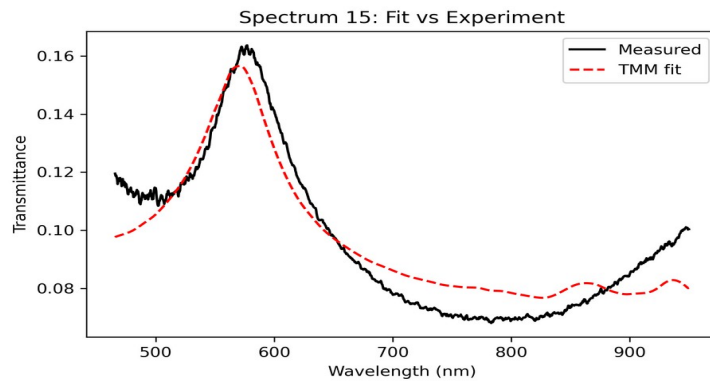
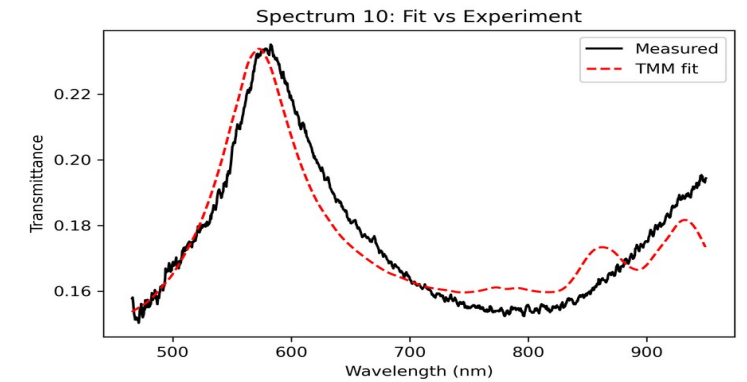
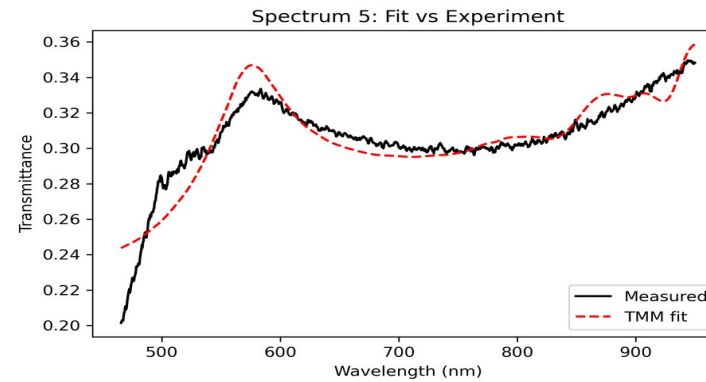
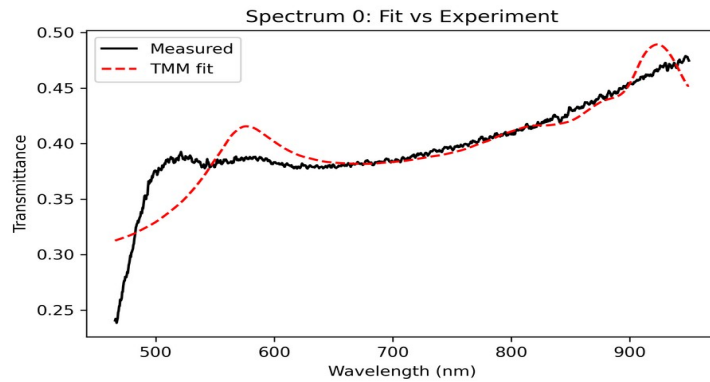
Local minimum



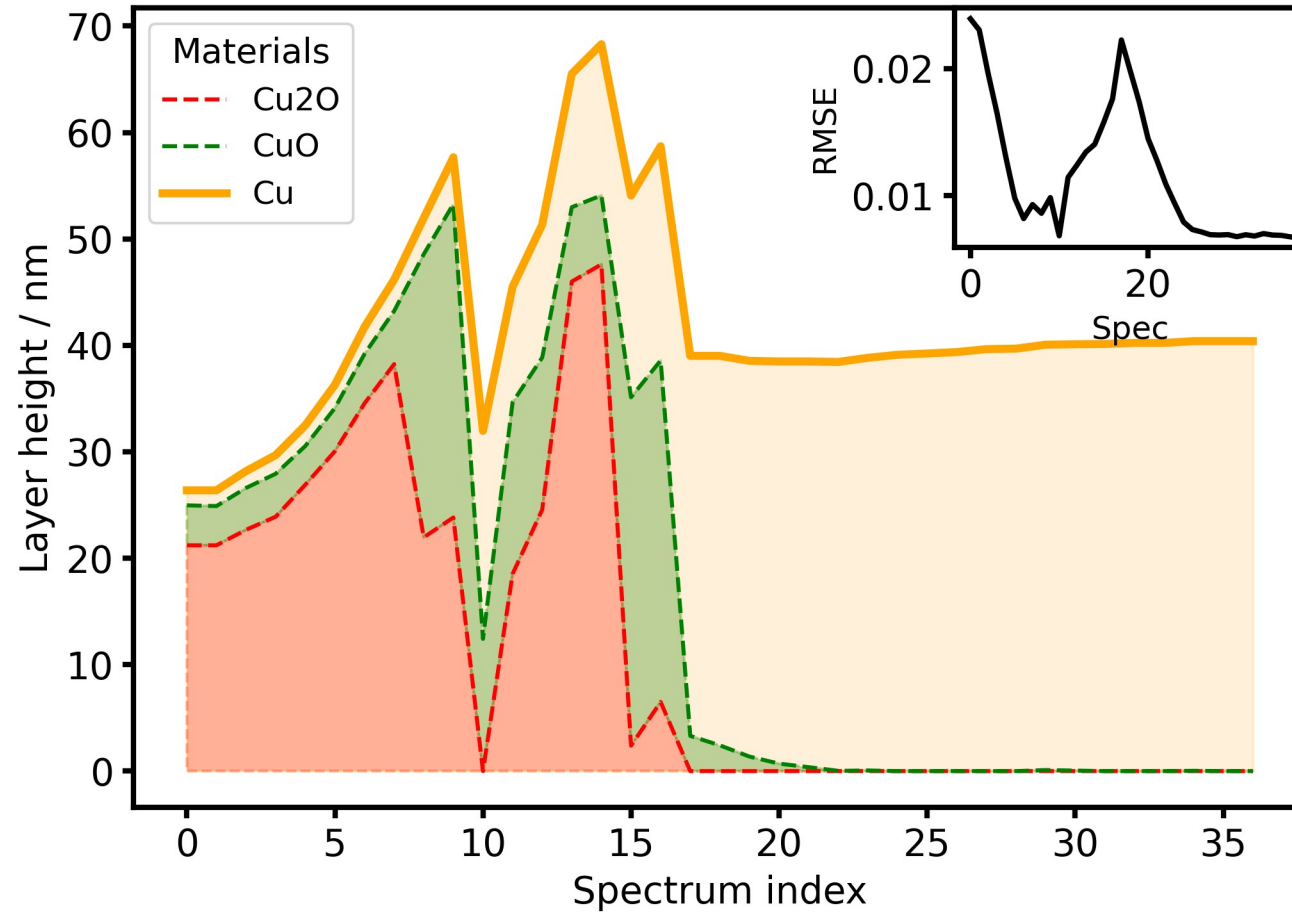
Cu – Cu + Cu₂O (sphere) + Vacuum (sphere) – CuO



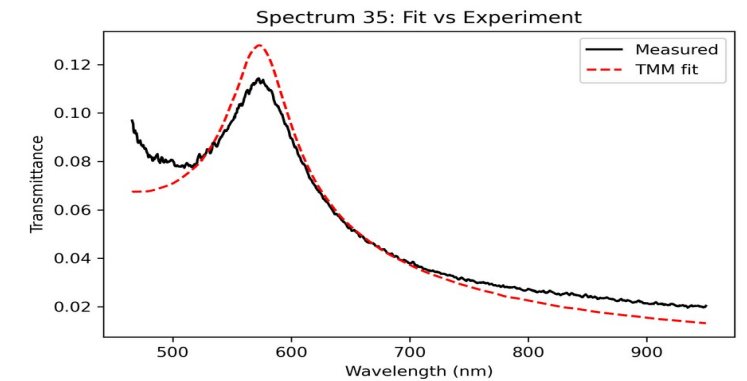
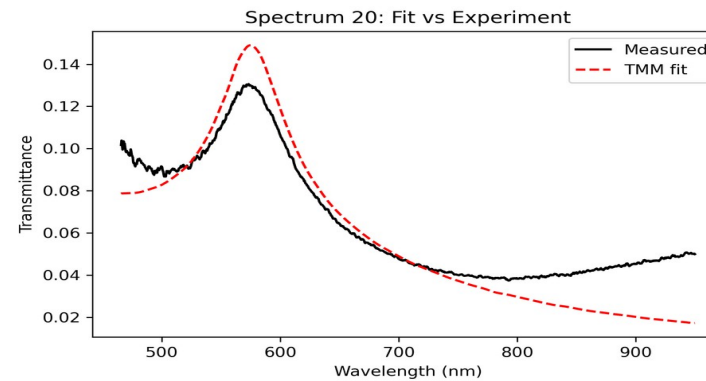
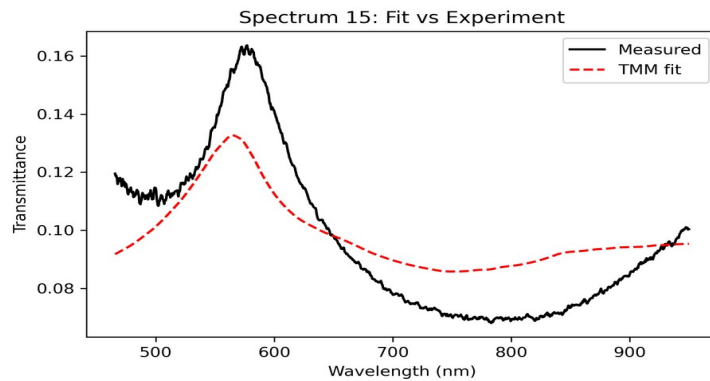
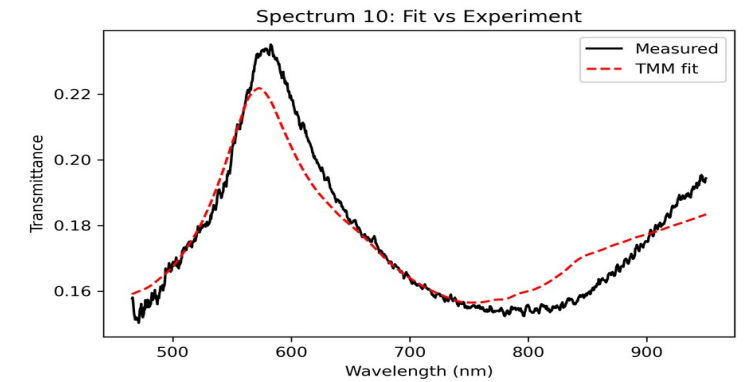
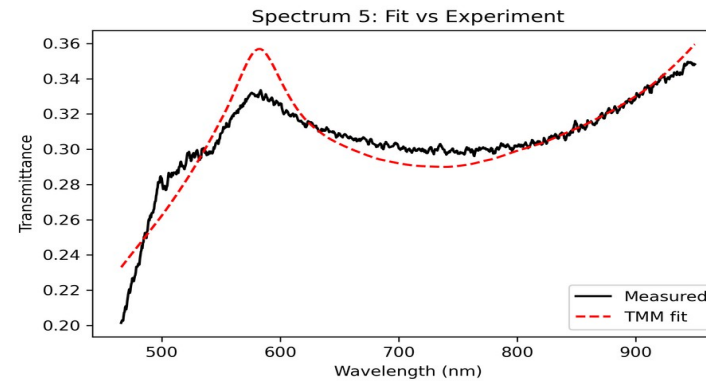
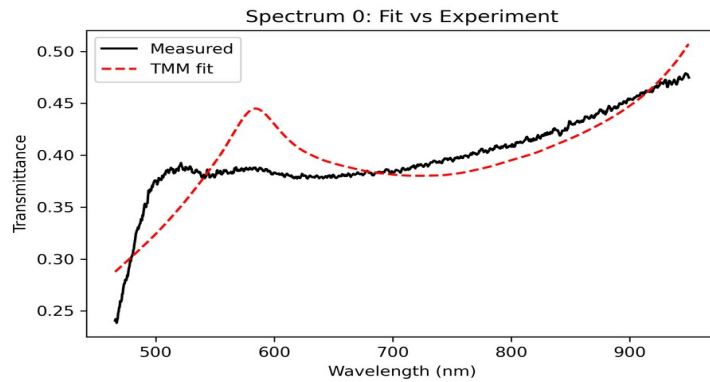
Cu – Cu + Cu₂O (sphere) + Vacuum (sphere) – CuO



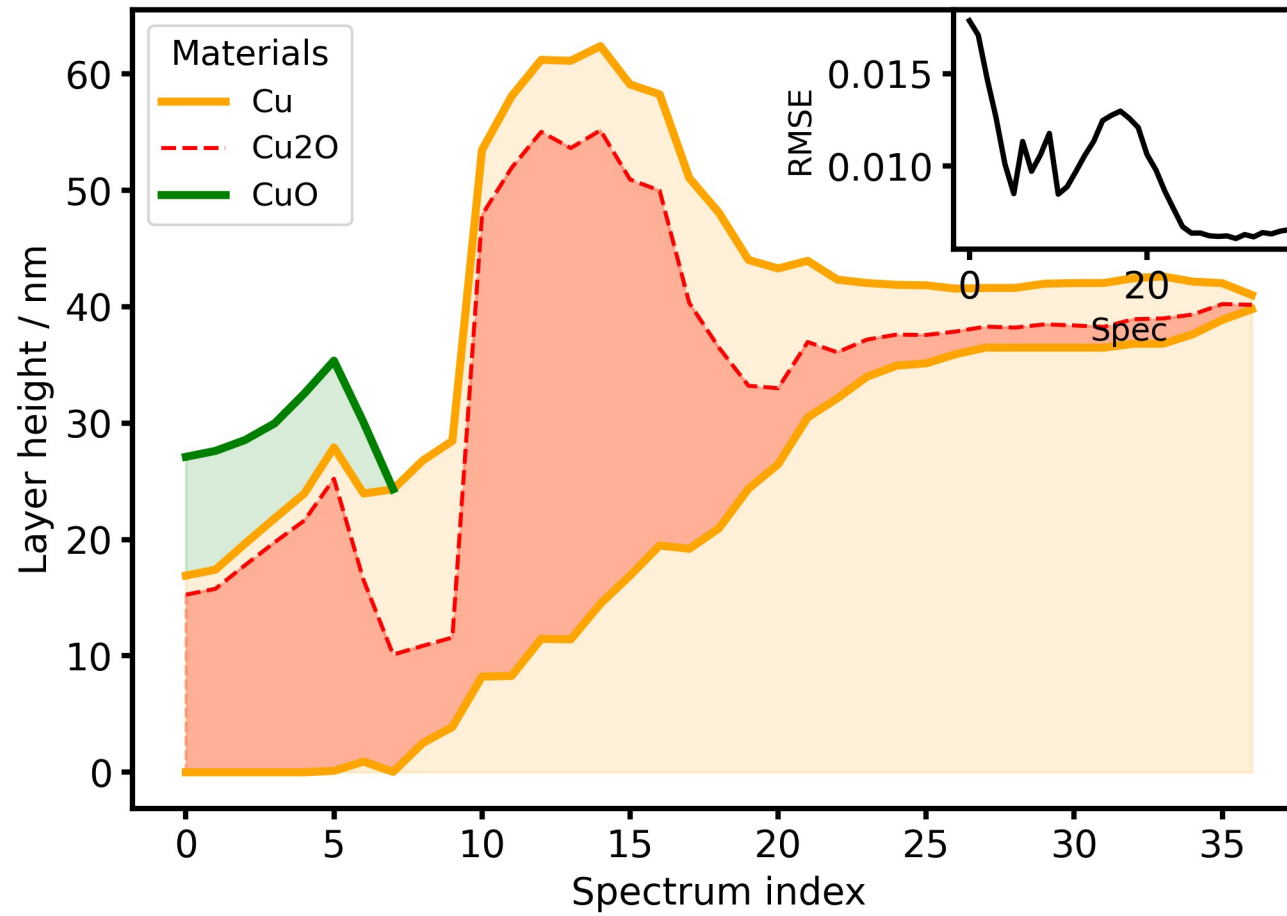
Cu + Cu₂O (sphere) + CuO (sphere)



Cu – Cu₂O + Vacuum (sphere) + CuO (sphere)

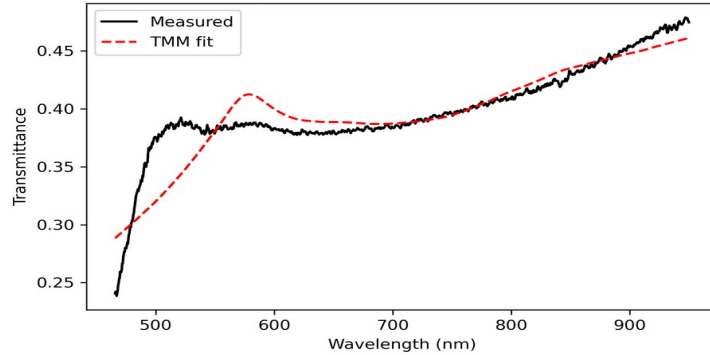


Cu - Cu + Cu₂O (sphere) - CuO

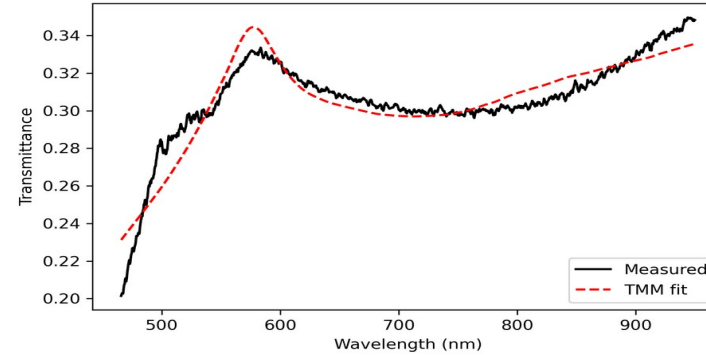


Cu - Cu + Cu₂O (sphere) - CuO

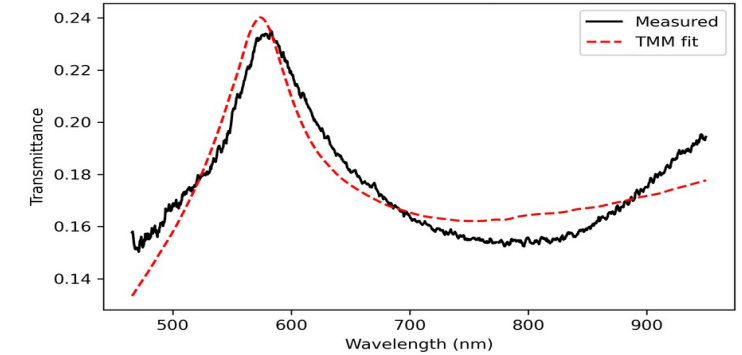
Spectrum 0: Fit vs Experiment



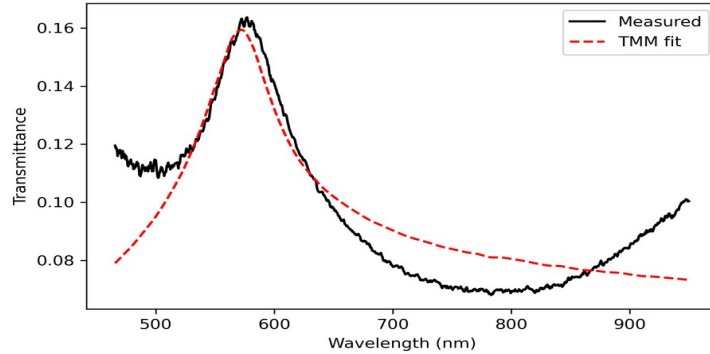
Spectrum 5: Fit vs Experiment



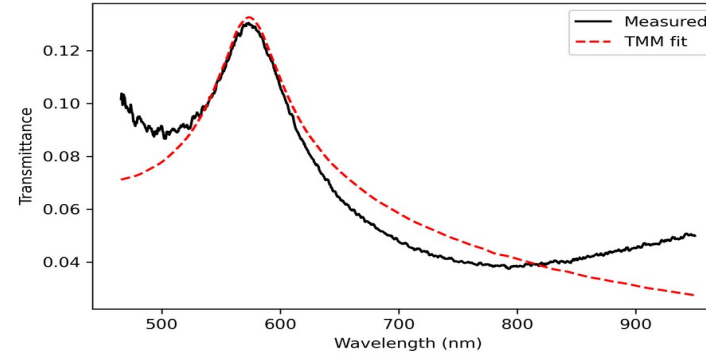
Spectrum 10: Fit vs Experiment



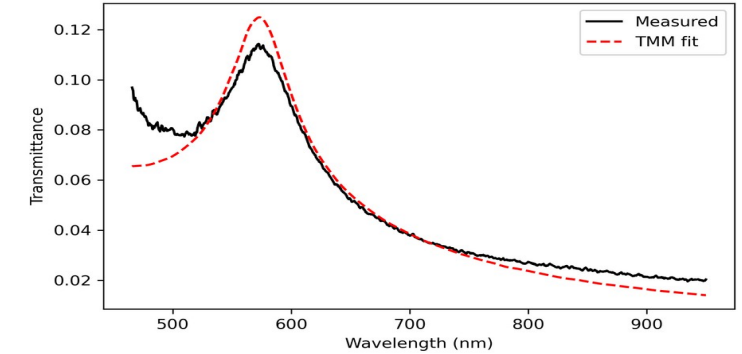
Spectrum 15: Fit vs Experiment



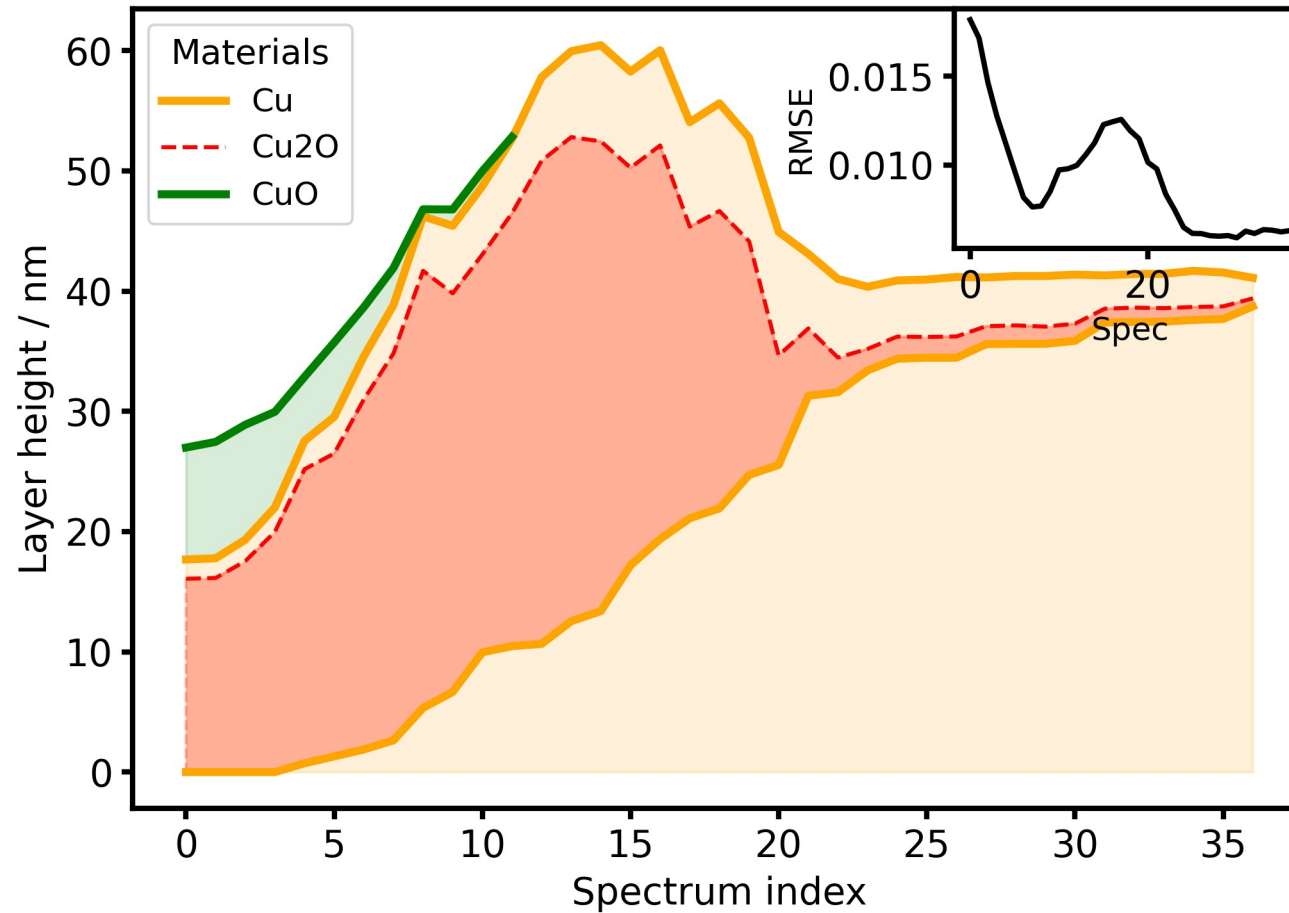
Spectrum 20: Fit vs Experiment



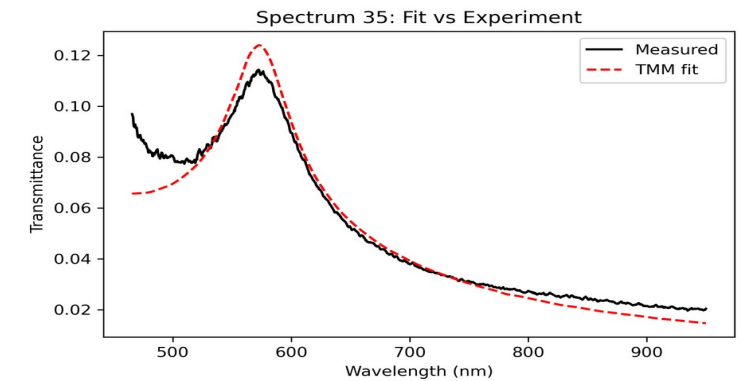
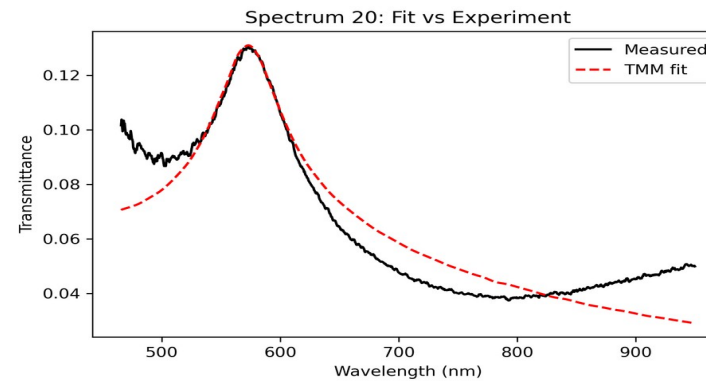
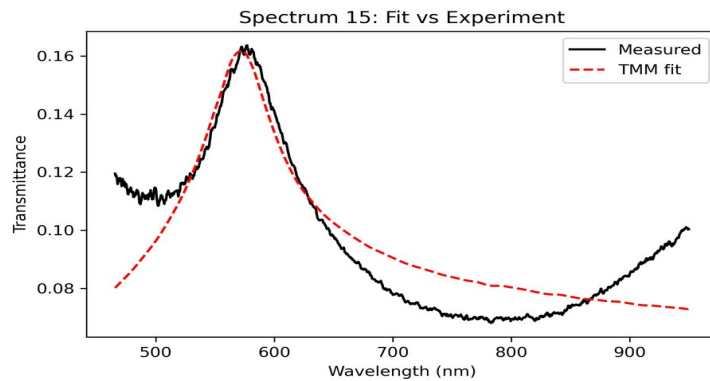
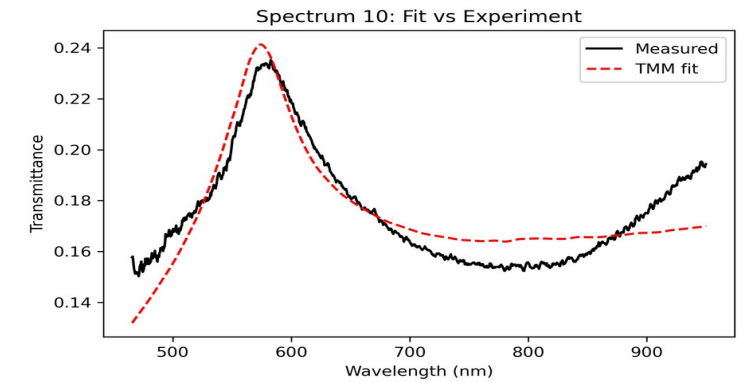
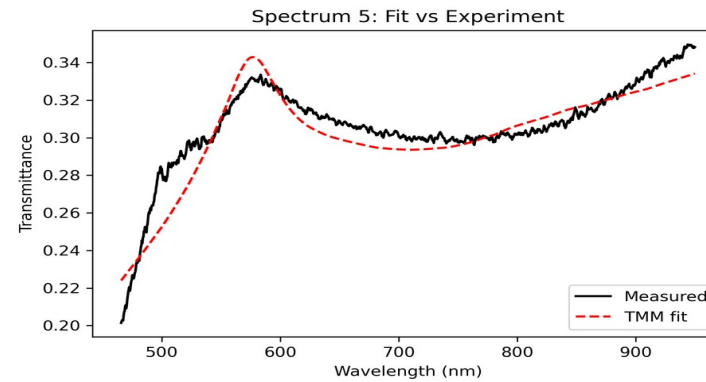
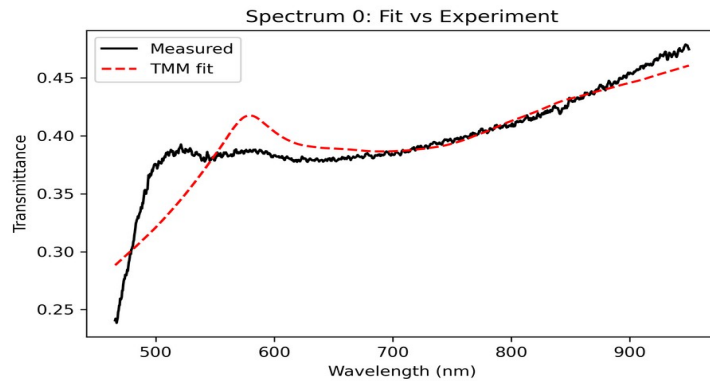
Spectrum 35: Fit vs Experiment



Cu - Cu + Cu₂O (chains) - CuO



Cu - Cu + Cu₂O (chains) - CuO



Lens 3W, 120s Fitting Results

